ReadSoft FORMS

FORMS 5-3

Handbook

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1 Introduction

What is FORMS?

ReadSoft FORMS is a software application for automatically capturing and managing information from forms. FORMS scans, interprets, and verifies the forms, then transfers the data to a target system.

The forms do not need to be specially prepared for optical reading, and they can be filled in with printed or handwritten information.

FORMS' flexibility makes it suitable for a wide range of production environments, from small single-user systems to large, multi-tasking, multi-user environments that process thousands of forms per hour.

The advantages of using FORMS

FORMS offers major advantages compared to manual forms processing. Many forms are captured automatically without any manual data entry at all. Other forms need some manual corrections, but in most cases FORMS has completed most of the work, and only a few characters per form must be entered.

Manual data entry is usually reduced by 90% or more.

When corrections are necessary, they are made in an efficient, user-friendly environment where the form being edited is shown directly on the screen.

Data capture is transformed into a faster, more accurate, cost-efficient process that absorbs variations in the workflow. FORMS' flexibility makes it suitable for both small and very large production volumes.

The five modules

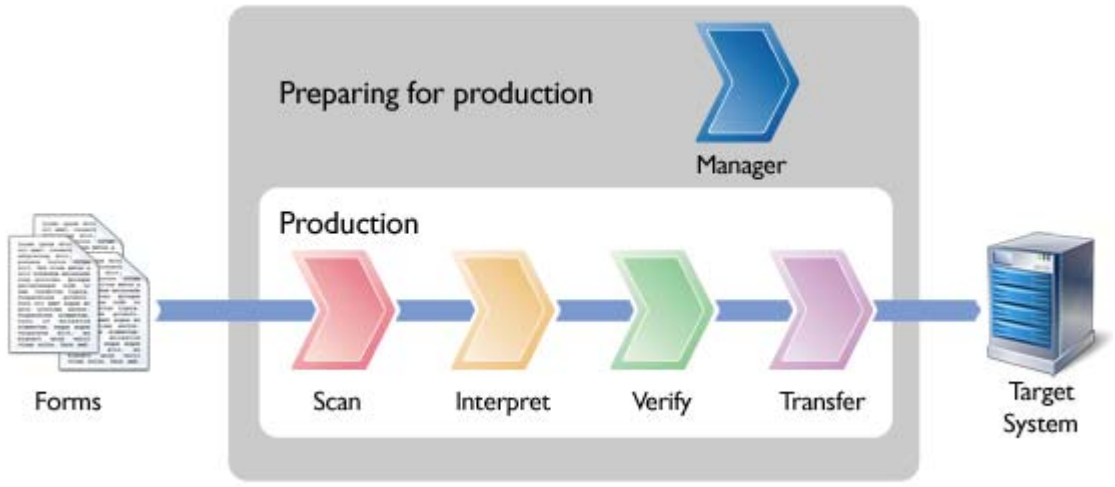
FORMS has a flexible modular design consisting of five separate modules with a user-friendly interface. The processes are independent of each other, and the modules can be run separately or simultaneously, on one computer or in a network. In this way, work can be planned to optimize computer capacity and personnel efficiency.

There are two basic types of work you do with FORMS:

- *Preparing for production* involves setting up FORMS before forms are processed. You do this using the Manager module, which is described below. Chapter 4 contains a tutorial that takes you through the steps in preparing for production.

- *Production* involves scanning, interpreting, and verifying forms, as well as transferring data from processed forms to an output file or target system. The production modules (Scan, Interpret, Verify, and Transfer) are described below. Chapter 5, the second part of the tutorial, teaches you how to process forms through the production modules.





The workflow in FORMS

Below is a description of each module and what you use it for. They are listed in the order in which you use them.

Manager

You use the Manager module for system administration and, most importantly, for creating and editing form definitions.

A *form definition* is a scanned image of a blank form, plus a digital template of the *fields* that will later contain information to be interpreted and processed. A *field definition* includes the position of the field, the type of information it will contain, controls for testing the validity of the data FORMS finds in the field, and more.

Each form definition has a unique name and can easily be saved to a form definition database.

The form definition also includes a *transaction description*, in which you specify the format and layout of the output data you want FORMS to generate when a form has been completely and correctly interpreted.

You can also set up *job descriptions* using the Manager. A *job description* is a set of instructions which describes what you want FORMS to do. It tells which form definitions you want to process, which module to use and, depending on the module, how you want FORMS to work.

Scan

Scan uses a scanner to scan and identify forms that match the form definitions listed in the job description that you open before scanning forms.

Scan recognizes the type of form that is scanned and saves an image of the form.

If FORMS finds a form that it cannot identify, it stops, displays the form, and tells you that the form cannot be identified.

You can have Scan run all your forms at once, or you can divide them into stacks of a specific number of forms to be processed together (*batches*).

You can indicate specific data (for example, date, time, and number) to be included with these batches so that you can easily find a particular form later, if you need to. Scan can also add this *batch data* automatically.

There are other functions in Scan, as well, such as exporting form or field images to files for archiving, and saving extra pages as appendices to the forms.

Interpret

Interpret interprets data from either forms which Scan has saved or from other forms stored in image files. It also validates that data in accordance with validation routines you specify in the form definition.

Interpret is easy to use. All you have to do is open or create a job description and click **Start**. Interpret only processes forms that match one of the form definitions listed in a job description.

The results of the interpretation are saved in FORMS' database. FORMS gives each field and each form a *status*. Statuses indicate how complete and error-free fields and forms are, and they determine the order in which data is presented to operators in the Verify module.

Verify

Using Verify, you manually type information in forms with uninterpretable characters, missing data, or errors detected by validation routines. You can also check data with no error status if you wish to confirm all information on a form.

You decide which forms you want to verify and correct. You can limit your work to forms and fields with a certain status and choose from many options for how you want FORMS to interact with you during the process. These settings are all made in the *Verify job description*.

For example, in *fast verify mode*, FORMS displays fields comprised of up to seven characters from different fields on the same form. This allows you to verify or correct them very quickly.

When FORMS' interpretation is uncertain, you can also choose whether or not you want to see tentatively interpreted data while you work.

Fields containing information which must be typed manually (called a *field complement*), such as a field with a handwritten name and address, are completed in a special dialog.

Transfer

You use Transfer to transfer the data from completely processed forms from the FORMS' database to an output file or target system. Data is transferred only from forms that match the form definitions listed in the job description that you open before starting Transfer.

The data is transferred in transactions. A *transaction* consists of a number of *records*, one for each form.



Transfer formats the output data according to the contents of *transaction descriptions*. A *standard transaction description* is part of every complete form definition, but you can also specify a *start transaction description*, *stop transaction description*, and/or *archive transaction description* in your Transfer job description.

Only data for forms with *Complete* status are transferred. As data from each form is successfully transferred, the form and all information attached to it is deleted from FORMS' image and form databases unless you specify otherwise.

Installation and setup

FORMS is simple to install, and can be installed on a local computer or in a network. The installation instructions are made available to you when you run the installation program contained in the FORMS CD.

Using this book

The *FORMS 5 Handbook* replaces both the *Reference Manual* and *Getting Started*, which were supplied with previous versions of FORMS. This book includes primarily overview information so that new and occasional users can orient themselves enough to be able to use *FORMS Help* effectively.

A tutorial is included for new users who want to learn the hands-on basics of FORMS. The tutorial is found in chapters 4 and 5.

If you are accustomed to using ReadSoft's previous books, take a moment to look at the table of contents and get an idea of how this book is organized.

- Chapter 1 explains what FORMS is, and the advantages of using FORMS.
- Chapter 2 gives you an overview of the program. It touches briefly on most areas of production and lists key words that you can look up in the online Help index for more information.
- Chapter 3 introduces you to version 5's new interface, as well as providing instructions on how to use the Help system.
- Chapter 4 runs through the steps that help you to get started in preparing for production. A sample form is included with the FORMS installation.
- Chapter 5 continues the tutorial with a presentation of the steps required to process forms in each of the four production modules.

While you are becoming acquainted with FORMS, the glossary in the back of this book and the one in Help can be helpful in explaining new terms. If you are looking for an explanation of a specific function, you will probably find it quickly and easily if you look in the index of the Help file.



After FORMS is installed, you can work through the tutorial in Chapters 4 and 5, using either the sample form included with FORMS or your own form. The tutorial shows you how to perform the necessary steps in the correct order, and you will learn enough about FORMS to be able to refer to Help for additional information that your specific circumstances may require.

Chapter 4 requires only your PC and the FORMS Manager module. For Chapter 5, you will use the other four modules and a scanner (if you are using paper forms).

Typographical conventions

The following conventions are used throughout product documentation:

Menu names, commands, and dialog selections appear in **boldface**.

Page numbers in the index that refer to a page with a definition appear in **smaller boldface**.

Names of keys on your keyboard appear in a special typeface with **SMALL CAPITAL LETTERS**.

Names of FORMS internal field statuses and form statuses are *italicized* (for example, *Interpretation error*). Occasionally, normal text is also italicized to add emphasis.

Field names and variable names appear in this typeface.

Variables that have to be replaced by the actual value appear in the same typeface as above but are *italicized* (for example, *Fieldname*).

Names of files and paths are written in `courier`.

2 Overview of FORMS

This chapter gives you a summary of some of the major tasks you will do in FORMS. It also provides the names of Help topics where you can find more detailed information.

Understanding form definitions

Before you start processing forms, you must describe the forms—what they look like and what type of information they contain—to FORMS. You do this by using the Manager to create form definitions. A *form definition* is a software-based description (like a template) of your form as it is to be processed by the other four FORMS modules.

The form definition is necessary for all other FORMS modules, and a carefully prepared form definition results in high interpretation rates and reduces verification work.

You will learn how to create a form definition in the tutorial (see “Create a form definition” on page 31).

A form definition is based on either of the following:

- A blank form (that is, a form that is not filled in) that you scan

  When you scan in a blank form, it is important that you use the same scanner or scanner combination and the same scanner settings that the filled-in forms will later be scanned with. The instructions below include how to adjust scanner settings.

  If only filled-in forms are available, choose one that has the least number of marks. You can erase these as you create the form definition.

- An image file which you read (open)

  A form definition based on an image should only be used to process images from files. Base your form definition on an image file only if you intend to always bypass Scan.

Hints on what to look for in Help

Five stages to prepare for production	Scanner settings
Adjusting an image after scanning	Appendices
Erasing marks from a form definition	Defining fields
Field validations	Form validation
Supplement filled-in data	Rescanning an image for a form definition
Checking the form definition for errors	Activate form definition

Determining the content and layout of the output

Understanding transactions and transaction descriptions is essential for effective use of FORMS. A *transaction description* is like a template for the output data from filled-in forms. It specifies the content, format, and physical layout of the information that Transfer will send to your target system after forms have been processed.

Of the three types of transaction descriptions, only the *standard transaction description* is required. The other two are the *start* or *stop transaction description* and the *archive transaction description*.

You can find detailed information about all three types in Help. Before you begin creating transaction descriptions for all the forms your organization will be processing, read also about *using a common layout for your transaction descriptions* in Help.

The tutorial covers the steps necessary to create a transaction description (see "Create a standard transaction description," starting on page 66).

Hints on what to look for in Help

Using a common layout for your transaction descriptions

Create standard transaction description

Transaction description – Three types

Sets of forms

A *set* is two or more form definitions that should be scanned, interpreted, verified, and transferred together, where the sequence of the forms is important.

FORMS treats the forms in a set as one logical unit. Examples of sets are multi-page market research questionnaires and forms with enclosures or attachments to be identified.

Before you create a set definition, create form definitions for all forms to be included in it. Then, select the form definitions to be included in the set definition and define the *set logic*. Set logic specifies, for instance, the order of the forms and the number of forms in the set.

A set definition can be selected in job descriptions (described in the next section) in every FORMS module, just as a form definition can. You can run only one set definition in a job, and you cannot combine a set definition with other form definitions in a single job.

Sets are processed differently than individual forms. Scan uses set logic to:

- Ensure that forms within a set are scanned in the correct sequence
- Ensure that a set contains the correct number of forms
- End one set and begin the next (called a *set break*)

Interpret processes forms set by set (rather than form by form) in the order scanned. However, the statistics compiled by FORMS are not affected.

FORMS checks set sums, if you specify them. In Verify, incorrect set sums are shown on the screen, including all the fields included in the set sum, even if they are from different forms. You can also use external validation routines for sets.

Transfer can process sets in two different ways: If a transaction description is defined for the set, one record *per set* is transferred. If no transaction description is defined for the set, one record *per form* is transferred, in the order scanned.

Hints on what to look for in Help

Set definition

Set logic

Sets

Transaction description for set

Managing production through job descriptions

A *job description* is a file containing your instructions to each of the production modules (Scan, Interpret, Verify, and Transfer). You can create and edit a job description using the Manager or any of the other modules. However, in the other modules you can only change those details that pertain to that particular module.

We often refer to job descriptions in connection with a production module name, such as the *Verify job description*. This simply refers to details specified for that module on the **Verify** tab of the **Job description** dialog. In fact, details for every production module are saved in each job description file.

A *Scan job description* tells Scan which forms to look for during scanning and how to handle the images.

An *Interpret job description* tells Interpret which forms to interpret and validate.

A *Verify job description* tells Verify which forms with which status you want to verify. You also have many options for how you want FORMS to interact with you during the verification process.

A *Transfer job description* tells Transfer where to send selected data from specific forms.

It is not necessary to use the same job description file for all forms of a specific type. You can create different job descriptions for the same form or set definitions in order to divide up your work in different ways.

You can create as many job descriptions as you like for different purposes, and you can edit them, using either the Manager or the respective module, right up until the time you use them.

Any time you create a job description, default settings are in effect unless you change them. That is, you can create a usable job description just by opening the dialog, choosing a form or set definition, closing the dialog and saving the job description when prompted.

“Create a job description” (page 74) walks you through the steps needed to create your own job description.

Hints on what to look for in Help

Scan job description

Interpret job description

Verify job description

Transfer job description

Job list, create and edit

Customizing job descriptions

Production

While you create form definitions and manage forms using the Manager, the four phases of production require the remaining four modules (Scan, Interpret, Verify, and Transfer). The phases of production are described below. The modules themselves are described on page 7.

You can do all phases of production on one PC, or a different PC can be used for each phase if the PCs share databases. To start any production process, you must first open or create a *job description* for that module, then start the job by choosing **Start** on the **Job** menu.

The tutorial provides details for running each production module (see chapter 5).

Hints on what to look for in Help

Flow of forms in the system

Job description overview

Form definition...

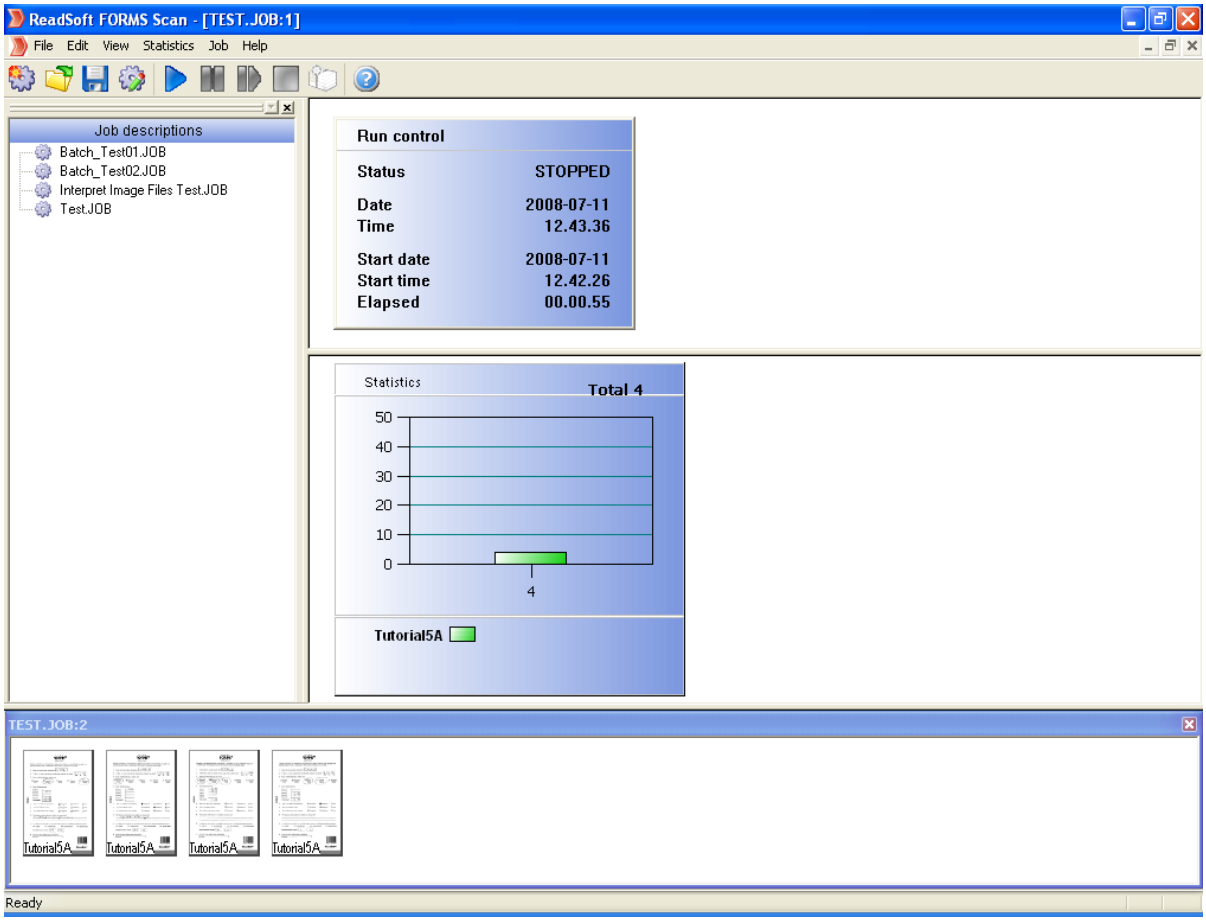
Transaction description...

Scanning and identifying forms (Scan)

In Scan, filled-in paper forms are scanned, and the images of the forms are captured and saved in a database. FORMS can identify many different forms in the same job, which makes it possible to mix different forms in the scanner's document feeder without presorting them.

Skip scanning if you are importing image files to FORMS.





Scan is used to capture the contents of filled-in paper forms.

When you start Scan, an information box displays the current system status, date, and time (the date and time formats in accordance with your Windows settings). When you start scanning, the start date, start time, and run (elapsed) time are also displayed in that box. The screen also displays a bar chart showing how many forms of each type have been scanned.

The lower half of the screen shows thumbnail images of the forms being scanned. You can view a form image at any time during scanning by double clicking its thumbnail. By right-clicking a thumbnail image, you can perform several operations.

During scanning, a number of events can require your intervention. For example, an empty or jammed document feeder causes messages to be displayed. You choose the required course of action to continue. When Scan cannot recognize a form (if it is upside-down, for example), a message that describes the event is displayed, and you can make corrections and continue or stop scanning.

Hands-on details for running Scan begin on page 87, “Scan and identify forms.”

Scan supports *batching* of scanned forms, which means you can process stacks of forms together in order to better keep track of them.

Hints on what to look for in Help

Scan	Scan job description
Scanning forms/sets/batches	Batch
Storage of scanned documents	Options for Scan job descriptions
Accept unidentified forms	Identify rotated/upside-down forms
Scanner cards and drivers	Scan statistics
Endorser	Appendices
Job control	

Interpreting and validating scanned forms (Interpret)

The Interpret module both interprets the data from forms and validates that data in accordance with the validation routines you specified in the form definition.

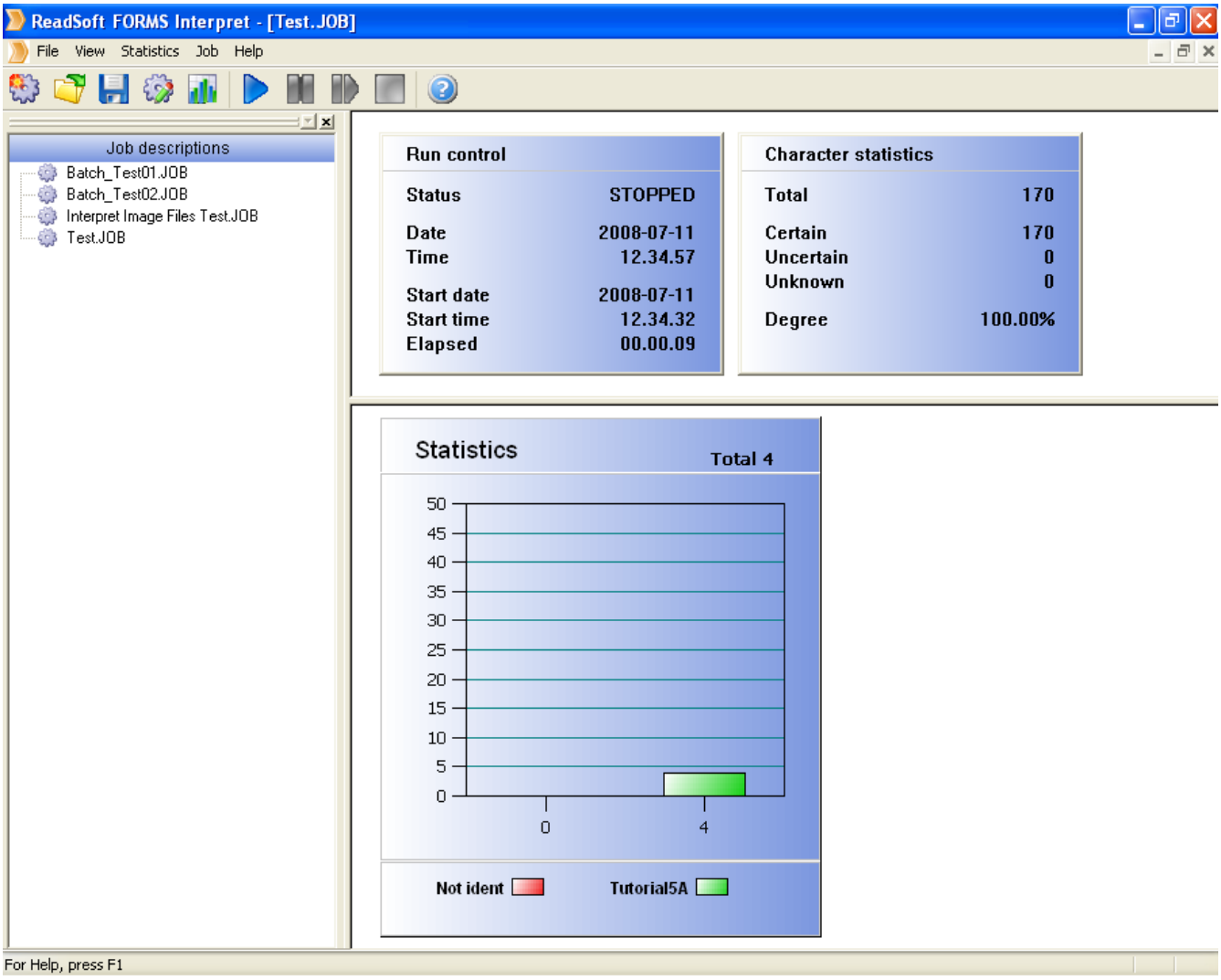
Interpreting a form means finding all defined fields on the form, deciding what the fields contain, and recording the field values in the FORMS database. *Validating* a form means comparing the contents of fields with validation rules that you have defined for the field.

There are no operator dialogs in Interpret (that is, there is nothing for you to do once the job is started), unless you are interpreting from imported image files and have indicated that you want to see a source configuration dialog each time a job is started.

When you start Interpret, an information box displays the current system status, date, and time (with the date and time format in accordance with your Windows settings). When you start an Interpret job, the start date, start time, and run (elapsed) time are also displayed. The screen also shows a bar chart indicating how many forms of each type have been interpreted.

Unless you deselected **Show character statistics** in the **Interpret job options** dialog, another control box shows the total number of characters examined, the percentage of those (**Degree**) that were interpreted, and the level of certainty for those characters that were interpreted.





Interpret is used to interpret the data from image files or scanned paper forms.

When Interpret has finished processing a form, it sets a status for the form and for each of its fields depending on the results of the interpretation, and saves this status information in the database. For example, if all fields on a form are interpreted and validated without error, and there are no fields for which you have requested complements or retyping, the statuses are all set to *Complete*. The *Complete* status is only given to forms in which every field is complete and correct.

If you interpreted forms from imported image files and **Accept unidentified forms** was not selected as an Interpret job option, check the directory you specified for unidentified forms. It may contain images that FORMS rejected. The most common reasons are:

- The form definition was not included in the job description.
- There was too much skew in the image. Normally FORMS tolerates up to 10 degrees, which covers the vast majority of cases.
- Adjustment fields defined in the form definition were not actually present on the form.

Try to determine the problem and then fix it. For example, if the form definition was not included in the job description, open the *Job description* dialog, select the additional form definition, and save the job.

After fixing the problem, use Windows Explorer to move these images back to the forms source directory for another try.

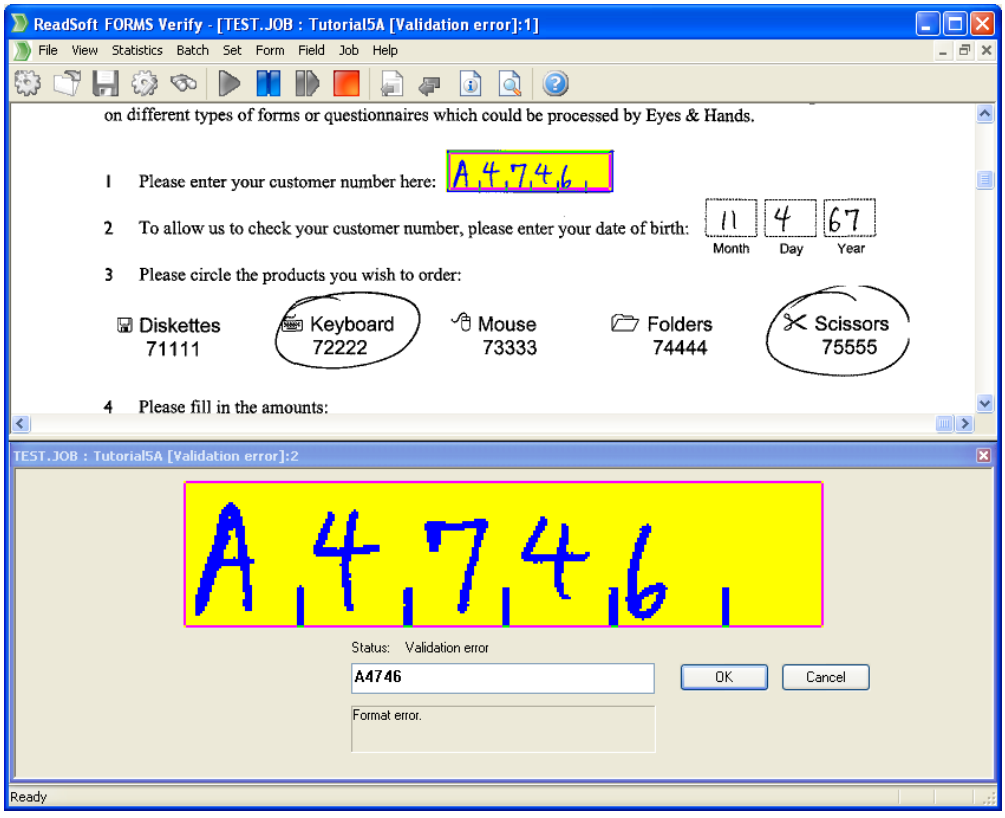
Hints on what to look for in Help

Interpret ation	Status
Interpret job description	Import image files
Interpret statistics	Unidentified forms
Identify rotated/upside-down forms	Accept unidentified forms
Job control	External interpretation engines

Verifying and supplementing interpreted forms (Verify)

Using Verify, you manually type data for forms with uninterpretable characters, missing data, or errors detected by validation routines.

If Interpret processed the forms with a 100% correct result, then these forms bypass Verify (unless you choose to Verify forms with the status *Complete*) and go straight to the Transfer module.



In the Verify module, you correct and confirm data from interpreted forms.

In the Help file, read about how to select the forms you want to verify and correct. You can limit your work to forms and fields with a specific status and choose from many options for how you want FORMS to interact with you during the process.

The image of the form that you see on the screen in Verify is normally a composite image: The background is from the form you used in the Manager to scan, create, and save the form definition. The fields are cut out from each individual form and pasted onto this background.

That is, the fields you see are images of the actual fields on the individual scanned form, while the background is just provided by FORMS based on the form definition.

However, when forms are associated with imported or exported images of entire forms, you can choose to display these full images instead of the composite. See "Verify from image files" in the Help file.

Working through the tutorial will give you a good sense for how the Verify module works (see "Verify forms," on page 94).

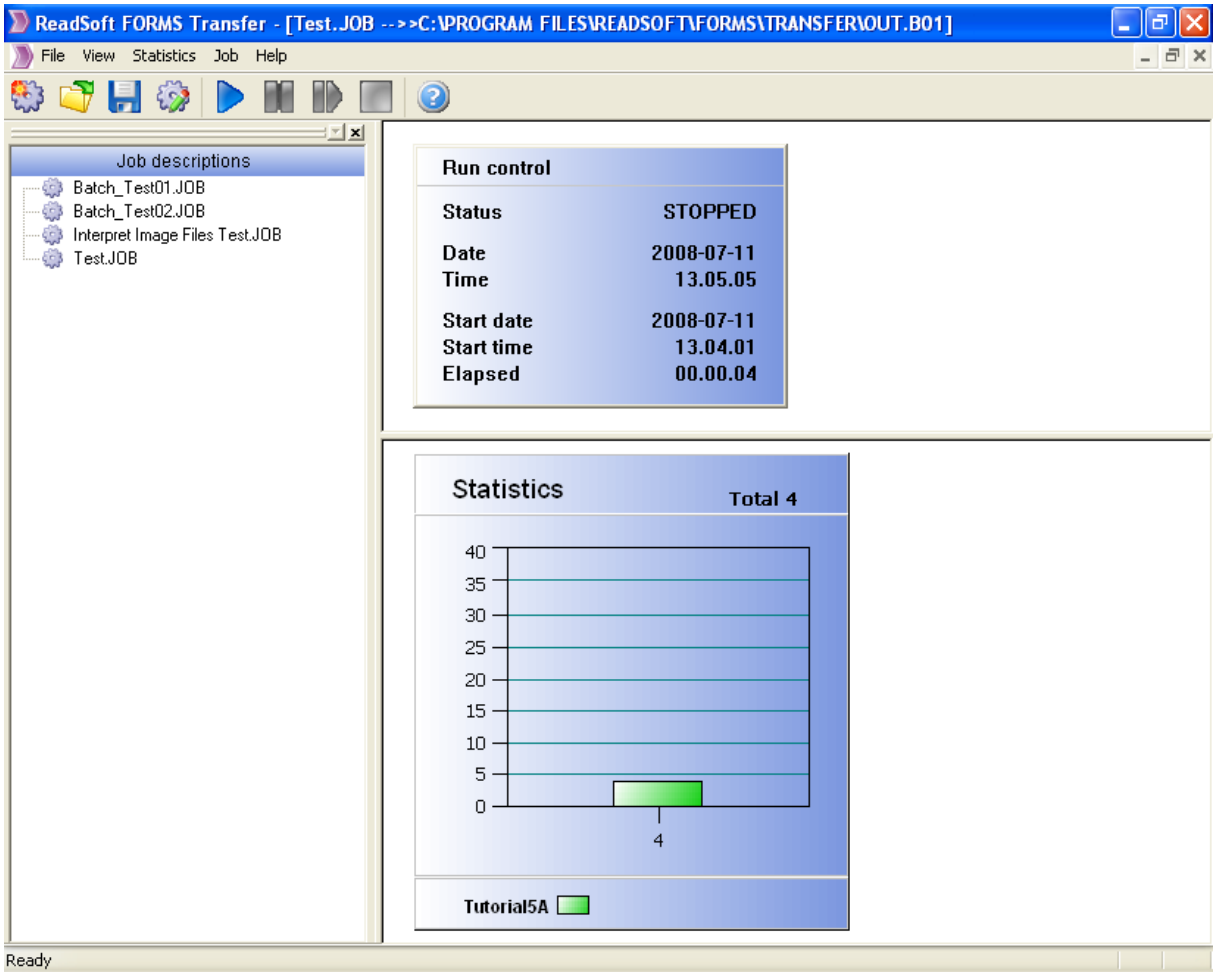
Hints on what to look for in Help

Verify	Status
Verify job description	Options for Verify job descriptions
Verify from image files	Verify statistics
Unidentified forms	Mass verify
Fast verify	Verification order
Buffering	Form information
Job control	Searching the form database

Transferring data from processed forms (Transfer)

You use Transfer to transfer data from processed forms from FORMS' database to a target system. In many cases the data is simply written to a text file. The data is transferred in transactions. A *transaction* is data resulting from processed forms that FORMS sends to an output file or another system after your forms are completely processed. A transaction usually consists of many *records*, where each record contains the data resulting from a single form. Data is transferred only from forms with the status *Complete*.





The Transfer module sends the completed data to the output file or target system of your choice.

When you start Transfer, an information box displays the current system status, date, and time (the date and time format are based on your Windows settings). When you start a Transfer job, the start date, start time, and run (elapsed) time are also displayed in that box, and you see a bar chart which shows how many forms of each type are transferred.

Since Transfer uses transaction descriptions, you can learn much about how Transfer works by reading about them (see “Create a standard transaction description” on page 66).

Hints on what to look for in Help

Transfer	Job control
Transaction description	Output data
Transfer job description	Options for Transfer job descriptions
Standard transaction descriptions	Transfer statistics
Using a common layout for your-transaction descriptions	

System administration

System configuration

FORMS can be used on one PC or in a network, with one or more scanners connected. Help contains information about network products, hardware requirements, troubleshooting, etc.

Databases and files

Form definitions, set definitions, batch definitions, and tables are stored in FORMS' database. Batch report layouts, job descriptions, and user-defined field types are stored in separate files.

Transaction descriptions saved to form definitions and set definitions are part of FORMS' database. Transaction descriptions saved to files are not.

Help describes the management of these items. Look in the index under Save, Copy, Move, Print, Delete, Import, Export, etc.

Hints on what to look for in Help

User authorization

User-defined field type

Import...

Export...

Network...

Using a common layout for your - transaction descriptions

Image files...

Backup routines

Monitoring performance

Database... (size, errors, etc.)

Troubleshooting

File locations

Integration

FORMS' open interface makes it easy to integrate FORMS with a variety of other systems.

You can customize FORMS to suit your requirements, modifying its functionality and adding more features. You do this with your own Visual Basic for Applications (VBA) code or DLL (Dynamic Link Library) functions (written in any language capable of producing DLLs), or by writing Component Object Model (COM) event servers residing in a DLL or EXE file. For example, you can add dialogs, COM automation calls, ODBC access, file manipulation, and much more.

FORMS offers you this access via the **Events** dialog or tab, which you can access from five different points in the Manager.

As forms and sets move through the FORMS production modules (Scan, Interpret, Verify, and Transfer), *events* are *fired* at predefined occasions. An event is a condition or the result of an activity at a specific point in a program.

When an event is fired, FORMS checks to see whether you have defined a handler for the event. An *event handler* causes execution of a predefined VBA function, DLL function, or both.

VBA and DLL programming is a complex task that requires knowledge of programming, Visual Basic, C, C++ or other programming languages, and simple COM automation issues. There is a description of the interface that FORMS uses for external communication, as well as a lot of other information and examples, in Help.

Hints on what to look for in Help

Event handlers

Customizing...

VBA

DLL

Designing forms

FORMS can handle most forms without any modifications at all. However, small changes on a form can, in some cases, significantly improve the efficiency with which FORMS handles the form.

Hints on what to look for in Help

Form design

Designing forms and fields

Paper quality

Field design

Field background

Print quality of forms



3 Getting around in FORMS

This chapter orients you to the FORMS environment and how you can navigate successfully to perform necessary functions and find the information you need.

On-screen basics

FORMS' new interface makes standard program operations easier and more natural to use.

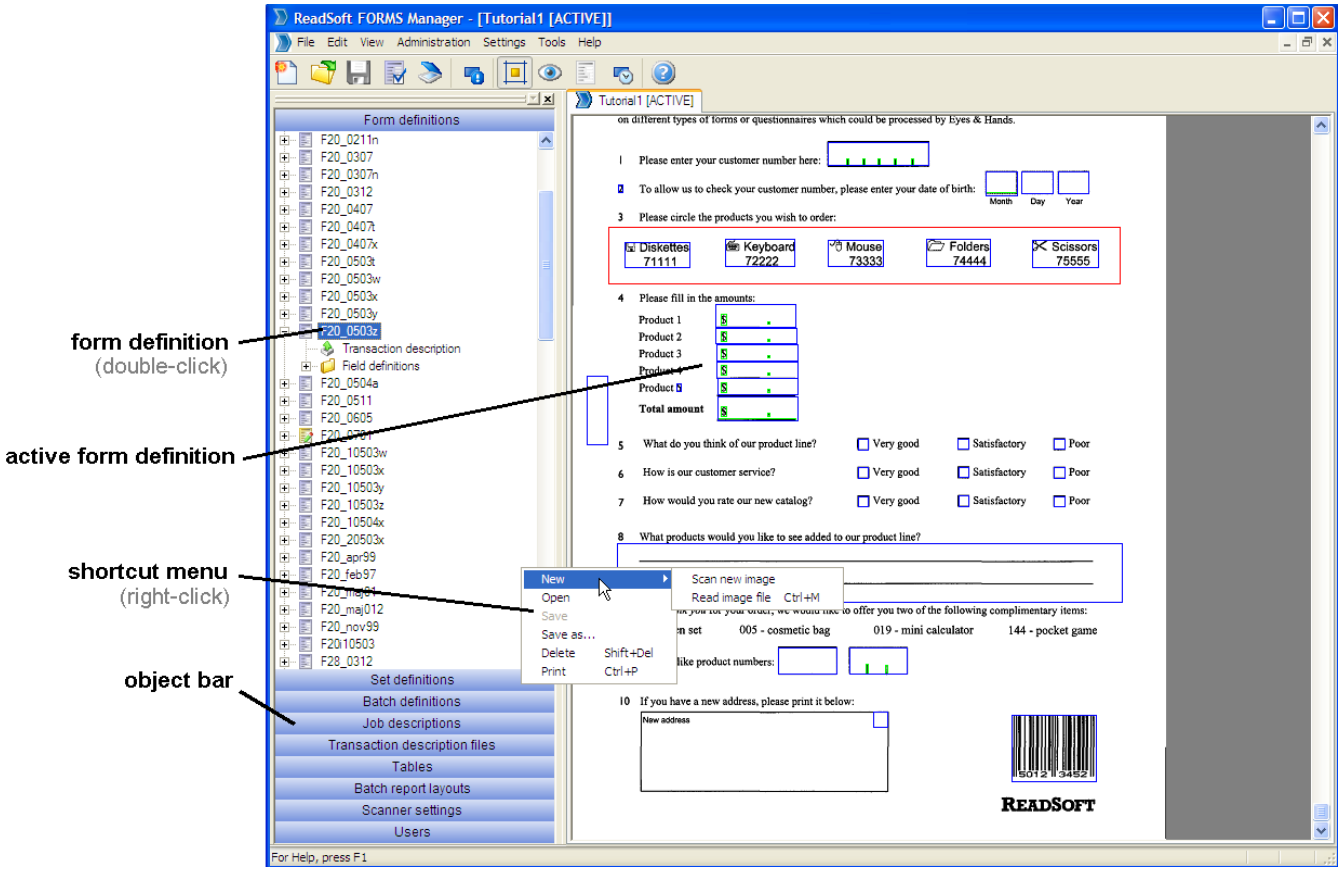
Most of the menu and toolbar button functions can also be accessed by clicking objects on the interface. For example, to create a new form definition in the Manager module, you can choose **New** (**File** menu), or you can right-click the white space under the **Form definitions** object bar, and select **New**.

Note that operations are context sensitive; in other words, commands are related to the object that is currently active.

In the example above, right-clicking the white space under the **Form definitions** object bar and choosing **New** gives you the option to create a new form definition. Accordingly, the same action when the **Job descriptions** object bar is open gives you the option to create a new job description.

Likewise, double-clicking a specific form definition displays it on-screen and activates several **Edit** menu commands that are inaccessible if no form definition is displayed.

Below is the Manager module with the **Form definitions** object bar open (and with a form definition displayed).



The Manager window

It will be helpful for you to take a look around in each module. By accessing menus and dialogs, you can get an idea of what types of options are available in FORMS. However, going through the tutorial in chapters 4 and 5 is the best way to orient yourself to the FORMS interface.

Using *FORMS Help*

FORMS Help contains the most up-to-date information available on the product. If you have a basic understanding of FORMS (having attended FORMS Basic Training or read this book and worked through the tutorial), it is easy to find the information you want in Help.

Help contains both overviews and very detailed information about:

- Form definitions, set definitions, batch definitions, and tables
- Working with image files
- Job descriptions (including all the options for verifying forms)
- Transaction descriptions
- How to process forms
- Designing forms for FORMS

- Program databases, system administration, and troubleshooting
- Customizing FORMS (events and event handlers)
- Integration with other systems
- And much more

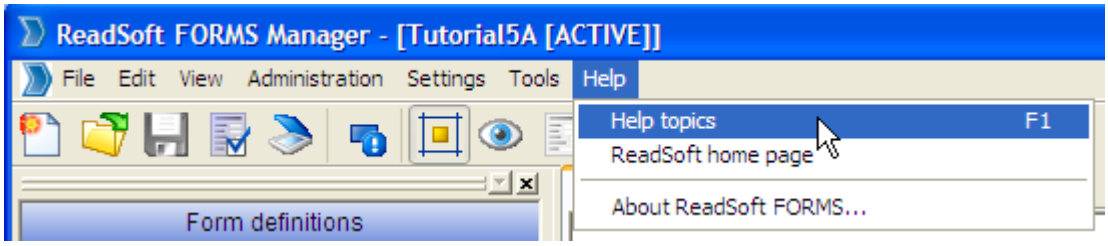
Accessing Help

F1

The easiest way to access Help is to press **F1** on your keyboard. You can do this at any time while you are using FORMS. **F1** is context sensitive, which means that the topic most closely related to whatever you are doing is displayed.

The Help menu

Each of the FORMS modules has a **Help** menu. Choose **Help topics** to open the Help system.



*Choose **Help topics** to open the Help system.*

Accessing Help from outside FORMS

You can access *FORMS Help* even if no FORMS window is open. Using Windows Explorer, locate the main Help file, `forms53_eng.chm`. It is located in the `\Documentation` directory. Double-click this file and use Help as you normally would.

Finding the information you need

There are several ways to find specific information in *FORMS Help*.

F1 (information about the current window or dialog)	The most direct way to get information about the current window or dialog is to press **F1** on your keyboard to open Help. **F1** is context sensitive, which means that the topic most closely related to whatever you are doing is displayed.
Index (often the fastest way of finding information)	When Help is open, click the **Index** tab or click the **Index** button. Use the Help index just as you would use the index in any book. Tip: Look for a word that describes either the action you want to take (for example, Open, Save, Delete, Update, etc.), or the object of an action (Table, Transaction description, Database, etc.). If the word is not in the index, try to think of other words with similar meanings. The index contains many synonyms. For example, if you do not find "remove", try "delete" or "erase". If you do not find "make", try "compile" or "create". You can also try using a full-text search (see below).
Table of contents	Click the **Contents** tab or click the **Contents** button. The books that are displayed are comparable to chapters in a real book. The books are arranged in a sequence similar to the FORMS workflow. Double-click the book that most closely relates to the information you want. Continue to double-click books until you see a page that has the topic you want. Each page in each book is a Help topic. Double-click the topic you want to read.
Full-text search	Click the **Search** tab and type the word or phrase you want to search for. Click **List topics** to display those topics that contain the desired search words. Then select the item in the list displayed and click **Display** to see it. Use the options at the bottom of the window to expand or limit the search. • Search previous results • Match similar words • Search titles only
Links in related topics	Even if you do not find the information you want immediately, you might find something related to your question. Almost all topics contain hyperlinks (underlined blue text) which lead to related topics. By clicking one of them, you may find just what you need.*
Glossary	Click the **Glossary** button while you are viewing any topic to view a list of FORMS terms and concepts.

* A few links lead to other Help files. However, this does not affect how you use Help. The **Index** and **Search** buttons, for example, still lead to *FORMS Help*.



Getting around in Help

Several buttons are available while you are viewing topics.

Button	What you can do with it
Contents	Use a familiar table of contents approach to searching for topics in the Help system. The books that are displayed are comparable to chapters in a real book. Each page in each book is a topic.
Index	Locate the information you want using an alphabetical listing of key words.
Search	Search for topics containing specific words and phrases.
Favorites	Keep a list of favorite topics for later reference. • **Add** – click this button when you want to save the current topic to the favorites list. The link is given the name of the topic unless you type a name of your own. • **Remove** – select an item in the list and click this button to remove it. • **Display** – select an item in the list and click this button to display the topic.
Hide/Show	Toggles the display of the navigation pane.
Back	Return to the previous topic viewed. You can click **Back** repeatedly if you followed links to several topics. (Available when a topic is displayed.)
Forward	Return to a topic(s) that you left by clicking **Back**. The button can be clicked as many times as the **Back** button was used to get to the current topic.
Print	Print the current Help topic to a connected printer. (Available when a topic is displayed.)
Options	Offers another way to use various help options.
Glossary	View a list of FORMS terms and concepts. (Available when a topic is displayed.)

In addition, there are a number of helpful commands:

To do this...	Select this command
Create a link so you can easily go to the same topic later	**Favorites** ⇨ **Add** to add the current topic to the list
Go to a topic marked as a favorite.	**Favorites** ⇨ ***Link name*** ⇨ **Display**
Print the current Help topic to a connected printer.	Click **Print** button

VBA Help

In FORMS' Visual Basic for Applications (VBA) window, there are two types of Help available from the **Help** menu:

- VBA programming language Help is also provided. The file name is `Cypress Enable Language Reference.pdf` and it can be found under `\Documentation\Other` in the installation.

- Help on methods and events contains information about events in FORMS, as well as a reference to all of the methods and attributes contained by the classic and the new API. The Help file name is `FORMS_API.chm`, and it is found in the `\Documentation` folder in the installation.

4 Getting started: Preparing for production

Preparing for production involves using the Manager module to prepare and set up FORMS before any forms are processed.

The instructions in this chapter take you through all of the basic steps in setting up FORMS for use in production.

We suggest that you follow the steps described in chapters 4 and 5 using the sample form provided with FORMS as a guide the first time. Alternatively, you can follow the same general instructions using your own form rather than the sample file.



READSOFT DOCUMENTS FOR FORMS

Sample form 5A

Examples on this form are not meant to be consistent or logical; they are models of fields that might occur on different types of forms or questionnaires that could be processed by ReadSoft DOCUMENTS|FORMS.

1 Please enter your customer number here: [| | | | |]

2 To allow us to check your customer number, please enter your date of birth: [ ] [ ] [ ]
Month Day Year

3 Please circle the products you wish to order:

Diskettes	Keyboard	Mouse	Folders	Scissors
71111	72222	73333	74444	75555

4 Please fill in the amounts:

Product 1	$.
Product 2	$.
Product 3	$.
Product 4	$.
Product 5	$.
Total amount	$.

IT436M-2

5 What do you think of our product line? ☐ Very good ☐ Satisfactory ☐ Poor

6 How is our customer service? ☐ Very good ☐ Satisfactory ☐ Poor

7 How would you rate our new catalog? ☐ Very good ☐ Satisfactory ☐ Poor

8 What products would you like to see added to our product line?

_____

_____

9 As a *thank you* for your order, we would like to offer you two of the following complimentary items:

001 - pen set 005 - cosmetic bag 019 - mini calculator 144 - pocket game

I would like product numbers: 019 144 ← Or please send me two of these, if possible

10 If you have a new address, please print it below:

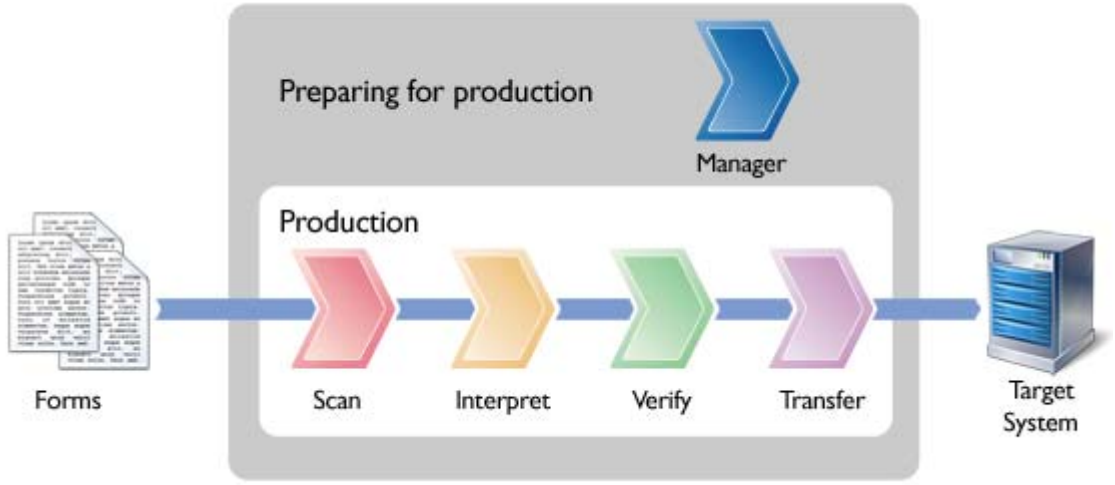
New address



READSOFT®

FORMS' sample form, included in Microsoft Word format (`Sample.doc`) and as an image file (`Sample.tif`).

Five stages to prepare for production



You use the Manager module to prepare for production.

Here are the stages needed to prepare a form for production (along with the page number where the explanation of each begins):

- Create a form definition (page 31)
- Create a standard transaction description (page 66)
- Activate and resave the form definition (page 74)
- Create a job description (page 74)
- Test the form definition before using it in production (page 78)

You use the Manager module to perform all of these stages, which are explained in detail in the rest of this chapter. Once you have finished this chapter and understand more about preparing for production, you may find it useful to refer to page 112 ("Five stages to prepare for production"), where you will find an expanded outline of these five stages.

Create a form definition

As stated earlier, a form definition tells FORMS what your form looks like and what type of information it contains. See page 11, “Understanding form definitions,” to review form definitions.

In this tutorial, you will define a one-page form. A *set* is two or more forms that belong together and should be processed together in a particular order. See Help for more information on using sets.

If you want to use FORMS’ sample form when working through the tutorial, you have two choices. Choose the method most similar to how you plan to use FORMS in practice:

- Use Microsoft Word (version 6.0 or later) to open and print the file `Sample.doc`, as well as several copies of the file `Blank.doc` (both files are located in `C:\Program Files\ReadSoft\FORMS\Samples\Tutorial` in Windows XP and Windows 2003 R2, or in `C:\Users\Public\ReadSoft\FORMS\Samples\Tutorial` in the newer operating systems). Do this now. Scan `Sample.doc` as described below on page 34. When you get to Chapter 5, you will fill in copies of the `Blank.doc` form as a customer might and then use these forms as you continue through the tutorial.

- Use FORMS to read the image file `Sample.tif` as described on page 34, and base your form definition on this image. When you get to Chapter 5, instructions will tell you to skip “Create some forms to use with the tutorial” on page 86 and interpret your data from `Filled1.tif`, `Filled2.tif`, `Filled3.tif`, and `Filled4.tif`, which are copies of `Sample.tif`. These have been filled in and provided for your convenience (these files are located in `C:\Program Files\ReadSoft\FORMS\Samples\Tutorial` in Windows XP and Windows 2003 R2, or in `C:\Users\Public\ReadSoft\FORMS\Samples\Tutorial` in newer operating systems).

Creating a form definition can be quite detailed and usually takes the most time of all the stages in preparing for production. This process requires the following steps:

1. Adjust scanner settings (page 33)
2. Scan a blank form or read an image file (page 34)
3. Save the form definition (page 35)
4. Define adjustment fields (page 35)
5. Define a recognition field, if needed (page 37)
6. Define all other fields (page 38)
7. Specify options and validation routines (page 60)
8. Check the validity of the form definition (page 66)

You will now follow instructions using the Manager module to create a form definition.

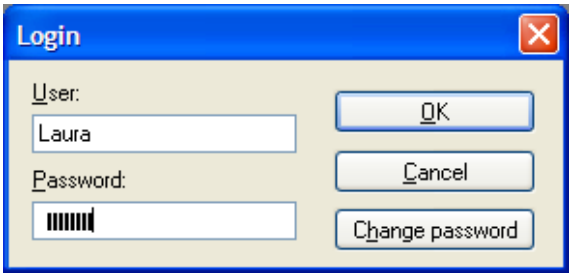


Step 1 – Adjust scanner settings

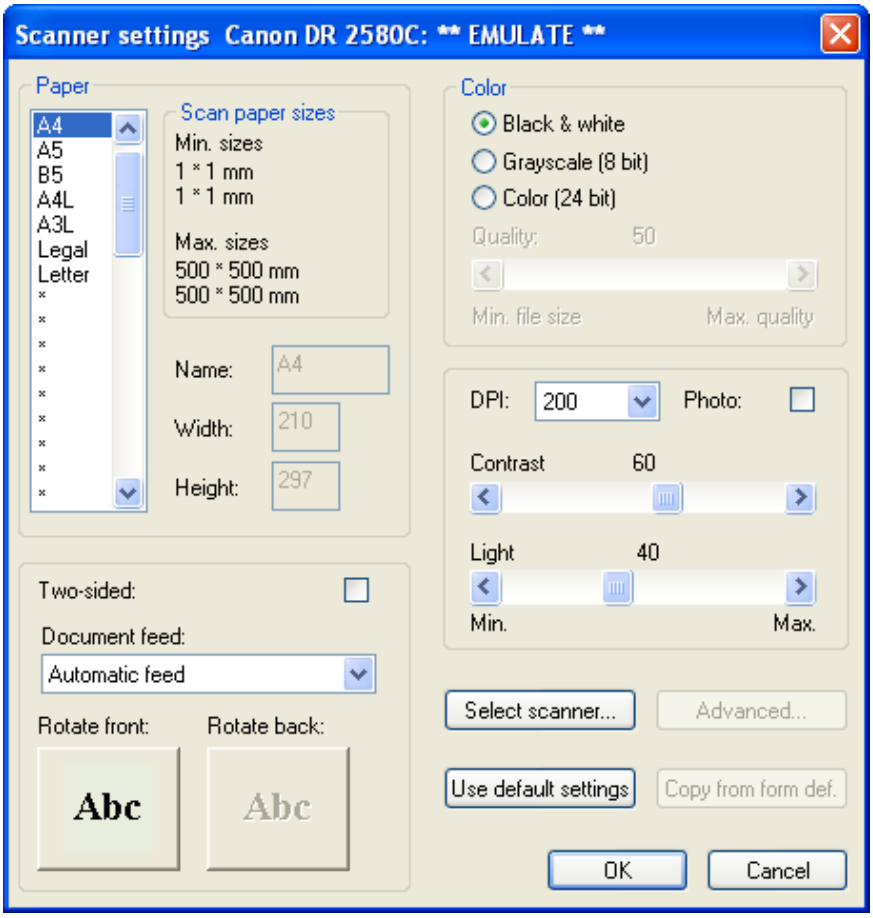
In this step, you will start the Manager and adjust the scanner settings.

1. Start the Manager by double-clicking the Manager icon (or by double-clicking `Efmngr.exe` in FORMS' `Bin` directory).

2. If the **Login** dialog is displayed, enter your user name and password, and click **OK**. The **Login** dialog is displayed only if the administrator has activated the login function.



3. If you just started the Manager for the first time, the **Scanner settings** dialog is displayed automatically to allow you to choose a scanner. If you want to use the FORMS sample TIFF file (or any other image file), you can close this dialog and proceed to "Step 2 – Scan a blank form or read an image file" on page 34.



The Scanner settings dialog.

- Click **Select scanner** and select your scanner model from the dialog.

- Click **OK** to close the **Select scanner** dialog.

If the **Scanner settings** dialog does not appear automatically, click the **Scanner settings** button on the tool bar (or **Settings** ⇨ **Scanner settings**). Select a paper size. FORMS has a number of predefined paper sizes. The sample form is A4 format, which is selected by default.

If none of the predefined sizes fit the form you want to define, you can either choose a larger paper size than the actual form or define a new paper size.

If you define a new paper size for your own form, make it about 1 cm larger than the actual form on all sides. However, note that only form definitions with the same paper size can be included in the same job description.

4. Select the DPI. In most cases, we recommend that you consistently use 200 DPI, which is the default setting, because scanning is faster and less disk space is used.

5. Change the scanner settings **Light** and **Contrast** only if necessary, after consulting Help.

6. Click **OK** to close the **Scanner settings** dialog.

The scanner settings associated with a form definition cannot be changed after you have scanned it, unless you use **Rescan**. See Help for more information.

Step 2 – Scan a blank form or read an image file

In this step, you will provide FORMS with a blank form of the type you want to process.

1. Click the **Form definitions** object bar on the left-hand side of the screen.

2. If you want to scan in a blank form, place a form (of the type you want to use FORMS to process, or you can place the `Sample.doc` form in the scanner) in the scanner's document feeder according to the instructions for your scanner. If you do not have a blank form, you can scan a filled-in one, following the instructions below, and then erase the filled-in information. (Erasing filled-in forms is explained in a later step, beginning on page 54.)

   If you want to use the sample TIFF file:

   - Right-click anywhere in the white area under the **Form definitions** object bar and choose **New** ⇨ **Read image file**.

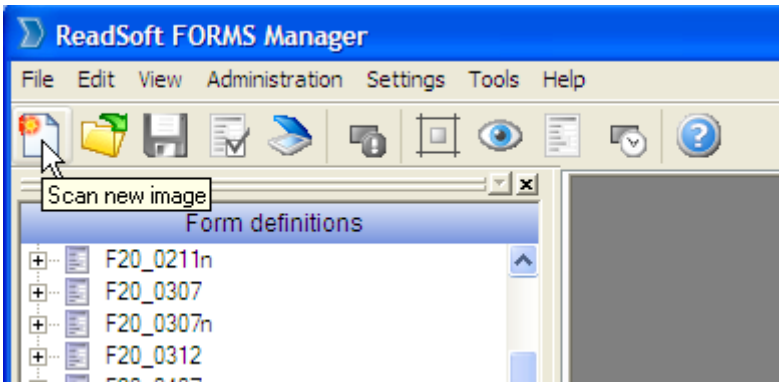
   - In the first dialog that is displayed, click **No** to indicate that the image is not two-sided.

   - In the second dialog that is displayed, navigate to `C:\Program Files\ReadSoft\FORMS\Samples\Tutorial` in Windows XP or Windows Server 2003 R2, or to `C:\Users\Public\ReadSoft\FORMS\Samples\Tutorial` in newer operating systems, and click `Sample.tif` to select the sample form included with the FORMS installation program. Then click **Open** to display it on the screen.

   - Skip ahead to "Step 3 – Save the form definition."



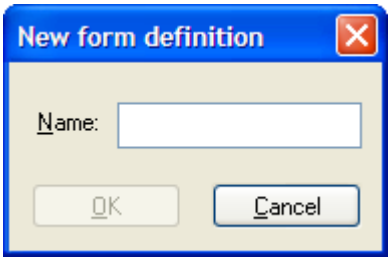
3. Click the **Scan new image** button on the toolbar. FORMS starts the scanner and displays the form on the screen.



Step 3 – Save the form definition

In this step, you will choose a name for the form definition and save it for the first time.

1. Click the **Save** button on the toolbar.



2. In the resulting dialog, type a name for the form definition. Note that form definition names are case-sensitive.

3. Click **OK** to save it in the form definition database.

Step 4 – Define adjustment fields

In this step, you will specify a few characters, symbols, or lines on the form that FORMS can later use to identify and straighten the images as they are scanned.

FORMS uses *adjustment fields* to identify and deskew forms and match them to the form definition. All the other fields have a fixed position on the form relative to the adjustment fields. Therefore, adjustment fields must be the first fields defined on every form.

There are two kinds of adjustment fields: FORMS' traditional adjustment fields and *flexible line maps*. Traditional adjustment fields are recommended for most form definitions. Therefore, we will define traditional adjustment fields here.

Ideally, each form definition should have one adjustment field in each corner (preferably at least 1 cm from the edges of the form) and one in the middle.

Adjustment fields, unlike most other fields when you define them, are not empty but contain one *adjustment object*. You choose the most suitable object in each area of the form as your adjustment object.

Angular (rather than rounded) objects are best. For example, the letter "T" is better than "C", and a "+" is best of all. A corner of a box or an intersection of lines is also suitable.

Since FORMS needs to see the adjustment objects each time a filled-in form is scanned, you must not choose adjustment objects that could be covered up by a sticker or written on by a respondent. In addition, since adjustment fields are used to match the form to the form definition, the adjustment objects you choose must be unique to the form. That is, do not choose any objects that are the same as objects on another form that will be processed at the same time.

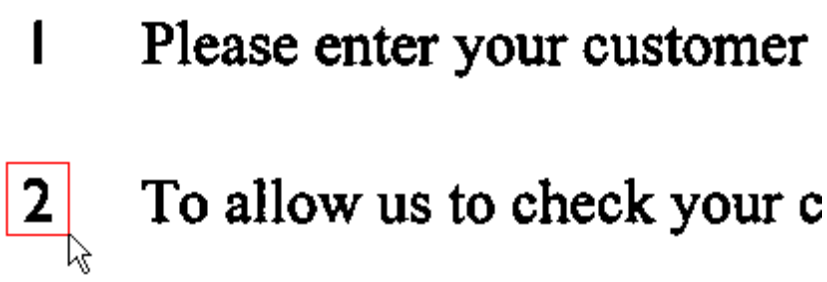
Similar objects may not be present nearby. If the Manager does not accept an adjustment object, select another and try again.

1. Familiarize yourself with the form before you continue. Use the **F3** and **F4** keys on your keyboard (or right-click anywhere on the form and choose **Enlarge** and **Reduce** on the shortcut menu), as well as **CTRL**+arrow keys or the scroll bars on the side and at the bottom of your screen, to adjust the image. Experiment to obtain the best view of the part of the form you are working with.

2. Now let's begin defining adjustment fields. On the sample form, the number "2", indicating the second question on the form, appears to be a good adjustment object. (FORMS might find "1" unsuitable as an adjustment object because, if the form were scanned slightly crooked, it could be confused with the letter "l" in "Please".)

   Adjust the image on the screen to see the "2" clearly. Use the **CTRL**+arrow keys to move that part to the middle of your screen. Then press **F3** several times to enlarge it.

3. Using your mouse, position the cursor above and just to the left of the adjustment object, "2".

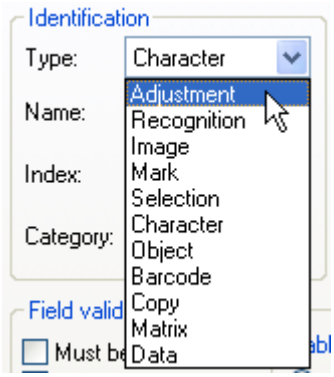


   Holding the left mouse button down, move the cursor down and to the right to draw a frame around the "2". The frame must only contain the adjustment object.

4. Release the mouse button. If you need to start again, press the **DELETE** key on your keyboard to erase the frame.

5. To define the field, double-click the frame or press **F5** on your keyboard to display the **Field** dialog.

6. On the **Standard** tab, select **Adjustment** from the **Type** box.



7. Click **OK**.

   The dialog closes and a blue frame appears around the adjustment object. The blue frame is usually not the same size as the one you drew, because FORMS automatically adjusts it to the ideal size for its purposes.

8. Repeat these steps until a total of five adjustment fields are defined.

   Tip: Keep adjustment fields a good distance from areas on the form where people might write, because an error occurs if an adjustment field has any unfamiliar marks on it when FORMS tries to identify it.

   Instead of double-clicking each subsequent frame after you draw it, you can press **F6** on your keyboard to save the step of indicating the field type (or right-click each new field area and choose **Define as previous** from the shortcut menu).

   On the sample form, the following would be good choices for adjustment fields (again, make sure that the entire character is contained within the adjustment field area):

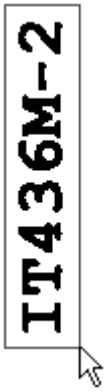
   - The “S” at the end of DOCUMENTS on the top right-hand side
   - The “5” in “Product 5” under question 4
   - The “9” on the bottom left-hand side
   - The second “4” in “144 – pocket game” on the bottom right-hand side

9. Resave the form definition by clicking the **Save** button on the toolbar.

Step 5 – Define a recognition field if necessary

Usually, Scan identifies forms by their adjustment fields. However, when you process forms that are nearly identical, Scan can use a preprinted number on the form to identify it.

The sample form has a preprinted identification number on it. Show this number to FORMS by defining a *recognition field*.



1. There is a recognition field area along the left-hand side of the sample form. Use the **CTRL**+arrow keys and **F3** and **F4** keys on your keyboard to bring the number "IT436M-2" into view.

2. Since the form number does not have a box preprinted around it, you must use your mouse to draw a frame around it, just as you draw a frame around the adjustment characters. Make the frame large enough to enclose the form number.

3. Right-click the frame and choose **Erase** from the shortcut menu. The number is erased from the frame, but the frame remains.

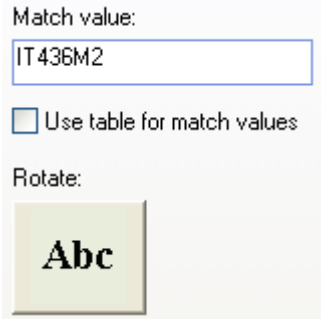
4. Double-click the frame (or press **F5**) to display the **Field** dialog.

5. On the **Standard** tab, select **Recognition** from the **Type** box.

6. Type the actual form number, IT436M2, in the **Match value** box. FORMS ignores dashes, commas, and dots in recognition fields, so do not specify these.

   You can also use an asterisk to replace characters that are insignificant. For example, if *all* of your form numbers started with IT, you could replace "IT" with "**", which would allow Scan to identify the form faster.

7. Because the field is sideways, it needs to be rotated before interpretation. Click **Rotate** once to rotate the field 90 degrees counterclockwise.



8. Click **OK** to close the **Field** dialog.

Step 6 – Define all other fields

Now that you have specified the form's identifying features, it is time to define the rest of the fields on the form. Every area on the form that contains information to be captured must be defined as a field.

This step is the bulk of creating a form definition, and in fact, it is the single task that takes up the most time in preparing for production. We will therefore go into substantial detail here in order to show you many possibilities for defining fields.

You will be shown the basic way to define several field types, including techniques that make defining fields more efficient.

Then, in step seven, we will go back to some of the fields and apply optional settings and validations, in order to give you an idea of some of the additional settings you can use ("Step 7 – Specify options and validation routines," begins on page 60).

The order in which you define fields affects the order in which you later verify them and the order that the data is transferred to the target system. Both of these can be edited, but it is to your advantage to define fields in consecutive or logical order.

If you are going to create several form definitions with some identical fields, we recommend that you decide on standard names for any fields that will be included in archive transactions. See Help about archive transactions and about using a common layout for your transaction descriptions.

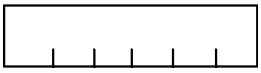
Note

While you are defining fields, keep in mind that most fields may not overlap. Only copy fields can overlap other fields. In addition, adjustment fields (which are defined first on the form definition) can be overlapped by other fields.

Define a character field contained in a box

Whenever information appears within boxes on the form, double-click the box to mark it. This is by far the most accurate and reliable method of indicating the perimeter of fields.

1. On the sample form, there is a box where customers write in their customer number. If you cannot see it, use **F3** and **F4**, as well as **CTRL**+arrow keys, to find and adjust your view of it.



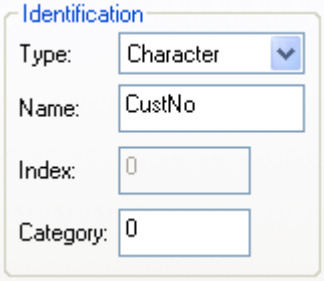
2. Double-click the box (a red frame appears around the box).

3. Double-click the frame (or press **F5**). The **Field** dialog is displayed.

4. In the **Type** box, there are many different field types to choose from. They are described in detail in Help.

   Since the field is to contain characters and numbers, choose **Character** (which is selected by default) as the field type.

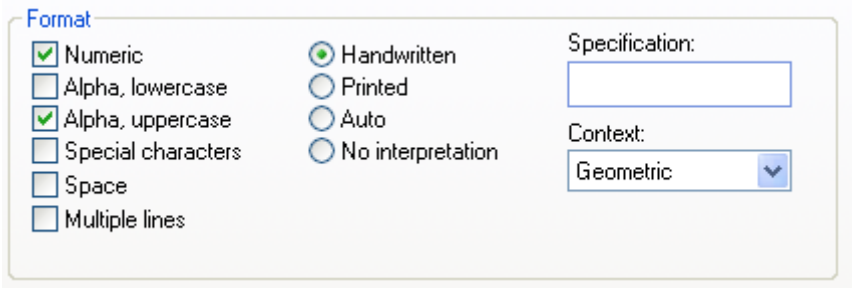
5. Type a suitable **Name** for the field, such as CustNo.



6. Now you need to tell FORMS what to look for in the field by specifying its format as accurately as possible.

Suppose that the customer number consists of one uppercase letter plus 5 digits.

Since the field is to contain both numbers and an uppercase letter, this is what you specify in the **Format** group box. **Numeric** is selected by default. Click **Alpha, uppercase** to add it as an acceptable format.



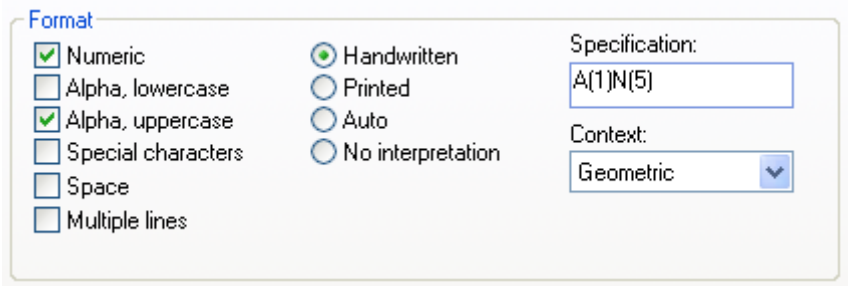
FORMS also needs to know whether the field will be handwritten, machine-printed, or an approximately even number of each (**Auto**). In this case, **Handwritten** (which is selected by default) is the correct choice because respondents will be writing in a customer number by hand.

7. The next step is to specify even more exactly the format of the information the field will contain. The more precise you can be, the better the interpretation result. Help contains details of all the possible ways of doing this.

   For this field, type the following in the **Specification** box:

   A(1)N(5)

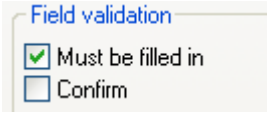
   because the field will contain one alphabetic character (A) followed by five numbers (N).



   Under the **Specification** box is a box from which you can choose one of three context types for character fields. If one of them applies, it is highly recommended because interpretation improves.

   However, none of the context settings apply to any of the fields we will define in this tutorial, and **Geometric**, which is the default option, does not have any effect on them. Therefore, you can disregard **Context** throughout this tutorial. If you would like to know more about this topic, consult the Help file.

8. You can specify a simple field validation; namely, that the field must be filled in. In the **Field validation** group box, select **Must be filled in**.



Later, when you are verifying forms, if CustNo is not filled in, Verify will draw it to your attention, and you can fill in a number or remove the form from the database.

9. Click **OK**.

   When you look at the field now, you see that there are small green boxes around the lines inside it. FORMS remembers these lines and does not try to interpret them as numbers or letters later on.

Define character fields with support lines

It is very important that field areas be defined as accurately as possible.

The most accurate way to define a field area on the form is by double-clicking the pre-printed box around it, if one exists. When you double-click a box, FORMS interprets the contents of the field on each scanned form after locating the pre-printed box. When you do not double-click a field, FORMS interprets the contents of the specific area of the form, as measured from the perimeter of the form, without checking for a pre-printed rectangle.

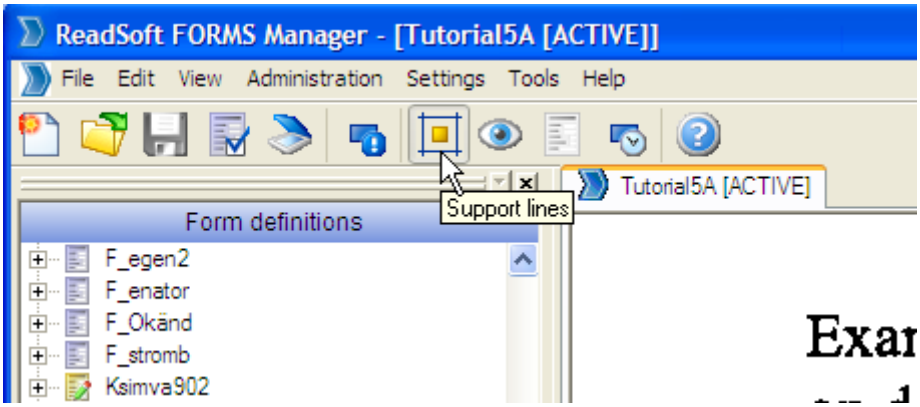
A compromise is possible if there is an incomplete box around the field, or even just a line underneath the field area or along one of the sides. In such cases, always use FORMS' **Support lines** function.

When you define the field area by drawing a frame around it with your mouse while **Support lines** is activated, FORMS locks the frame to nearby lines. This is the next-best method (after double-clicking a box) for indicating field areas.

1. On the sample form, question 2 has incomplete boxes in which customers are to fill in their date of birth.

   The **Support lines** function is automatically enabled when you start the Manager. If it is not, click the **Support lines** button on the toolbar. Alternatively, you can right-click anywhere on the form (outside of a defined field area) and then choose **Support lines** from the shortcut menu.

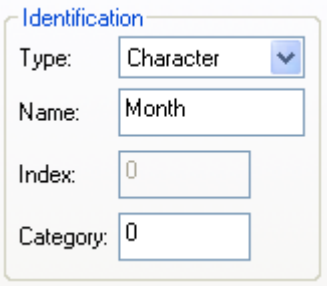


2. Draw a frame around the Month box with your mouse, keeping close to the lines provided. If you are close enough to those lines, you see that FORMS locks the lines of the box to them.

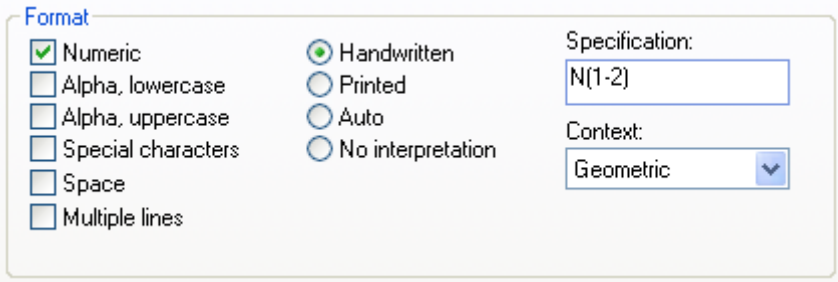
3. Double-click the frame (or press **F5**). The **Field** dialog is displayed.

   Since the field is to contain numbers, **Character** (which is selected by default) is the correct choice of field type.

4. Type a **Name** for the field. Month would be appropriate.



5. Again, you need to tell FORMS what to look for in the field by specifying its format. As the field is to contain a number, **Numeric** and **Handwritten**, which are selected by default, are the correct choices.

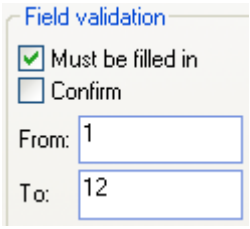


   You still need to tell FORMS how many digits to look for. Since you expect the customer to write in a number between 1 and 12, type the following in the **Specification** box:

   N(1-2)

   This tells FORMS to expect a one- to two-digit number.

6. There is a fixed range of valid numbers for this field (1 to 12). If you indicate this range to FORMS, its interpretation engine can rule out other numbers, making it more effective. Here is one way of indicating the range of valid numbers:



   In the **Field validation** group box, type 1 in the **From** box and 12 in the **To** box. FORMS will not accept anything outside of this range.

7. Click **OK**.

8. Define the Day field in the same way as the Month field. For **Field validation**, the valid range is **From** 1 **To** 31.

9. Define the Year field in the same way as the Month and Day fields, except type the **Specification** N(2) to indicate that the year must be a 2-digit number.

   Field validation is not appropriate for this field, unless you wish to limit the age group of the respondents.

10. Resave the form definition by clicking the **Save** button on the toolbar.

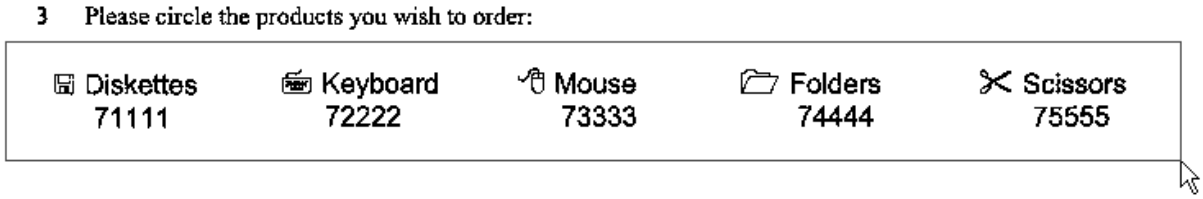
Define a selection field

A *selection field* is an area containing one or more words, characters, or symbols that are either circled or marked through (crossed) to indicate a response. They are sometimes referred to as "circled mark fields."

Question 3 on the sample form shows five different products that customers can order. Rather than defining five separate product fields, you can define the entire area as a selection field, and then tell FORMS where each object in the field is located:

1. Adjust the display using **F3** and **F4**, as well as **CTRL**+arrow keys, so that you can see all five products.

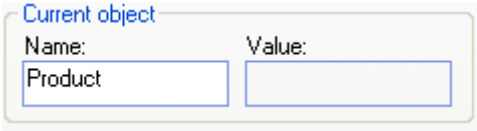
   Draw one frame around the group of objects, with plenty of space between the objects and the edge of the frame:



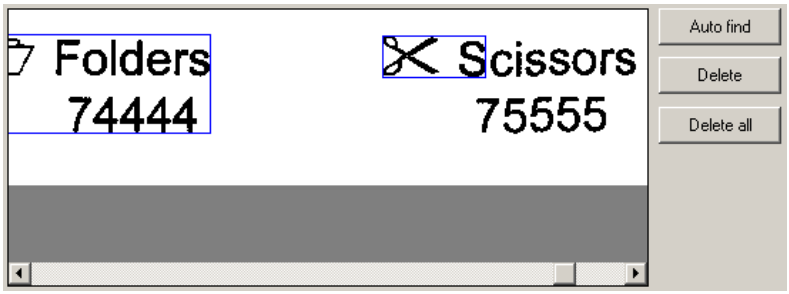
If you need to start again, right-click within the frame and choose **Delete** from the shortcut menu (or press the **DELETE** key to erase the frame).

2. Double-click the frame (or press **F5**). The **Field** dialog is displayed. Choose **Selection** as the field type. The settings of the **Standard** tab change.

3. In the **Current object** group box, type a new name for the subfields in the selection group (such as Product). This name, plus an index number, applies to each object inside the field area that you drew. When you define the fields in question 4 on the sample form, you will learn more about indexed fields and how they function.



4. FORMS automatically finds the markable objects (surrounded with blue frames), and displays them in a view box. If you are satisfied with the objects FORMS has found, go to the next step now. However, if you are using the sample form, you can see the "Scissors" subfield is inadequately marked. You put "Scissors" into view by using the scroll bar, **F3,** and **F4**.



You want the name, illustration, and number of each product to be *one* subfield. Therefore, you need to modify any object in the selection field that is insufficiently marked as one subfield:

- With the "Scissors" product in view, click the frame, and then click **Delete**, which is found to the right of the view box. This removes the frame partially covering the information in the "Scissors" subfield.

  Tip: In a situation where all, or the majority, of your objects are inadequately marked with frames, using **Delete all** will remove *all* of the frames within the selection field. You then manually mark all the objects from the beginning.

- Draw one frame surrounding the name, illustration, and product number, leaving as little extra space as possible between these objects and the edge of the frame.

- When frames are deleted and added, FORMS adjusts the index of each object.

5. FORMS can now determine which of the products a respondent has chosen from the selection field, but now it needs to know what value to assign to each product. You specify this in the **Format** group box.

   Almost every field needs a value that can be put in the transfer file when the field is filled in. However, a response that is either circled or marked through (crossed) has no inherent value. Therefore, you must give it one.

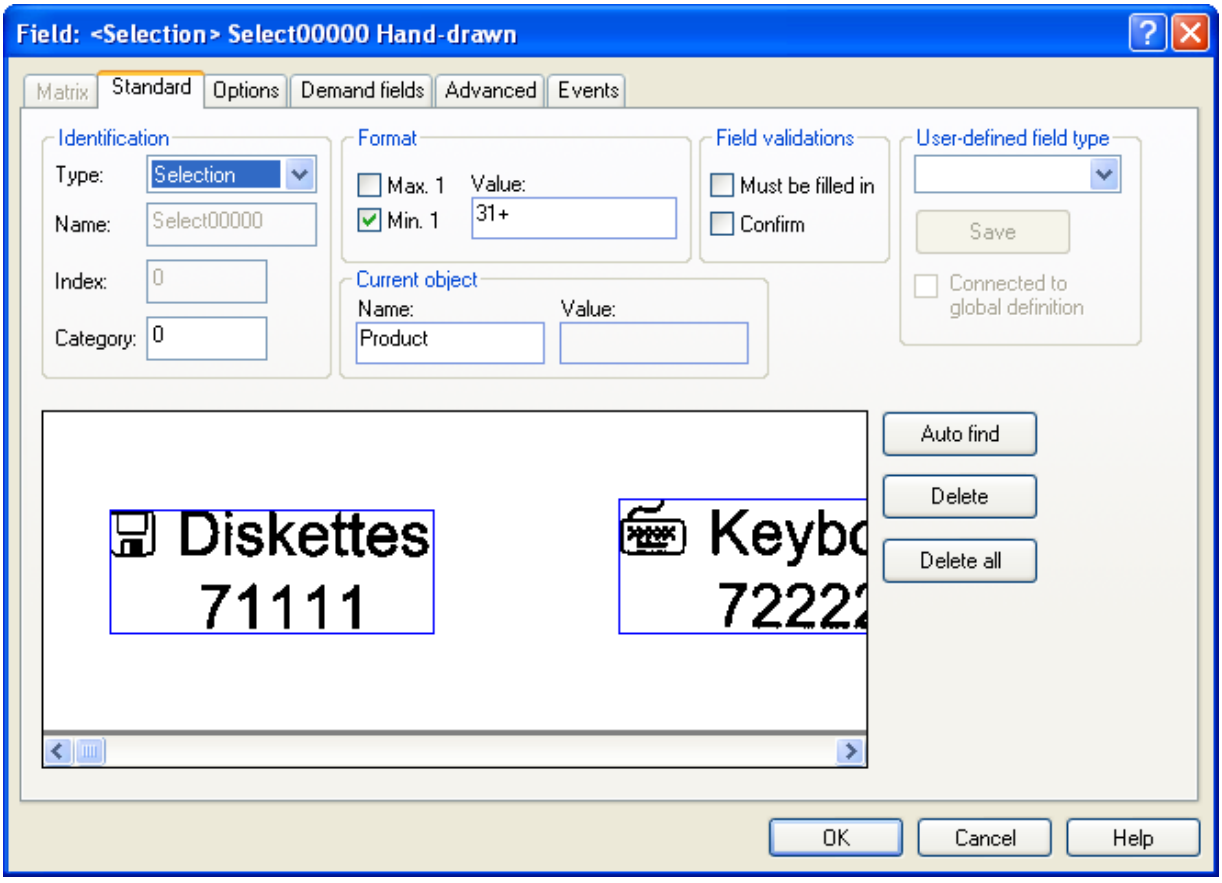
   In the **Value** box, you can type any number that will represent the field when it is filled in. When you look at the transfer file in this case, you might appreciate a system where the values are 2-digit numbers, with the first digit indicating the question number and the second digit indicating which response was given.

   In the **Value** box, type:

   31,32,33,34,35 or 31+

   Either of these result in the products having values of 31, 32, 33, 34, and 35 respectively, if circled or marked through on a form.

6. In the **Format** group box, select **Min. 1**, because at least one of the products must be circled by the customer. If this condition is not met on a scanned form, then the Verify module will alert you to this later.



7. Click **OK** to close the **Field** dialog.

8. Resave the form definition by clicking the **Save** button on the toolbar.

Define a series of character fields (indexed fields) with decimals

You can define two or more fields at the same time, if the fields contain exactly the same type of information and are to be processed identically.

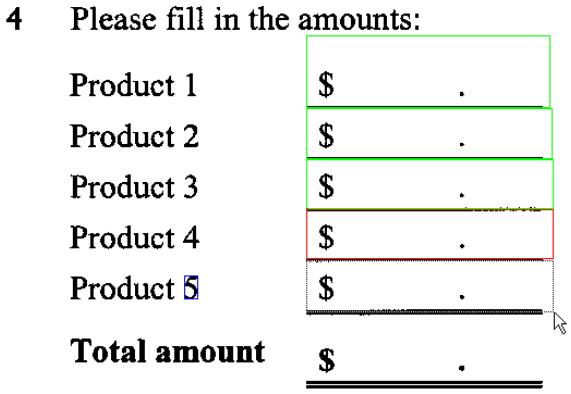
Such fields are called *indexed fields*. Simply double-click (if possible) or draw each of the field areas before you define them. For field areas that can be double-clicked, there is a faster way of drawing indexed field areas (see page 50, "Define three series of mark fields," for details).

Question 4 on the sample form contains a series of character field areas in which customers can write amounts. The amount field areas for products 1 through 5 are to be handled identically, so you will define them as a series.

These field areas also contain preprinted decimal points that will be defined as a part of each field.

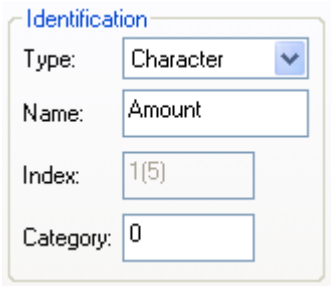
1. Adjust the display using **F3** and **F4**, as well as **CTRL**+arrow keys, so that you can see all the field areas in the series below question 4.

   Using the line under each field area as a support line, draw five frames for the amounts. Include the dollar signs ($) and the decimals in the frames, and try to make each frame the same horizontal length. Also, begin the first frame from just under the word "amounts", in case respondents use all of the available space to fill in product 1's amount. Starting from the second frame, use the line above each field area as a second support line.



2. Double-click any of the field areas (or press **F5**) to display the **Field** dialog.

3. Since the field is to contain numbers, **Character** (which is selected by default as the field type) is the correct choice. Type a name for the fields (such as Amount) in the **Name** box.



All of the fields in the series receive the same name, but each has its own index. The first Amount field you defined is 1 of 5 (as you can see under the name), and from now on, FORMS refers to individual Amount fields with their indexes, like this: Amount[1], Amount[2], and so on.

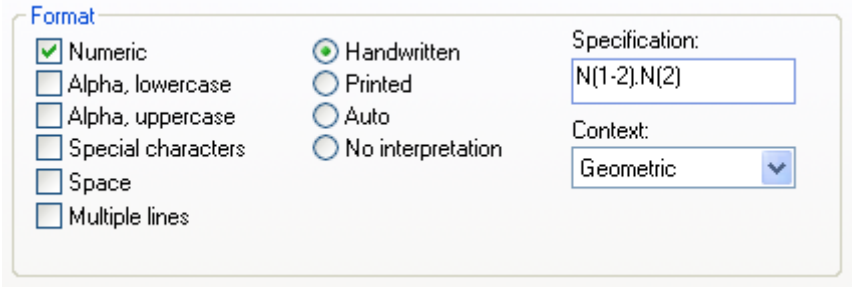
All of the fields receive the same attributes that you select in the **Field** dialog.

The Amount fields are to contain dollar amounts, so **Numeric** is correct. **Handwritten** is also already selected by default.

4. Let us suppose that every available Product costs between $4.99 and $49.99. In the **Specification** box, type:

   N(1-2).N(2) or N(1-2).NN

   to indicate that all amounts are to consist of a 1- or 2-digit number followed by a decimal and another 2-digit number.



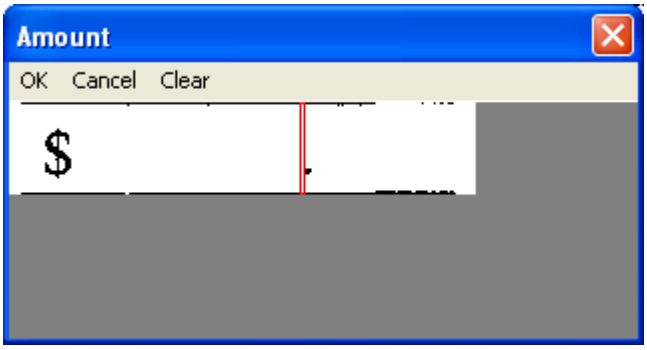
   Although there is a fixed range of valid amounts (from 4.99 to 49.99), do not type these amounts in **From** and **To**, because no decimals are allowed there. However, you could enter 4 and 50 in **From** and **To**.

5. When a numeric character field includes a decimal point that is pre-printed on the form, you must show FORMS the decimal.

   The decimal is already printed in the Amount fields, so this must be shown to FORMS.

   Click **Decimal**. A new dialog is displayed, showing the field.

6. In the new dialog, click the decimal point. A vertical line appears to separate the non-decimal from the decimal number area.



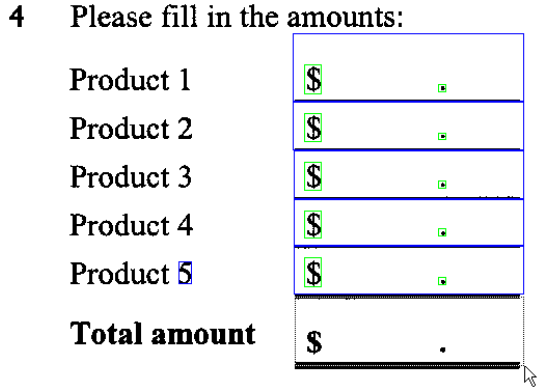
If you click the wrong spot, simply click a different spot or click **Clear** to start again.

7. Click **OK**. FORMS saves the same decimal area for all of the fields in the series.

8. Click **OK** to close the **Field** dialog.

9. There is one more decimal field to define: The field that will to contain the total amount.

   Using the line provided for the total field as a support line, draw a frame for the field area. Include the dollar sign ($) in the frame.



10. Now you can use the **Define as previous** shortcut again to define this field area (just as you did with the adjustment fields), because it is to have the same characteristics as the Amount fields. Press **F6** on your keyboard (or right-click on the field area and choose **Define as previous**).

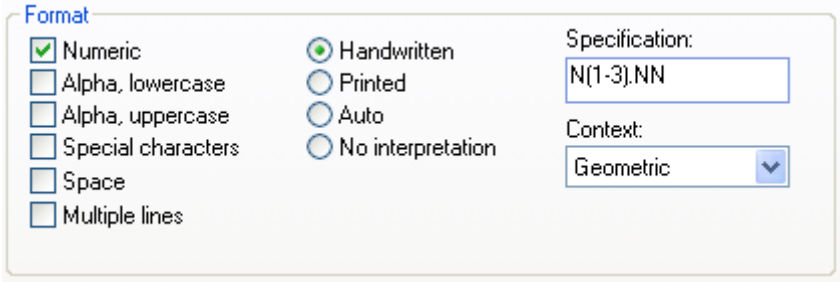
11. On the **Standard** tab, type a **Name** for the field (such as Total).

    Since the field is to contain numbers, **Character** (which is selected by default as the field type) is the correct choice.

    The Total field is to contain a dollar amount, so **Numeric** and **Handwritten**, which are selected by default, are also correct.

12. Since the prices of the products range between $4.99 and $49.99, then the total could be anywhere between $4.99 to almost $250 dollars. Thus, the **Specification** copied from Amount needs to be changed from N(1-2).N(2) or N(1-2).NN to:

    N(1-3).N(2) or N(1-3).NN



13. The decimal must be redefined for the new field. Click **Decimal** and define the decimal as described above in steps 5 and 6.

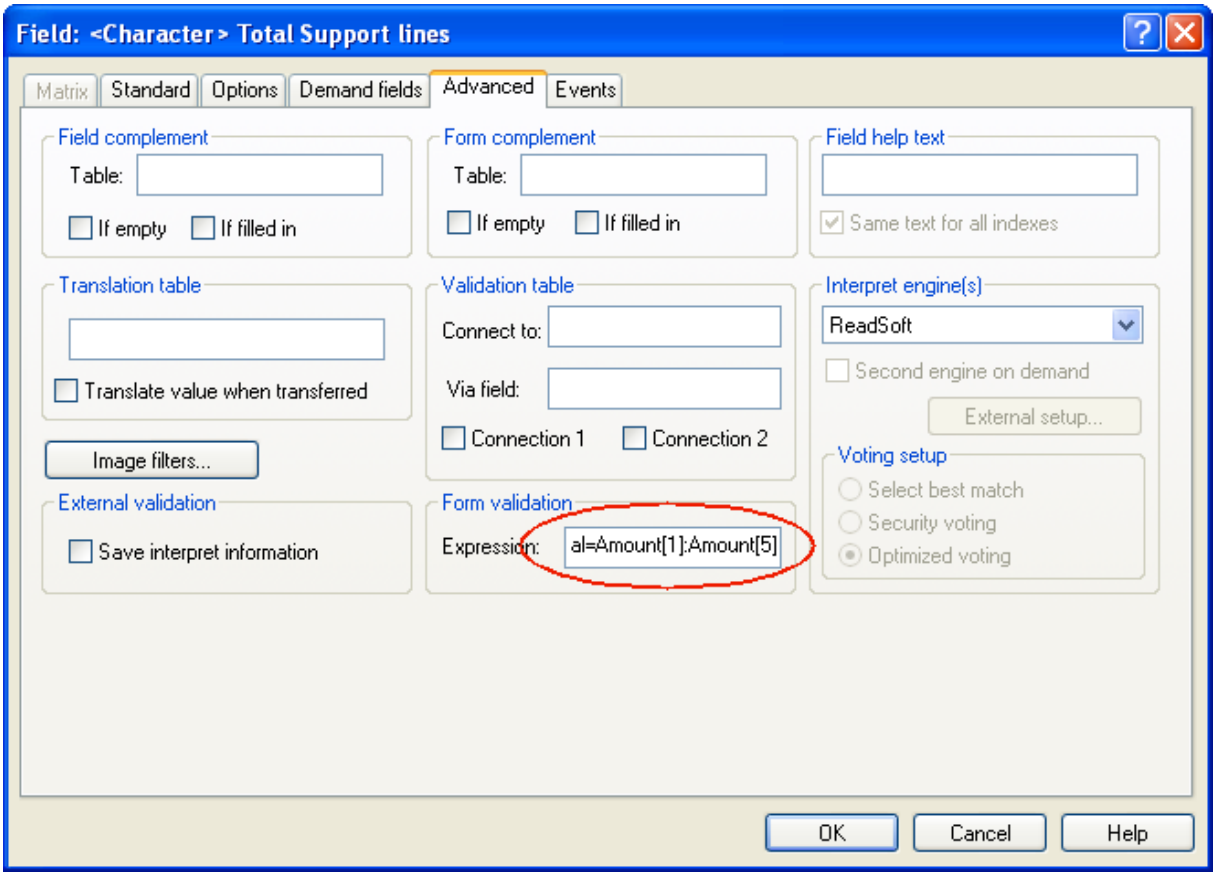
14. The Total field should be the total of all the Amount fields. FORMS can check this for you.

    Click the **Advanced** tab at the top of the **Field** dialog.

15. In the **Form validation** group box, next to **Expression**, type:

    Total=Amount[1]:Amount[5]

    to indicate that Total is to equal the total of Amount[1] through Amount[5]. Ensure you use square brackets in this box.



16. Click **OK** to close the **Field** dialog.

17. Now let us specify that if any one of the Amount fields is filled in, then Total must also be filled in. Double-click any Amount field.

18. Click the **Demand fields** tab.

19. By **Demand field 1**, type Total. This means that the Total field is somehow related to the Amount fields.

20. Click **Requirements/Actions**. A grid is displayed in which you can specify action to be taken in specific circumstances.

21. In the upper right-hand corner, select **Verify Total**. This means that if Amount is filled in and Total is empty, you want to check the total.

22. Click **OK** twice to exit the **Demand field conditions and actions** dialog and the **Field** dialog.

23. Resave the form definition by clicking the **Save** button on the toolbar.

Define three series of mark fields

The sample form contains three questions that require customers to respond by marking one box in a series of three.

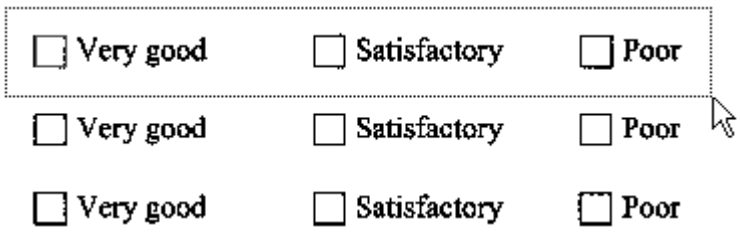
1. Adjust the display using **F3** and **F4**, as well as **CTRL**+arrow keys, so that you can see all the boxes in the series next to question 5.

2. There are two ways to select the boxes. If you want to try both ways, press **DELETE** three times after you try the first alternative.

   - Double-click each box in the series, in the order you want FORMS to number them (usually from left to right).

   or

   - Double-click the first box in the series, and then draw one large frame around the *entire area* that includes *all* the boxes in question 5 (including the first one that is double-clicked). FORMS searches for boxes of similar size within the area and automatically selects the ones that it finds.



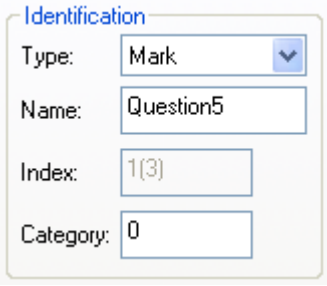
   The way you draw the search area determines how the resulting fields are indexed (numbered). The most common and recommended way of drawing the search area is from the upper left to the lower right.

3. Double-click any of the boxes (or press **F5**) to display the **Field** dialog.

4. Since the field areas consist of boxes to be marked or ticked, select **Mark** as the field type on the **Standard** tab.

5. Type a **Name** for the fields (such as Question5).

   All the fields in the series receive the same name, but each has its own index, as described previously. All the fields receive the same attributes you select in the **Field** dialog.



6. As explained earlier, since a check mark or "x" has no inherent value, you must give it one in the **Value** box (located in the **Format** group box).

   Type a number that will represent the field when it is filled in. Even if you do not want any number to appear at all in the transfer file when the field is filled in (which you can indicate later), you must type in a **Value** for FORMS' use.

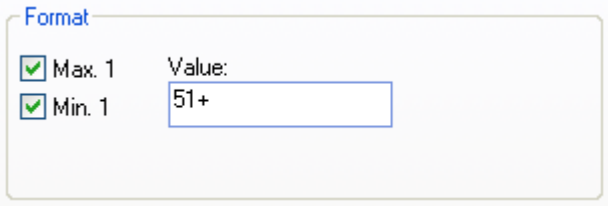
   Using the same logic as used in question 3, you can type in 2-digit numbers for the values, with the first digit indicating the question number and the second digit indicating which response was given.

   In the **Value** field, type:

   51,52,53 *or* 51+

   Either of these result in the boxes having values of 51, 52, and 53 respectively, if checked (that is, if filled in).

7. In the **Format** group box, select both **Max. 1** and **Min. 1**, because one and only one of the boxes in the series must be filled in by the customer.



   If you click the **Options** tab, you see that **Select manually** is selected by default under **If too many** and **If too few**. This means that whenever the **Max. 1** or **Min. 1** condition is not met on a form, the Verify module later shows you the boxes and prompts you to choose which one to count as filled in.

8. Click **OK** to close the **Field** dialog.

9. There are still two rows of mark fields to define. Double-click the first box in each row.

   Then choose **Edit** ⇨ **Field** ⇨ **Copy field pattern** or press **F7.** FORMS finds the two rows of mark field areas that have the same format as the first row.

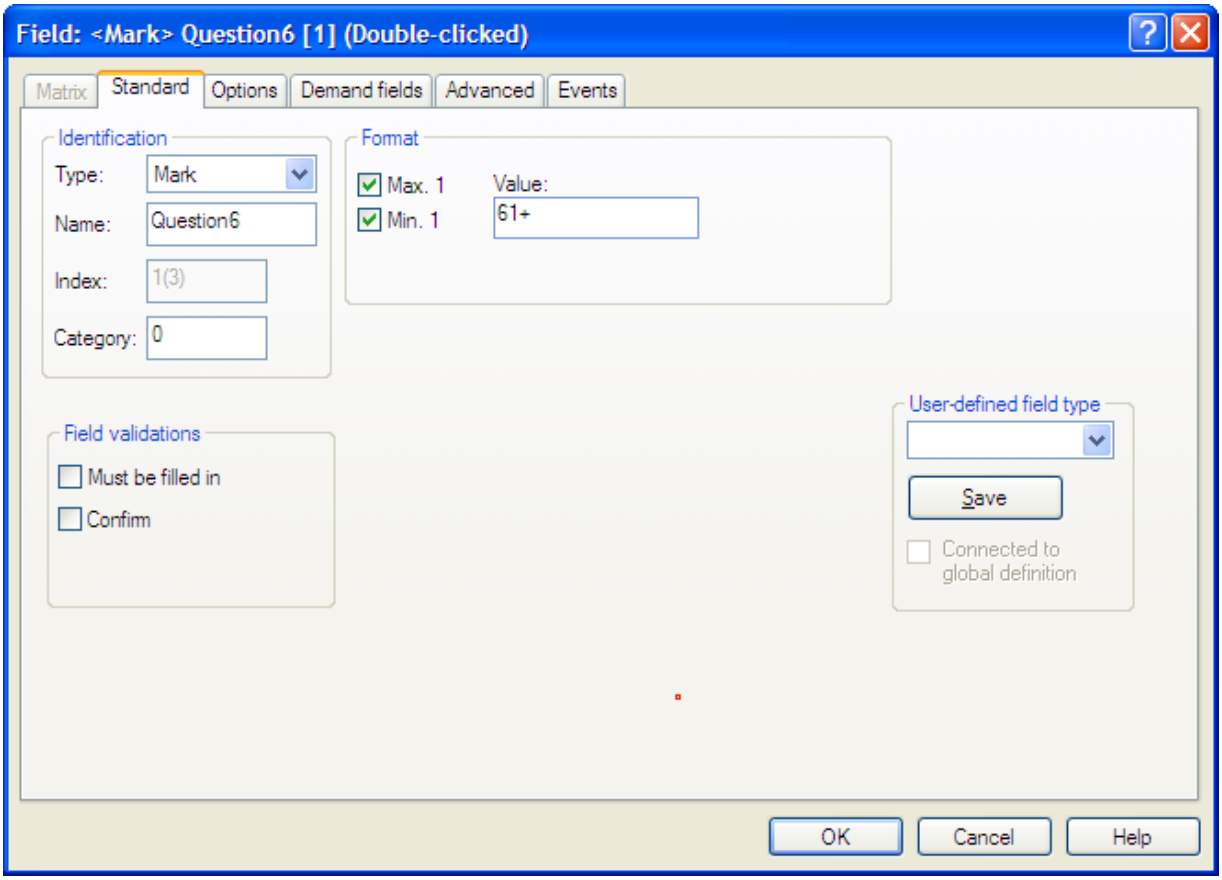
10. The **Field** dialog is displayed, showing the same selections you made for the Question5 series.

    If you gave the previous series a name ending with a number (such as Question5), then FORMS copies the **Name** of that series, as well, but increases the number. That is, the **Name** of the second series is Question6. You can change this name if you wish.

11. If the second series results in the same values as the first, the output data could be confusing. In the **Value** field, we suggest you replace the copied values with:

    61,62,63 *or* 61+

    The first digit (6) indicates the question number and the second digit (1, 2, or 3) indicates which response was given.



12. Click **OK** to close the **Field** dialog. Another **Field** dialog is displayed automatically. This one is for the third series of mark fields. FORMS named the field Question7 (unless you changed the field name in step 10).

13. For the third series of mark fields, we suggest you type in the **Value** field:

    71,72,73 *or* 71+

14. Click **OK** to close the **Field** dialog.

15. Resave the form definition by clicking the **Save** button on the toolbar.

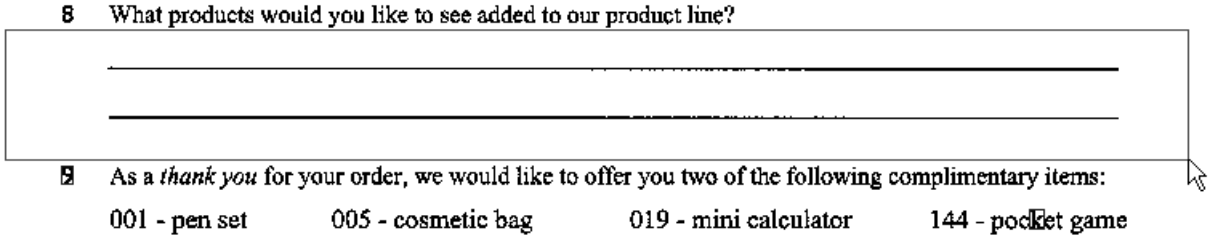
Define a hand-drawn image field for unspecified text

The next field area on the sample form is less structured. It contains two lines on which customers can write which additional products they would like to see offered.

There is no way to validate such a field (that is, to predefine what information can be expected to occur there), and the variety of responses is limitless. Thus, let us say that you wish to look at this field area (when it is filled in) and type into the database any sensible suggestions for new products that you find there.

1. Adjust the display using **F3** and **F4**, as well as **CTRL**+arrow keys, so that you can see the entire field area below question 8.

2. Draw a frame around the area, making it extend from just under the text of question 8 to just above the first line of text in question 9. Extend the frame into both the left and right margins, as well, in case people write there. In other words, draw the field area in such a way that it contains the entire area where customers might write their responses.



   You can change the size of the frame by pressing and holding the **SHIFT** button, clicking anywhere inside the frame, and then using the mouse to move any of the sides. You can also move a field that is already defined, except for selection and matrix fields.



Using the mouse to resize and move a field.

   Resize or move only those field areas that have no support lines and cannot be double-clicked (like this one); fields that can be double-clicked may lose important properties if you use the mouse to move them or change their size.

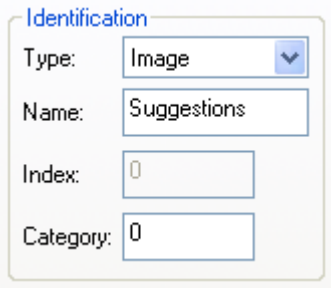
   If you need to start again, press the **DELETE** key on your keyboard to erase the frame.

3. Double-click the frame (or press **F5**) to display the **Field** dialog.

4. On the **Standard** tab, select **Image** as the field type. Information in an *image field* is not interpreted by FORMS, but the presence of filled-in data is detected. You use image fields to provide an image of data you want to manually type as a field or form *complement* (see below).

   As you can see, the **Format** group box disappears when you select **Image**.

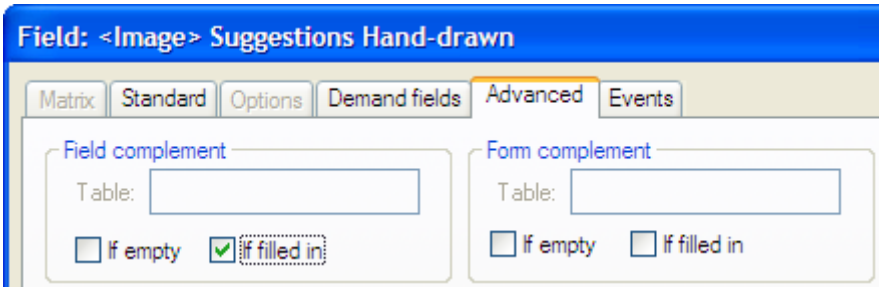
5. Type a **Name** for the field (such as Suggestions).



6. In order to get any data from this field into your transaction (output), you must use this image field as a prompt for typing in supplementary text (max. 512 characters) during verification.

   To specify that you want to do so, click the **Advanced** tab at the top of the **Field** dialog.

7. In the **Field complement** group box, select **If filled in**. Later, Verify will prompt you to type text into a dialog if there is any writing in the Suggestions field.



8. Click **OK** to close the **Field** dialog.

Erase the contents of filled-in forms, define a series of character fields

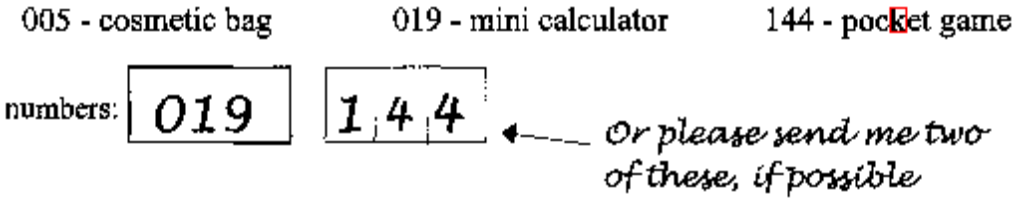
As previously mentioned, a form definition should be created from a blank form (that is, a form that is not filled in). The exception to that rule is a form that includes matrix fields. Read more about matrix fields in Help.

However, if no blank form is available, you can show FORMS where to erase information from the form.

It is important that you erase only writing and marks that are not actually part of the form. Do not erase any characters or lines that are preprinted on the form.

On the sample form, there are two character field areas that are filled in with 3-digit numbers. The contents of these field areas must be erased before you define them.

1. Display a good view of the filled-in boxes by using **F3** and **F4**, as well as **CTRL**+arrow keys.



2. Select the box on the left by double-clicking it.

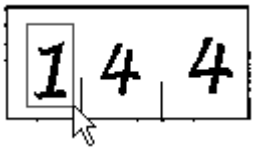
3. Since the field area does not have any preprinted marks or text in it, erase its contents by right-clicking inside the frame and choosing **Erase** from the shortcut menu. The filled-in information disappears.

4. You could proceed to defining the field immediately. However, since there is another field area right next to it that is to be handled in the same way, let us define them as a series.

   Press **DELETE** on your keyboard in order to deselect the frame which FORMS left behind after erasing the contents of the box.

5. The second box has small preprinted lines in it that are part of the form, so you must not erase them. Therefore, even though it is a double-clickable field area, do not double-click it before erasing its contents. Also, deactivate the **Support lines** function before erasing filled-in information from field areas that have preprinted lines in them. This ensures that FORMS will not erase part of the original form.

   Use the **F3** key to enlarge the box until it covers a large part of your screen. Use the mouse to draw a frame around the "1". You can resize the frame as described on page 53.

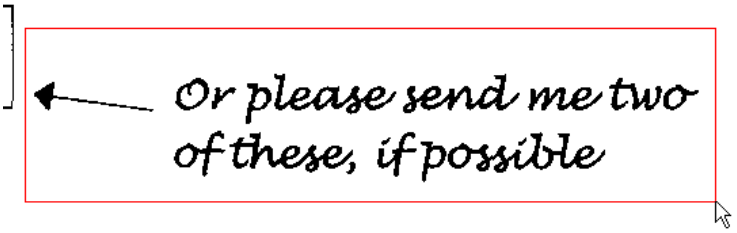


   Since erased marks cannot be reinserted (that is, there is no undo command), make sure that no permanent part of the form is included in the frame.

6. Right-click the frame you just drew and choose **Erase** from the shortcut menu. The "1" disappears. The frame surrounding it remains. Press **DELETE** on your keyboard to remove the frame.

7. Next, draw frames around each of the 4's and erase them in the same way. Press **DELETE** on your keyboard twice to remove the remaining frames.

8. In addition, the customer wrote a little note outside of this field area. We recommend that you erase all marks that occur on a form that are not preprinted.



   Use the mouse to carefully draw a frame around the customer's request and the arrow.

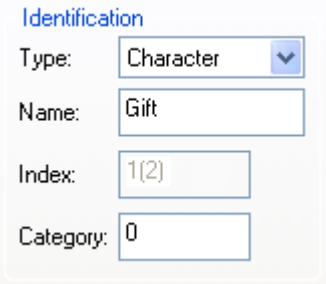
9. Make sure that no permanent part of the form is included in the frame. Then right-click the frame and choose **Erase** from the shortcut menu.

   Press **DELETE** on your keyboard to remove the remaining frame. Click the **Support lines** button on the toolbar to reactivate this function.

10. Double-click each of the two boxes that are to contain 3-digit product numbers.

11. Double-click one of the frames (or press **F5**) to display the **Field** dialog.

    Since the fields are to contain numbers, **Character** (which is selected by default) is the correct field type.

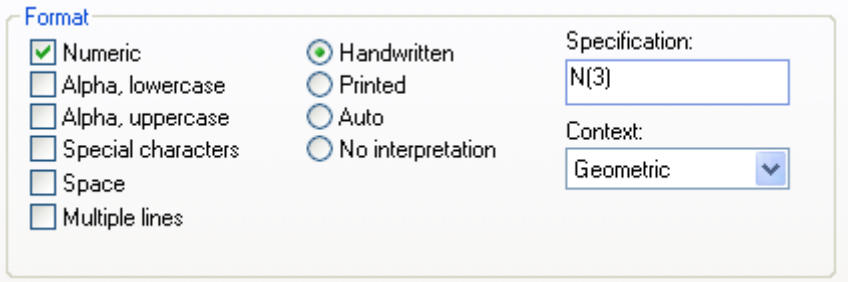


12. Type a **Name** for the fields. Gift would be appropriate.

    Both of the fields in the series receive the same name, but each has its own index, as described previously. Each field receives the same attributes you indicate in the **Field** dialog now.

13. Again, you need to tell FORMS what to look for in the field by specifying the format. Since the field is to contain a handwritten number, **Numeric** and **Handwritten**, which are pre-selected, are the correct choices.

    You still need to tell FORMS how many digits to look for. Type N(3) in the **Specification** box. This tells FORMS to expect a three-digit number.



Generally, the more specifically you indicate what the field can contain, the more efficient the interpreter becomes. If you wanted to, you could be very specific in this case, since there are four valid values for this field (001, 005, 019, and 144). In a field specification, characters within square brackets represent *actual characters* allowed to occur in the field. The pipe character (|) means OR. (Help contains more instructions on how to specify field formats and contents in detail.) This means you could type the following specification:

{N[0](2)N[15] | N[0]N[1]N[9] | N[1]N[4](2)}

This tells FORMS to expect one of the three-digit numbers provided. Another correct way to specify the format would be as below:

{N[01] N[014] N[1594]}

This indicates that the first digit must be either 0 or 1, the second digit must be 0, 1 or 4, and the third digit must be 1, 5, 9 or 4. Note that no pipe character (for OR) is included.

However, there are other, more versatile ways of validating field values, which you will see later. For now, N(3) is a good enough field specification.

14. Click **OK** to close the **Field** dialog.

15. Resave the form definition by clicking the **Save** button on the toolbar.

Define a barcode field

The barcode in the lower right-hand corner of the form needs to be defined. Barcodes were originally designed to be read using a laser beam. However, barcodes read by a scanner are easily interpreted by FORMS.

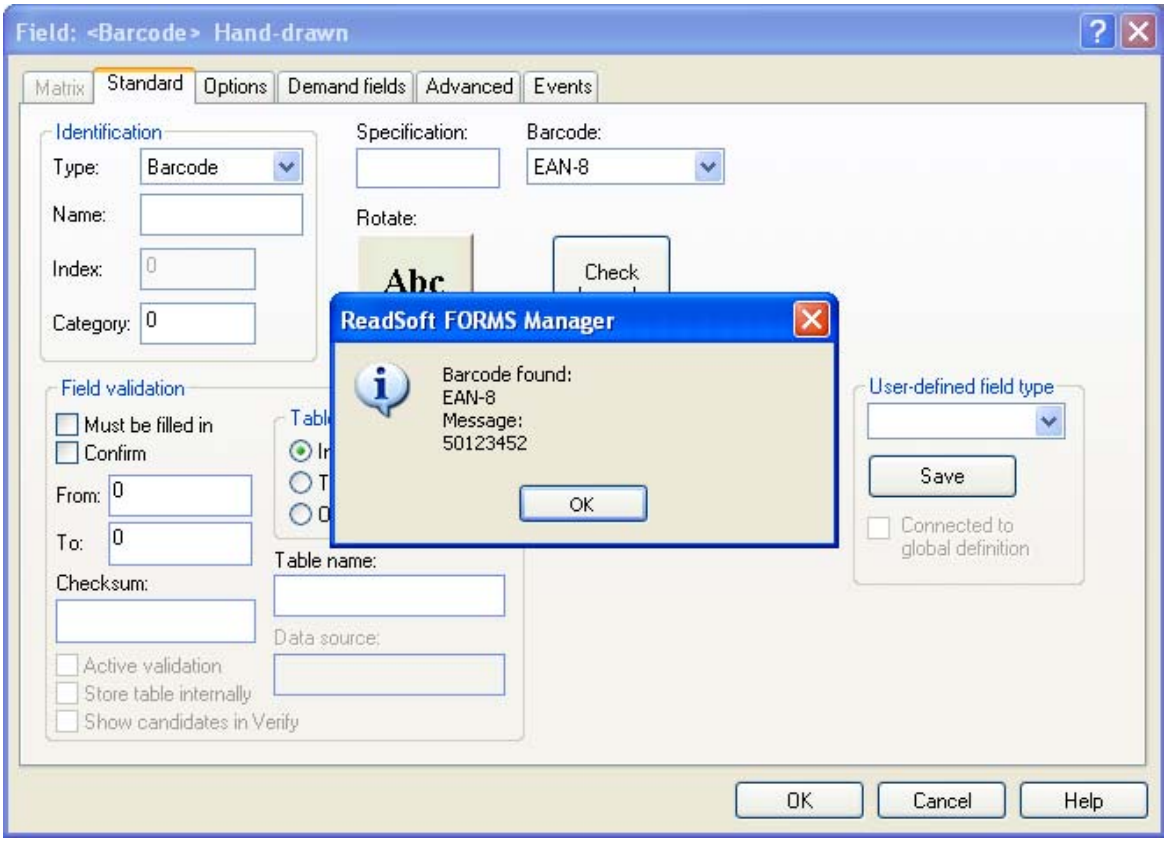
1. Adjust the display using **F3** and **F4**, as well as **CTRL**+arrow keys, so that you can see the entire barcode field area.

   Draw a frame around the entire barcode, leaving a small amount of space between the barcode and the edge of the frame:



2. Double-click the frame (or press **F5**) and select **Barcode** as the field type in the **Field** dialog.

3. Click **Check barcode**. If you are using the sample form, then FORMS identifies it as an **EAN-8** barcode. Now you know what type of barcode to define in a later step.



4. Click **OK** to close the dialog.

5. Click **Cancel** to close the **Field** dialog.

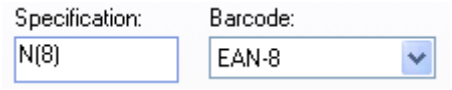
6. Select **Edit** ⇨ **Field** ⇨ **Erase**. The barcode disappears, but the frame you drew remains on the screen.

7. Double-click the frame (or press **F5**) and select **Barcode** as the field type in the **Field** dialog.

8. Type a field name (such as Region). (We can imagine that this form was specific to a particular geographical region, and that the barcode number reflects this.)

9. Select **EAN-8** in the **Barcode** box.

10. You need to tell FORMS how many digits to look for when the barcode is interpreted. Type N(8) in the **Specification** box. This tells FORMS to expect an eight-digit number.



11. Click **OK** to close the **Field** dialog.

12. Resave the form definition by clicking the **Save** button on the toolbar.

Define an address field

The final field to define on the sample form is another example of a field that might contain unspecified text. (The first example was on page 53.) However, in this field you can expect the text – a new address – to have a predictable format.

For the purposes of this tutorial, you do not have to define this field. The rest of the tutorial will work fine whether you define it or not. However, continue reading to learn about some different ways you can define address fields.

There are three possible ways you can define an address field in FORMS:

- By using an external address interpreter
- By defining it as an image field containing unspecified text
- By using VBA code to create a customized solution

Using an external address interpreter

A separate product called the ADDRESSES is available from ReadSoft. It is used with FORMS to significantly improve interpretation and validation when processing address-related information. ADDRESSES supports address interpretation for various countries (see your ReadSoft representative for details).

Standard image field definition for an address

You can define the address field as an image field for unspecified text, just as you did earlier with the customer suggestions field. Although defining the address field in question 10 this way is not a necessary part of this tutorial, you can go back to page 53 and follow the directions again (before continuing on below to "Step 7 – Specify options and validation routines"). The only difference is that you can double-click the new address box (instead of hand drawing it). Plus, you have to give the field a different name (such as NewAddr) than the field in question 8. If you choose not to define this field, then read about the third alternative before moving on to step 7 below.

A VBA solution for unspecified text

The third alternative for processing unspecified text involves writing some VBA code to handle the data. This would work well not only for the address field in question 10, but also for any field that will contain text in some other predictable format.

Using the built-in VBA editor is an advanced topic that is not included in FORMS Basic Training. Thus, users new to FORMS are not expected to define an address field in this way. However, those readers who would like to see how VBA is used in FORMS can try a simple VBA solution to define this field by going to the Appendix on page 104.

Step 7 – Specify options and validation routines

Now that you have defined all of the fields on the sample form (with the possible exception of the new address field in question 10), it is time to look at some additional settings that will enhance the interpretation of these fields.

A *validation routine* allows FORMS to check the validity of a value it interprets for a field, in accordance with conditions you specify. Some options also give FORMS more power to check and format the data, while others help you to verify or format the field yourself. In this step, you will set some options and validation routines.

The more validation routines and options you can specify, the more accurate and customized the output data will be. However, these settings should be applied judiciously: Applying unnecessary settings can slow down interpretation (while not necessarily improving interpretation results).

There are *many* options and validation routines available for most types of fields. They are all described in Help.

Options and validation routines can be specified either when you define each field or after you have defined and saved some or all of the fields in the form definition.

If you followed the tutorial up to this point, you already have some practice at this, because you specified the following options and advanced characteristics:

- A field must be rotated before interpretation (page 38)
- A field must be filled in (see page 40)
- Two numeric values must fall within certain ranges (page 43)
- The sum of a series of fields must equal the value of another (page 49)



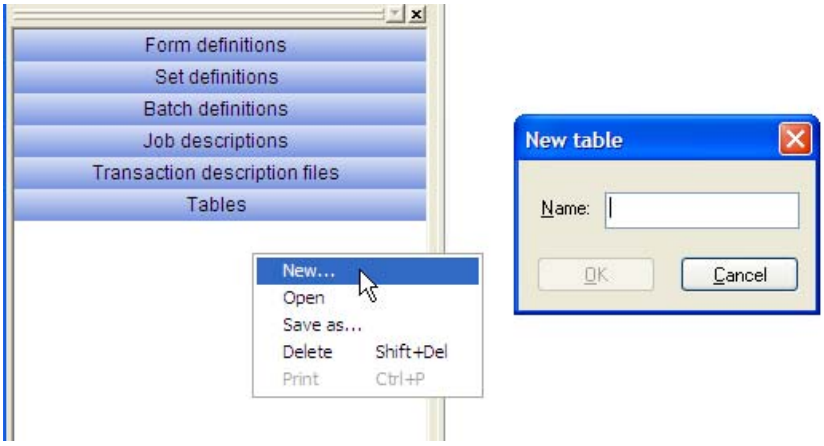
- From a series of mark fields, one and only one of the boxes must be filled in (page 51)
- You want to type extra data (a field complement) when a particular field on the form is filled in (page 54)

Now we will describe two more.

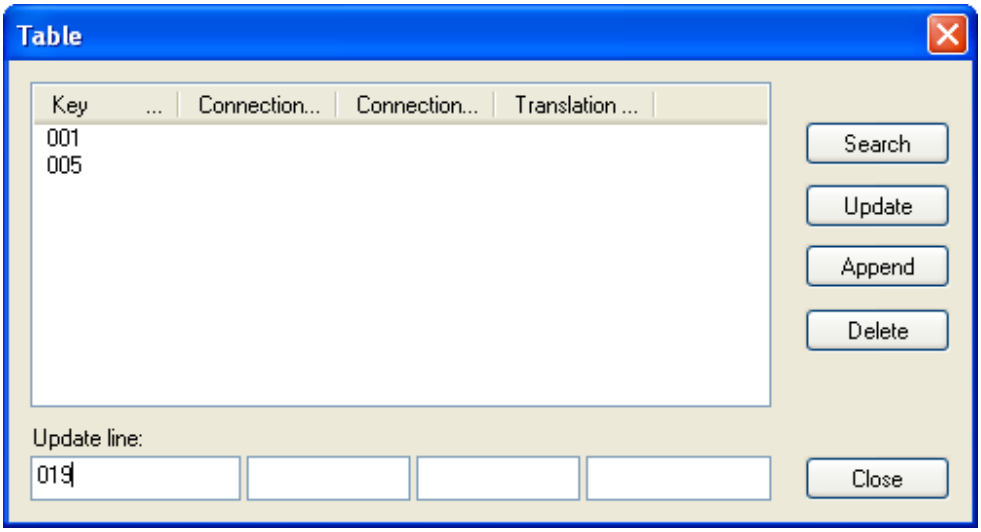
Create and use a table of valid gift numbers

A table of valid gift numbers for the Gift field series defined in question 9 on the sample form would increase interpretation accuracy.

1. Click the **Tables** object bar.
2. Right-click anywhere in the white area under the **Tables** object bar and choose **New**. The **New table** dialog is displayed.



3. Type a suitable name (for example, GiftNo).
4. Click **OK** to close the **New table** dialog. The **Table** dialog is displayed.



*The **Table** dialog lets you create tables for many different uses.*

5. In the **Update line**, type the *first* gift number below.

   001 005 019 144

   Then click **Append**.

6. Continue by replacing the text in the **Update** line with each gift number, one at a time, clicking **Append** after each one until you have entered all four of the valid gift numbers into the table.

   If you insert an incorrect value by accident, select the value in the table. Then click **Delete**. The value is deleted from the table.

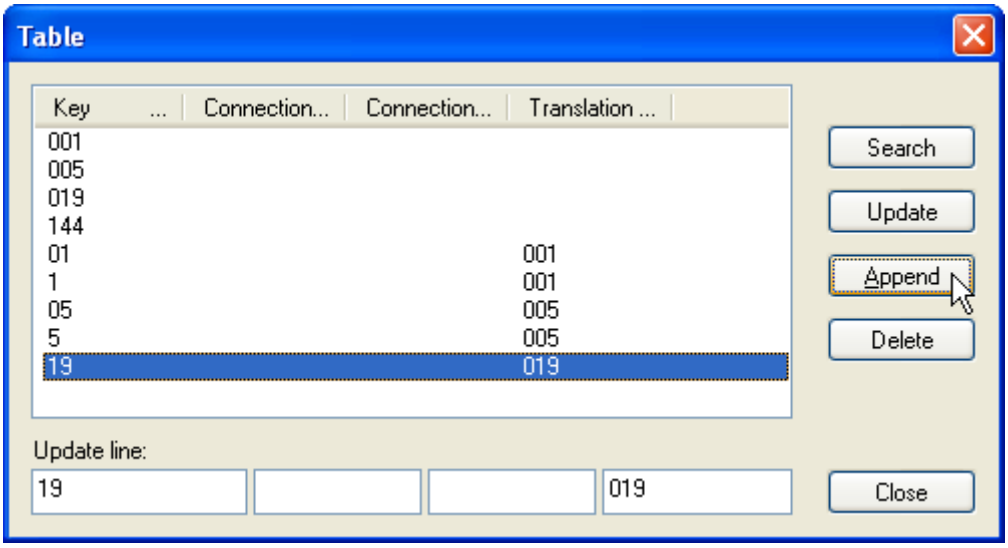
7. Since some customers might not write the initial zeros in the gift numbers, this offers an interesting opportunity to translate values. For example, you know that 19 and 5 are valid gift numbers, but FORMS does not, unless you insert these values into the table.

   In addition, suppose you always want a 3-digit number in the transfer file, even when customers do not write initial the zeros.

   Proceeding as you did above, type each of the following valid gift numbers individually in the **Update line**:

   01 1 05 5 19

   After you type each gift number in the first **Update line**, press **TAB** three times. In the **Update line** box all the way to the right, type a 3-digit **Translation value** for each of the incomplete numbers (clicking **Append** after each one). FORMS will change each incomplete number to its 3-digit form.

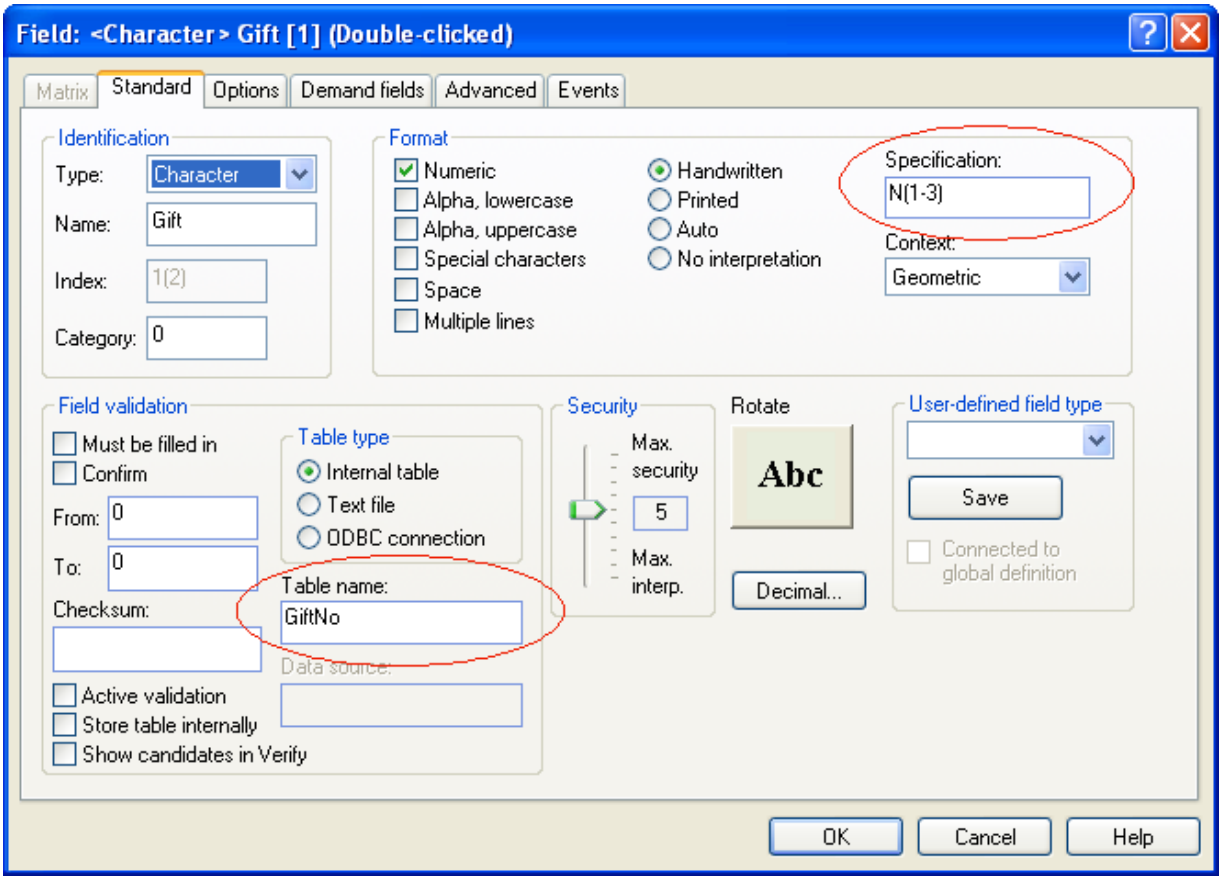


*The **Table** dialog as it appears when you create the table described above.*

8. Click **Close** to close the **Table** dialog. FORMS saves the table at this time.

9. The GiftNo table is ready for use. Going back to the sample form, double-click either of the Gift fields.

10. In the **Specification** box, change the specification to N(1-3).

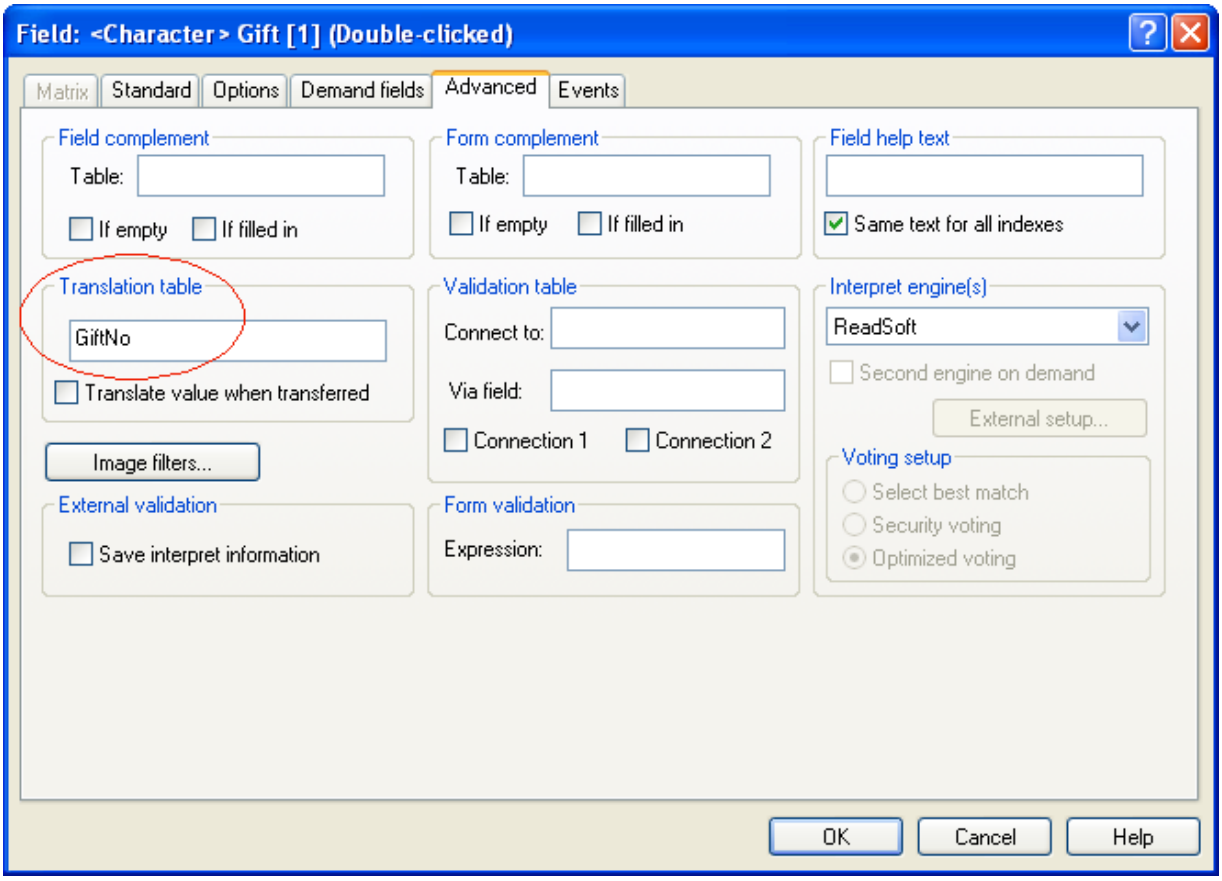


11. In the **Field validation** group box, type the name of the table, GiftNo, below **Table name**.

    For the two Gift fields in question 9, FORMS' Interpret module will only allow gift numbers that are present in the table called GiftNo. If an invalid gift number is found, FORMS will present the field to you during the verification process so that you can correct it accordingly.

12. Click the **Advanced** tab at the top of the **Field** dialog.

13. In the **Translation table** box, type the name of the table (GiftNo) again. Note that FORMS will use the same table for both validating the field and for changing incorrectly formatted values to the correct 3-digit format.



14. Click **OK** to close the **Field** dialog.

    If you double-click the other Gift field, you see that the GiftNo table now applies to it, too.

15. Resave the form definition by clicking the **Save** button on the toolbar.

Detect strikeouts on mark fields

Suppose that customers often check (tick) a mark field and then change their minds, scribble over their checked mark field, and check another one.

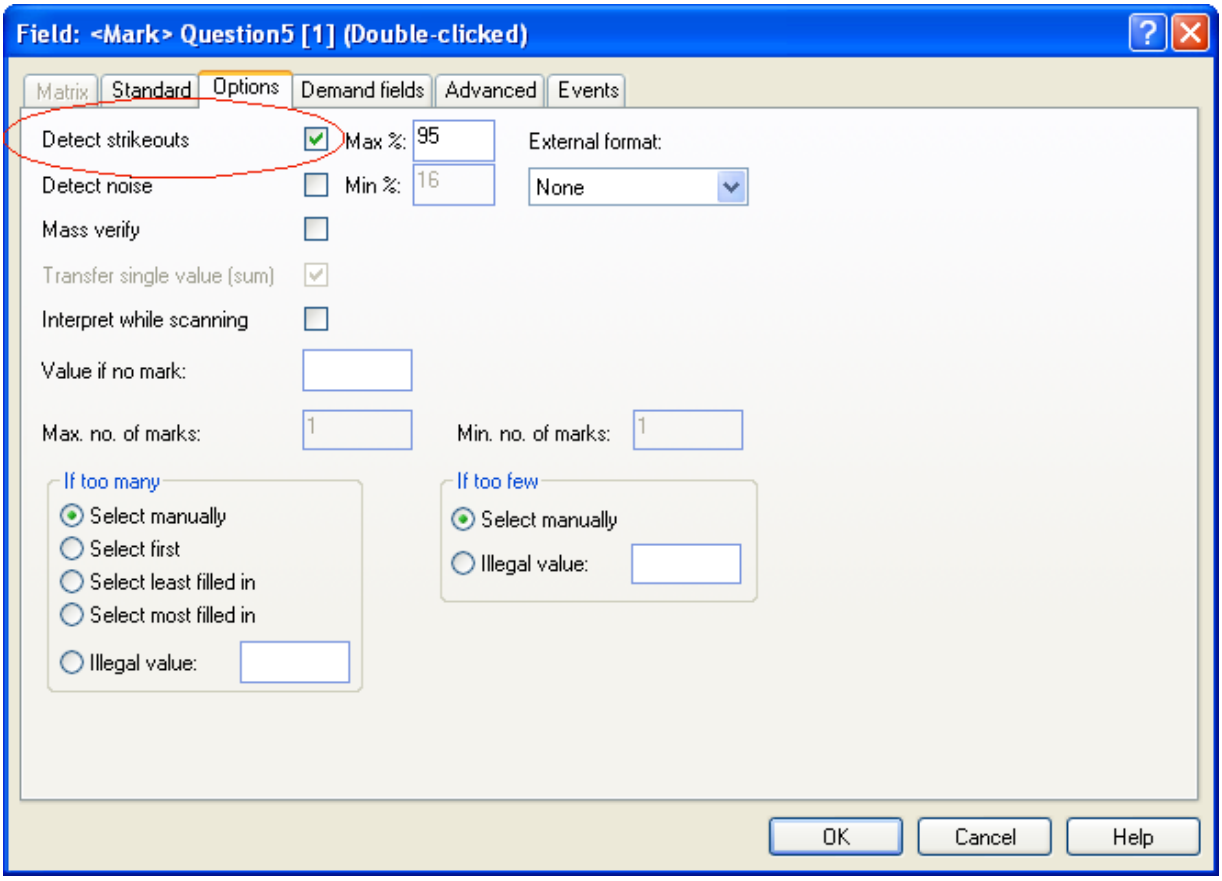
FORMS can detect such crossed-off fields.

1. Double-click one of the mark fields in the first row of three (question 5). The **Field** dialog is displayed.

2. Click the **Options** tab.

3. Select **Detect strikeouts**.

   The **Max %** number next to the right stands for a percentage of the field that can be occupied by pixels. If the field area contains more pixels than the indicated percentage, FORMS ignores the mark field. That is, the field is treated as if it were not filled in.



4. Type 95 next to **Max %**, if it does not appear automatically. Ninety-five percent is probably a good value for this mark field. However, in practice it is very important that you test and adjust the percentage based on test runs.

5. Click **OK** to close the **Field** dialog.

6. Double-click one of the mark fields in the second row of mark fields and select **Detect strikeouts** in the same way, except with a different percentage, such as 90.

7. For the third row of mark fields, select **Detect strikeouts** with still another percentage, such as 50.

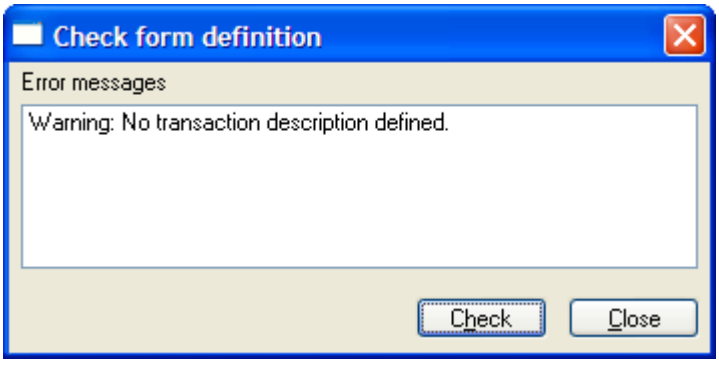
   Later in test runs, you can check which of the three percentages (95, 90, or 50) works best on this particular form definition.

8. Resave the form definition by clicking the **Save** button on the toolbar.

Step 8 – Check the validity of the form definition

When all the fields on a form definition have been defined and their validations specified, check the validity of the form definition. This is the final step in creating a form definition.

1. Choose **Check** (**Edit** menu). FORMS checks the formal validity of your field definitions and displays the results in a dialog.



2. If you followed the instructions above, this dialog now indicates that only the transaction description is missing; it will be defined in the next stage. If there are any other errors, correct them and check the form definition again.

If you wish, you can keep the **Check form definition** dialog open while you are defining and editing a form. Whenever you click **Check**, FORMS updates the information.

First stage is now complete

The first stage in preparing for production is now complete. You have created a form definition that will be used in production. This stage is the main task in preparing for production.

Nevertheless, there are four other pre-production stages to complete before you can run forms through the production modules.

Create a standard transaction description

Now that you have created a form definition, it is time to use the Manager module to create a transaction description for it.

See page 12, “Determining the content and layout of the output,” to review transaction descriptions.

Creating a standard transaction description requires the following steps:

1. Open the **Transaction description** dialog and view a standard transaction description (page 67)
2. Remove unnecessary transaction fields (page 68)
3. Edit transaction fields (page 69)
4. Insert additional transaction fields (page 71)

5. Change the order of transaction fields (page 72)

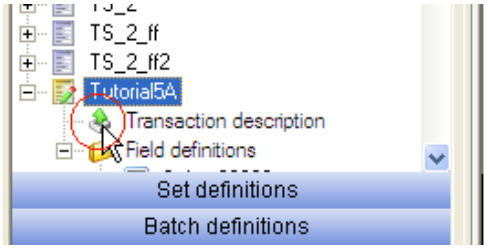
6. Determine the physical layout of the output data (page 73)

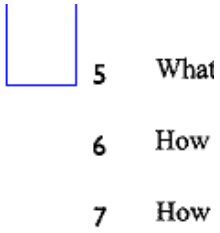
7. Save the standard transaction description (page 73)

Step 1 – Open the Transaction description dialog and view a standard transaction description

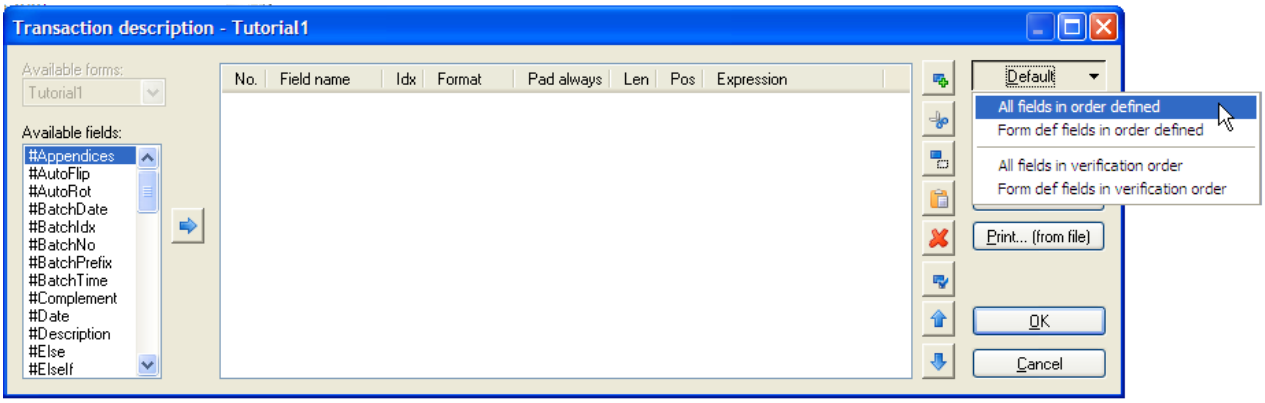
1. Click the **Form definitions** object bar if it is not already displayed. Click the plus sign (+) to expand the form definition for which you want to create the standard transaction description.

2. Double-click the **Transaction description** icon. The **Transaction description** dialog opens.





3. Click **Default** and select **All fields in order defined** to obtain a standard transaction description for the form definition. FORMS creates a number of *transaction fields* that include both *system variables* (which begin with the "#" character) and all the fields you defined in the form definition.



We will examine this transaction description and edit it in various ways.

4. Again, notice that some fields begin with "#". The data that results from these system variables is not related to how the form is filled in but comes internally from FORMS.

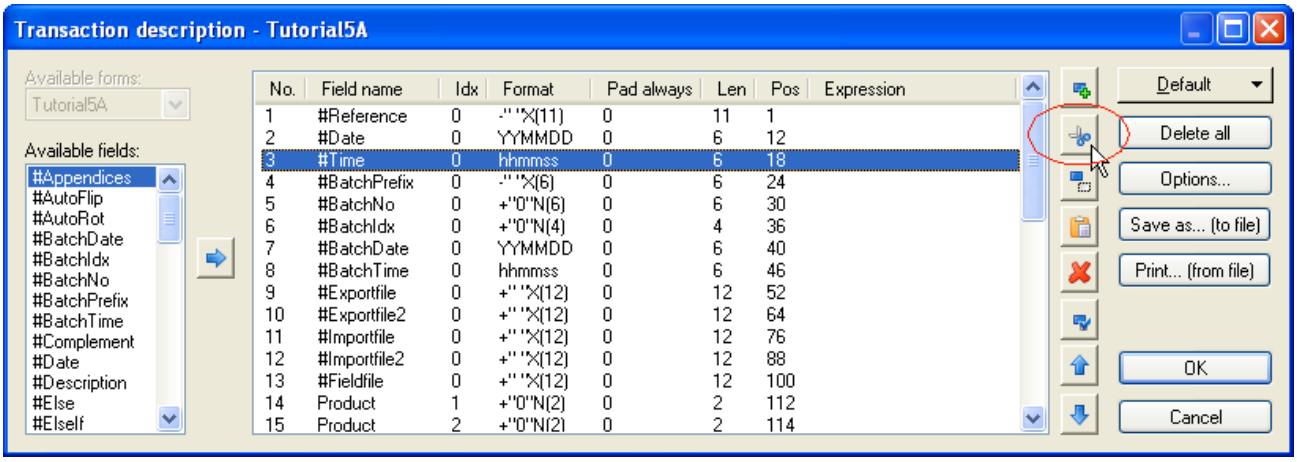
5. Use the scroll bar on the right-hand side of the dialog to move down the list of transaction fields. Notice the transaction fields that have the same names as the fields you created. This means that data from those fields will be included in the output.

6. Notice that one or more fields begin with an ampersand (&). These are for *field complements*. For example, &Suggestions is a transaction field for the field complement of the Suggestions field (see page 54). The Suggestions field itself is not included, because it is an image field. Information in an image field is not interpreted by FORMS.

Step 2 – Remove unnecessary transaction fields

1. The details in the dialog that you see may be slightly different from how it appears below, depending on the settings in your `Ehglobal.ini` file. This is nothing to be concerned about.

   Assume you do not want the system variable #Time to appear in your output file.



   - Click anywhere on the #Time line to select it.

   - Click the **Cut** button. (Do not click **Delete all**, as this deletes *all* the transaction fields.) The #Time field disappears from the transaction description.

2. Also assume that you do not want the system variables #Exportfile, #Exportfile2, #Importfile, #Importfile2 or #Fieldfile to appear in your output file.

   - Click anywhere on #Exportfile to select it.

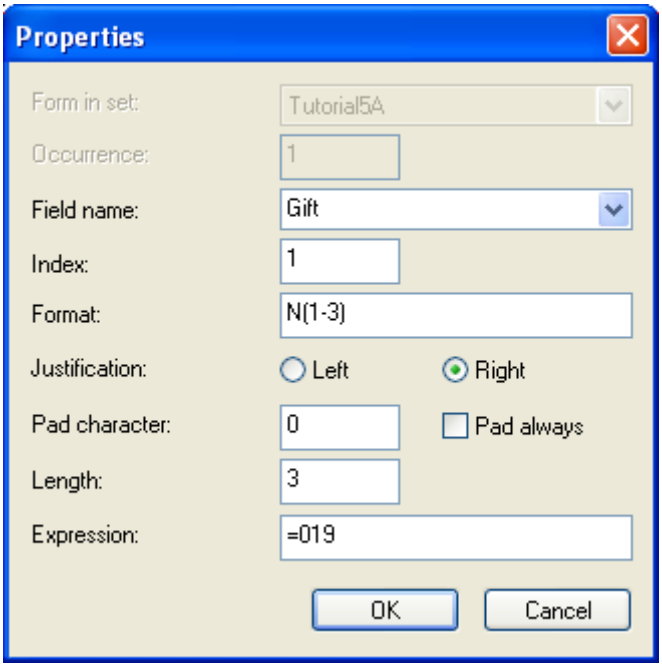
   - Holding your keyboard's **SHIFT** key down, click the #Fieldfile line. All five lines are now selected.

   - Click the **Cut** button. All five transaction fields disappear.

Step 3 – Edit transaction fields

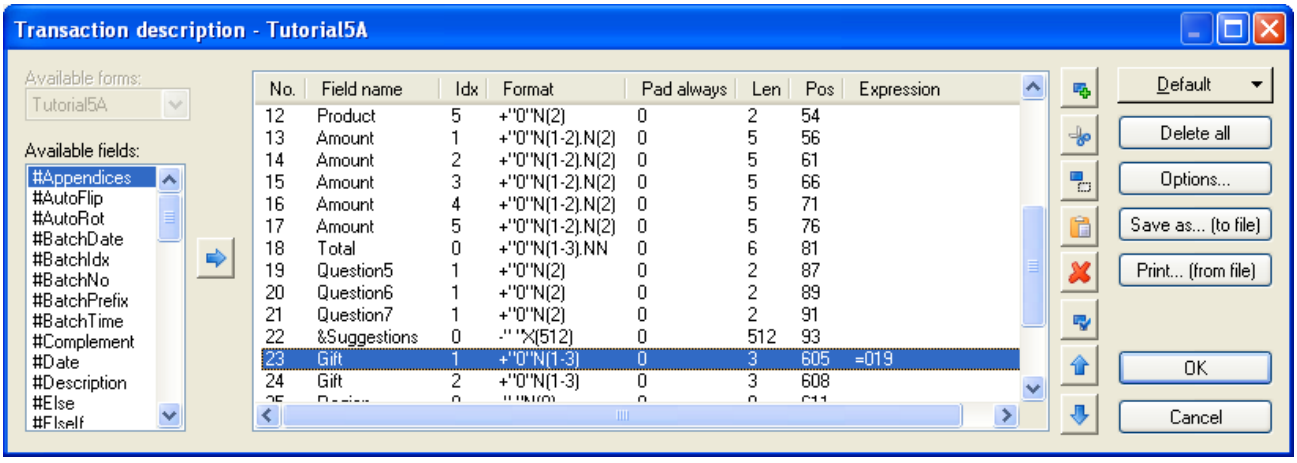
On the sample form, Gift is a series of fields in which the customer can choose two free gifts. But what if the customer failed to specify a gift? Assume that you want to send a particular gift to customers who failed to choose.

1. Double-click anywhere on Gift[1] (that is, the Gift line with "1" as the index). The **Properties** dialog is displayed.



2. Click the **Expression** box and type one of the gift numbers, preceded by an equal sign. For example, =019.

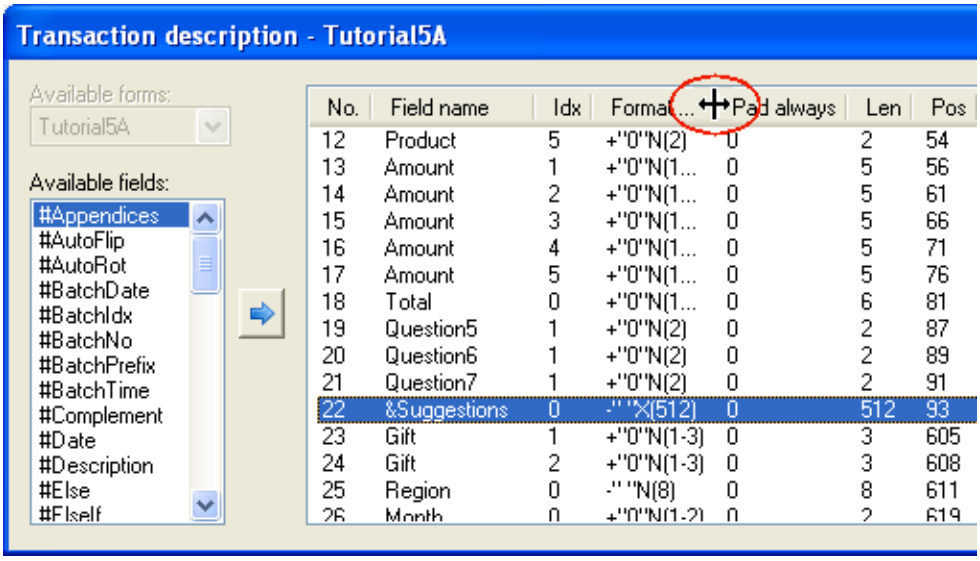
3. Click **OK** to close the **Properties** dialog. The expression is added to the Gift[1] transaction field.



The result of this is that you have defined a default value for Gift[1]. If Gift[1] has no value after verification, the value "019" is assigned.

Now take a look at &Suggestions. &Suggestions is the field complement for Suggestions (which is not included as a transaction field because it is an image field). It represents the customer suggestions that will be typed into a dialog in Verify, if the field is filled in.

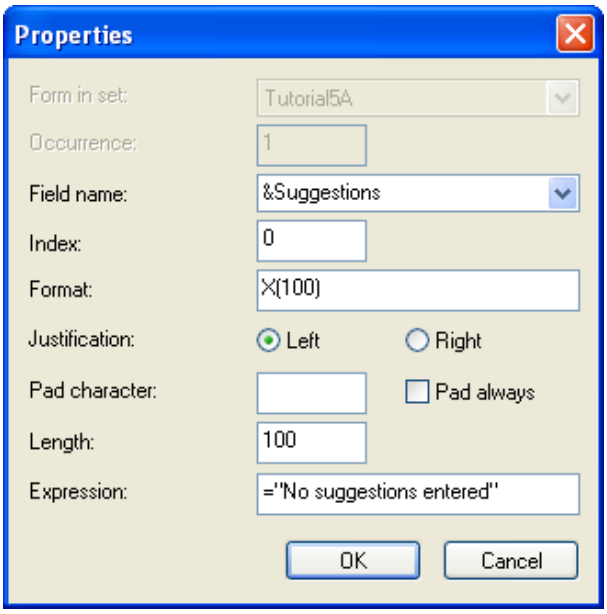
You can see that according to the format, the length of this field is 512 characters. If you cannot see the entire format, resize **Format** as shown below.



*Resizing the **Format** part of the **Transaction description** dialog.*

You can change &Suggestions and other field complements to a more reasonable length.

1. Double-click anywhere on &Suggestions. The **Properties** dialog is displayed.
2. In the **Format** box, change X(512) to something more reasonable, for example X(100).
3. In the **Length** box, change 512 to 100, as well.



4. If you prefer, you can enter ="No suggestions entered" (including the quotation marks) as a default value for &Suggestions. This text will show up in the transaction if the field is not filled in. Type this in the **Expression** box.

5. Click **OK** to close the **Properties** dialog. You can see that the length is changed.

6. If you are not continuing to the next section immediately, save your work by clicking **OK**. This closes the **Transaction description** dialog and saves the transaction description to the current form definition, replacing the old transaction description (if any). Then resave the form definition by clicking the **Save** button on the toolbar.

   To continue, re-open the **Transaction description** dialog by double-clicking the **Transaction description** icon.

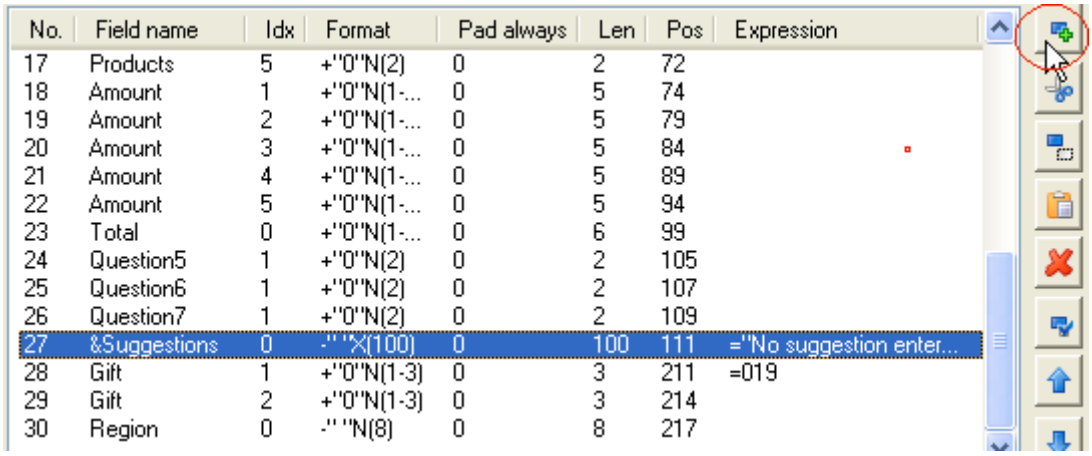
Step 4 – Insert additional transaction fields

You can manually insert transaction fields with any strings or filler characters, text comments, etc. that you want to have in the transfer file.

Here we will insert a header into the transaction description.

1. Select &Suggestions, because the filler is to be inserted just before &Suggestions.

2. Click the button on the top right that has the small plus sign (+) in order to add the new transaction field. The **Properties** dialog is displayed.



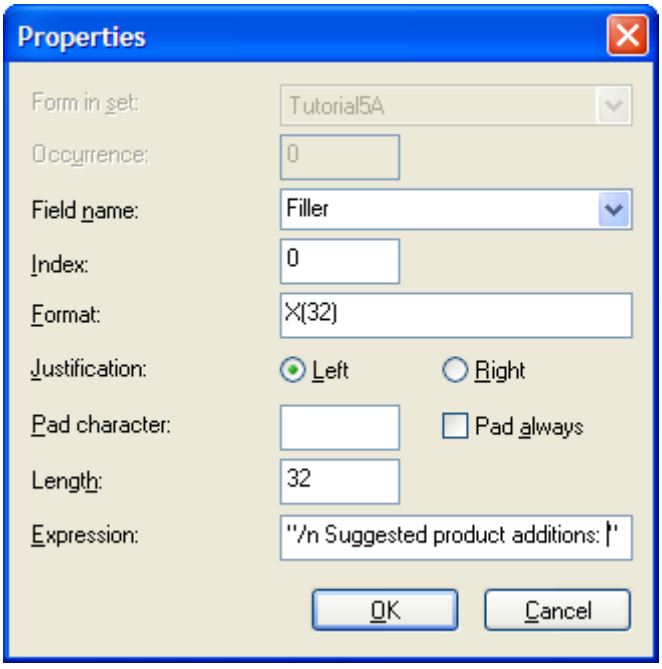
No.	Field name	Idx	Format	Pad always	Len	Pos	Expression
17	Products	5	+''0''N(2)	0	2	72	
18	Amount	1	+''0''N(1-...	0	5	74	
19	Amount	2	+''0''N(1-...	0	5	79	
20	Amount	3	+''0''N(1-...	0	5	84	
21	Amount	4	+''0''N(1-...	0	5	89	
22	Amount	5	+''0''N(1-...	0	5	94	
23	Total	0	+''0''N(1-...	0	6	99	
24	Question5	1	+''0''N(2)	0	2	105	
25	Question6	1	+''0''N(2)	0	2	107	
26	Question7	1	+''0''N(2)	0	2	109	
27	&Suggestions	0	-'' ''X(100)	0	100	111	='No suggestion enter...
28	Gift	1	+''0''N(1-3)	0	3	211	=019
29	Gift	2	+''0''N(1-3)	0	3	214	
30	Region	0	-'' ''N(8)	0	8	217	

3. In the **Field name** box, type Filler or some other recognizable field name.

4. Press **TAB** twice or click in the **Format** box. Type X(32) for an alphanumeric field with 32 characters.

5. Press **TAB** twice or click in the **Pad character** box. Type a space. This means that any unused portion of the format is to be filled with spaces.

6. Press **TAB** twice or click in the **Length** box. Type 32.



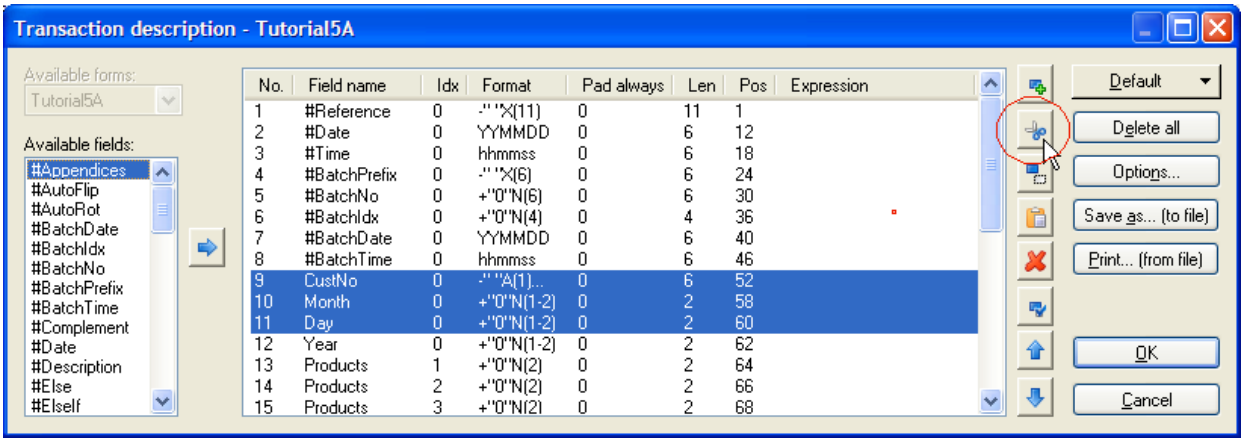
7. Press **TAB** once more or click in the **Expression** box. Type "/nSuggested product additions: " (including the quotation marks and space). "/n" is the symbol that tells FORMS to enter a new line before the text. The result in the transaction (the output data) will be a new line, then the text Suggested product additions, then a colon, and a space.

8. Click **OK** to close the **Properties** dialog. You can see that the new transaction field is inserted.

Step 5 – Change the order of transaction fields

1. Suppose you decide you want the customer's date of birth to appear before any of the other form data, but after the system variables. Click anywhere on Month to select it.

2. Holding your keyboard's **SHIFT** key down, click the Year line. The Month, Day, and Year transaction fields are now selected.



3. Click the **Cut** button.

4. Click CustNo to select it.

5. Click the **Paste** button to paste the three fields above CustNo.

6. If you are not continuing to the next section immediately, save your work by clicking **OK**. This saves the transaction description to the current form definition, replacing the old transaction description (if any). Then resave the form definition by clicking the **Save** button on the toolbar.

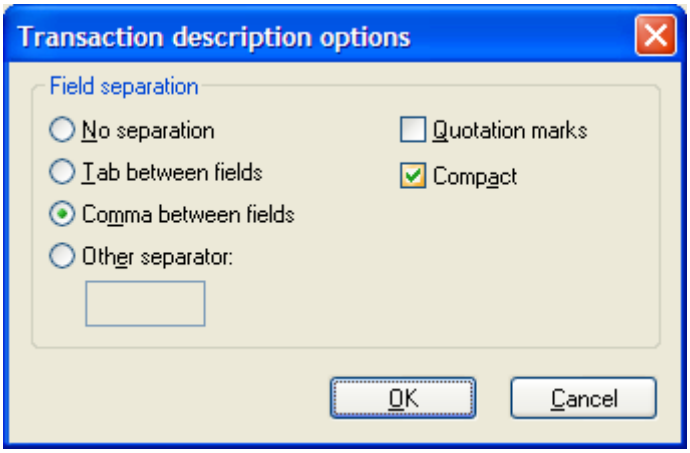
Tip: You can also use the arrows in the lower right-hand corner of the **Transaction description** dialog to move a selected field up or down in the window.

Step 6 – Determine the physical layout of the output data

One way to change the layout of the output data is by adding headers and text (described on page 71). Another way is to add formatting in the **Transaction description options** dialog:

1. Click **Options** in the **Transaction description** dialog. The **Transaction description options** dialog is displayed.

2. Select **Comma between fields** to have FORMS insert a comma between each output field.



3. Select **Compact** to have FORMS remove empty spaces from the output file. Commas will remain. If you do not select **Compact**, the fields will be filled to their maximum length with zeros or spaces.

4. Click **OK** to save and close the **Transaction description options** dialog.

Step 7 – Save the standard transaction description and resave the form definition

1. Close and save the **Transaction description** dialog by clicking **OK**.

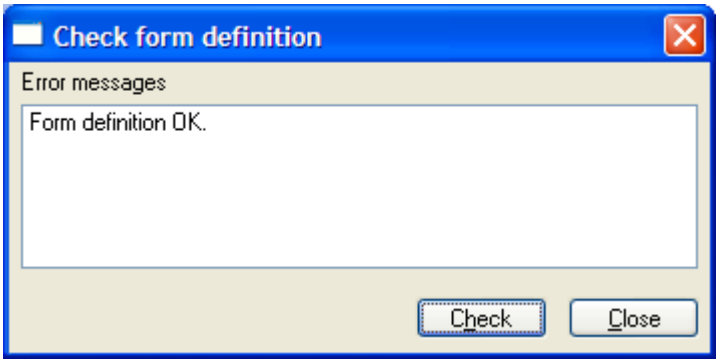
2. Resave the form definition by clicking the **Save** button on the toolbar. (You must resave the form definition if you want to save your changes after editing a standard transaction description.)

Activate and resave the form definition

In this stage of preparing for production, you will re-check the form definition and activate it for use in testing or production. To do this, the form definition must be open.

If the form definition is not open, click the **Form definitions** object bar in the Manager. Then double-click the form definition you want to open. The form definition is displayed.

1. Click the **Form definition settings** icon on the toolbar. FORMS checks the technical validity of your field definitions and displays the results in a dialog. (The result is similar to when you choose **Check** on the **Edit** menu.) If you followed the instructions above, the dialog now indicates that the form definition has no errors.



   Note that FORMS cannot check for logical errors. You must ensure that tables used in the form definition are correctly defined, that external routines are correct, and that the transaction layout suits your purposes.

2. In the **Form definition settings** dialog, click **Active**. (If there are errors listed in the dialog, FORMS will not allow you to activate the form definition.)

3. Click **OK** to close the **Form settings** dialog.

4. Resave the form definition by clicking the **Save** button on the toolbar. The form definition is now complete and ready for use in other modules.

Create a job description

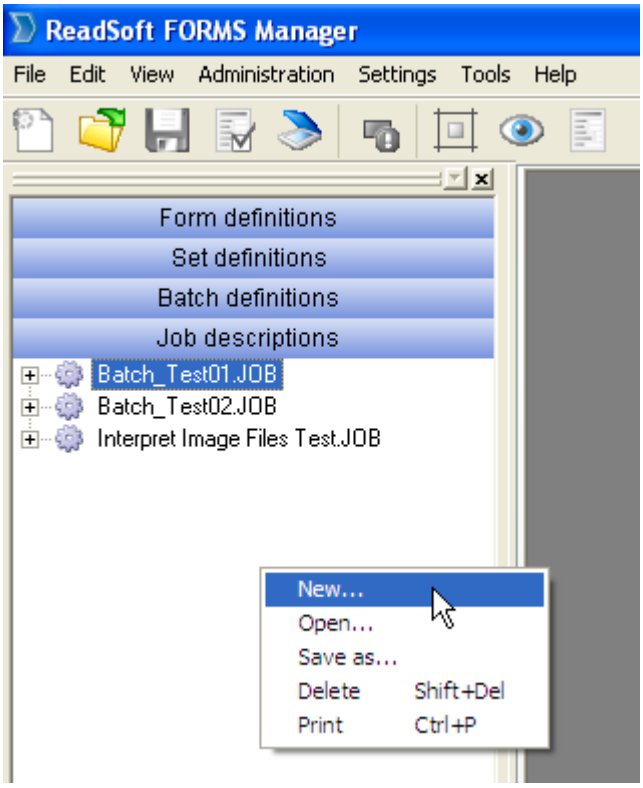
As stated earlier, a job description is a set of instructions that describes what you want FORMS to do. It specifies which form definitions or which set definition you want to process, which module to use and, depending on the module, how you want FORMS to work (for more details, see “Managing production through job descriptions” on page 13).

Your first step in Scan, Interpret, Verify, and Transfer is always either to open an existing job description or create a new one.

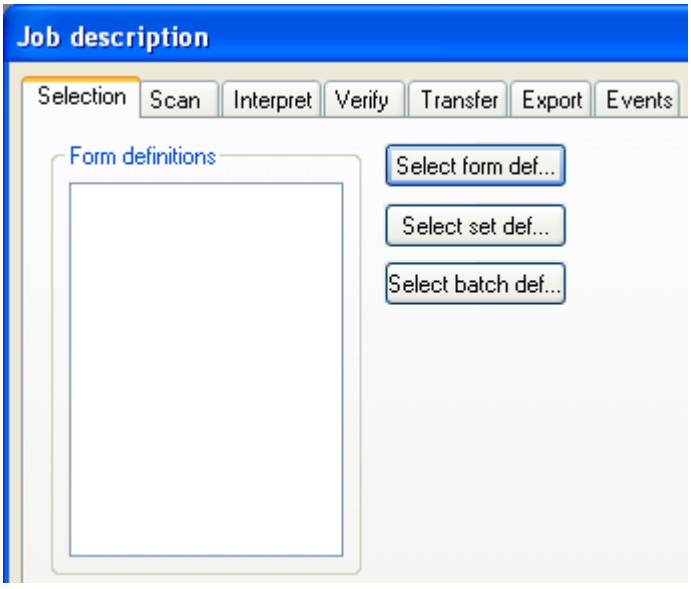
Here, you will use the Manager to create a job description that you can later open and use in all four production modules.

1. Click the **Job descriptions** object bar.

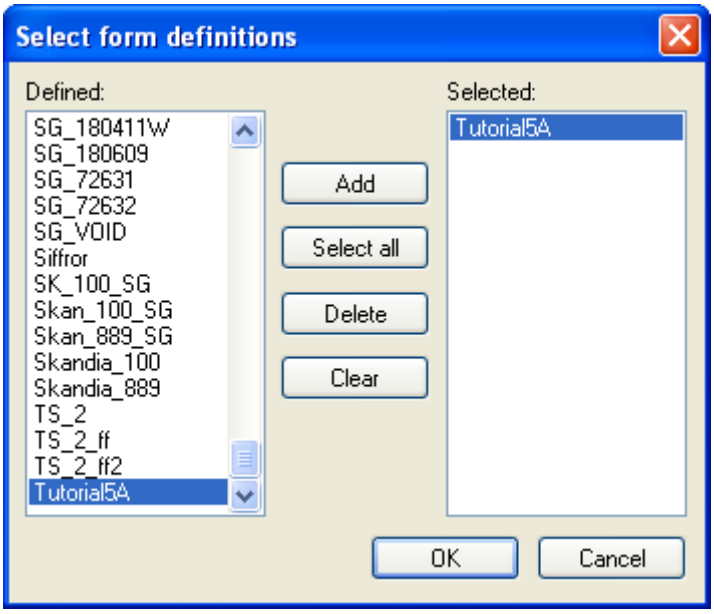
2. Right-click anywhere in the white area under the **Job description** object bar and choose **New**. The **Job description** dialog is displayed.



3. The first thing you need to do is select which form definition or definitions are to be included in the job description – that is, which forms you want to process. Click **Select form def**.



4. In the **Select form definitions** dialog, click the name of the form definition you created.



5. Click **Add** to add the form definition to the list of those included in the job description.

   More than one form definition can be included in a job description. The form definitions must have unique adjustment or recognition fields, and they must be based on the same paper size and scanner settings.

6. Click **OK** to close the **Select form definitions** dialog.

7. If you will be using the image files as you work through this tutorial, rather than actual paper forms, do the following:

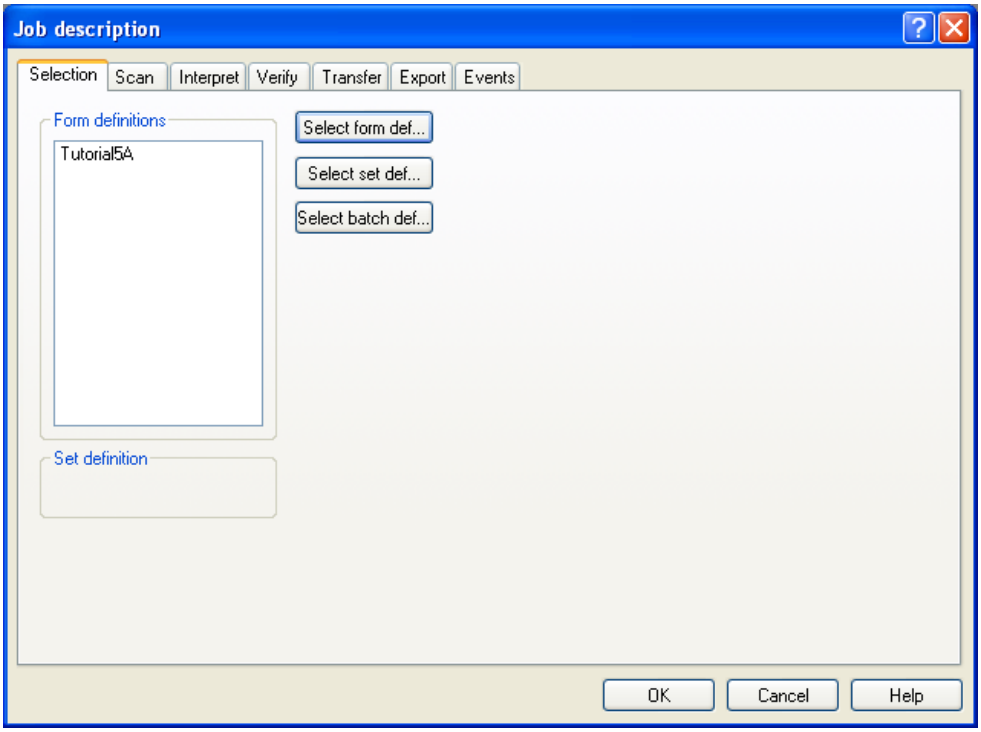
   - Go to page 81 and perform steps 12-16. Read through steps 18 and 19, as well, since you will need this information later for processing image files in the Interpret module.

   - Using Windows Explorer, move the TIFF files `Filled1.tif`, `Filled2.tif`, `Filled3.tif`, and `Filled4.tif` (which are copies of `Sample.tif` that we have filled in and provided as TIFF files for your convenience):

Your operating system	Move from	Move to another folder as chosen in step 15 on page 82, but NOT this folder*
Windows XP or Windows Server 2003 R2	`C:\Program Files\ReadSoft\FORMS\Samples\Tutorial`	`C:\Program Files\ReadSoft\FORMS\Img`
Newer operating systems	`C:\Users\Public\ReadSoft\FORMS\Samples\Tutorial`	`C:\Users\Public\ReadSoft\FORMS\Img`

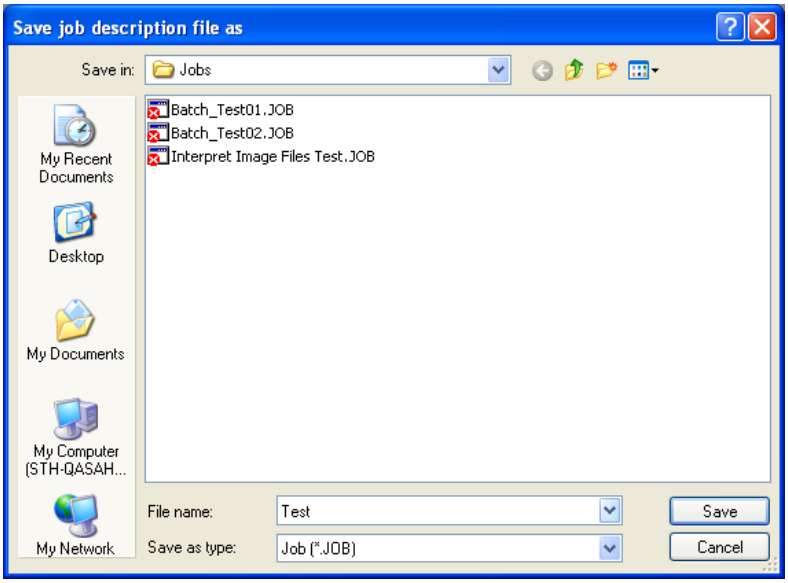
   * Do not move the files to the folder that images are exported to. This is specified in FORMS' **Directories** dialog, and the default folder is listed in the table.

8. Look at the other tabs of the job description dialog, if you wish. FORMS gives the job description standard settings by default, so it is ready for use as soon as you specify which form definitions are to be included. You can read about the various job description settings in Help.



Click **OK** to close the dialog.

9. After you click **Yes** to indicate that you want to save the job description changes, FORMS prompts you to name the job description file. In **the Save job description file as** dialog, type a file name, such as `Test`. FORMS adds the appropriate extension (`.JOB`).



10. Click **Save**. The job description is now ready for use.

Customizing job descriptions

The information above describes how to create a basic job description.

However, before you begin using FORMS in production, it is worthwhile to read in Help about the job description alternatives in the four production modules. By testing different settings, you can customize FORMS to your needs and make the verification process very efficient.

As far as the tutorial is concerned, you do not need to follow any of the instructions in the rest of this section. Nevertheless, by reading the following information, you can get a more complete idea about how to prepare completely for forms processing.

Do not hesitate to define a number of job descriptions with different settings to see which ones work best with different form definitions or for different users. Follow these instructions to base a new job description on an already existing one, using any of the production modules:

1. In the Manager module, click the **Job descriptions** object bar.
2. Click a job description you have already created.
3. Choose **Save as** (**File** menu).
4. In the **Save job description file as** dialog, type a new name.
5. Click **Save**.
6. Double-click the new job description you just saved. The **Job description** dialog is displayed.
7. Make any changes you prefer.
8. Click **OK** to close the **Job description dialog**.
9. FORMS asks if you want to save the edited job description. Click **Yes**.

Test the form definition and job description before production

At this point, you have done everything necessary to begin processing forms, which you will begin doing in the next chapter. Although you do not need to follow any of the steps in the rest of this chapter for the tutorial, the following information will give you a broader idea about how to prepare completely for processing forms in volume.

In fact, after you go through the next chapter on production, you may want to come back to this information in order to try alternative form definition and job description settings with the sample form or your own forms.



Modifying your form definition for testing

Before you begin using FORMS to process real forms in production, it is essential that you take the time to test the form definition and the job description. Testing reveals adjustments you can make that will increase FORMS' interpretation rate and make handling more efficient.

To test a form definition, use your first form definition in production with some representative test forms (as you will do in the next chapter), checking the statistics as you work and comparing the actual forms with the transfer file. Define several additional form definitions with different settings and validation routines to determine which work best for that particular form.

Use the following instructions in order to base one form definition on another:

1. To open an existing form definition, click the **Form definitions** object bar. Then double-click the form definition you want to open. The form definition is displayed.

2. Choose **Save as** (**File** menu).

3. Type a new name and click **OK**.

4. Choose **Form definition settings** (**Edit** menu).

5. Select **Inactive** as the form definition's status.

   Now you can make any changes you wish to the new form definition. You must reactivate and resave it before you can use it.

What to look for during testing

During testing, you will try out different settings to see how these affect FORMS' performance. You can also make sure that basic production functions perform as they should. Keep these points in mind (and see Help for details not covered here):

- Make sure that you have defined all of the areas on the form that need to be captured.

- Make sure that Scan (or Interpret, if you are importing image files) can identify the form reliably. If you get an error message that a form cannot be identified, then you should check adjustment fields (or recognition field, if you have one).

- If the interpretation rate is not satisfactory for your needs, then ask yourself questions such as: Are all of the fields defined properly? Is the format correct for all of the fields? Could the string in each **Specification** box be more detailed? Are the field validation and optional settings correct? (Read more about fine-tuning interpretation in Help.)

- Notice how your Scan, Interpret, Verify, and Transfer jobs function for this form definition. Do they run smoothly? Do you want to use a new view in Verify so that information is presented to you in a different way?

- Try processing forms that contain typical respondent errors (for example, entering invalid product numbers, writing outside of the field area, scratching out one mark field and marking another, and so on). Can you add more field validations and optional settings to compensate for respondents' errors?



You can either make one change at a time to a single form definition or define several additional form definitions with different settings and validation routines to determine which work best for your particular form.

See the next section, “Skip the Scan module during testing,” for details about how you can run forms through the system repeatedly for testing without having to rescan them.

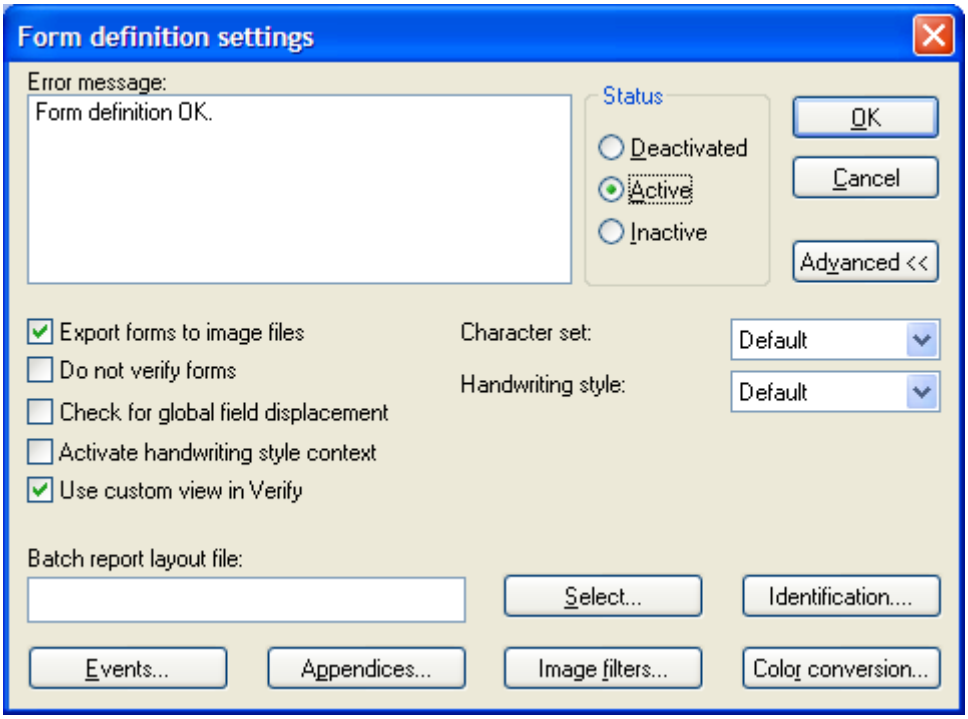
Skip the Scan module during testing

When you are testing the form definition or the Verify job description, you can export forms to image files the first time you scan them. Then, you can interpret those image files repeatedly, skipping the step of rescanning the forms for each test.

1. Open the form definition.

2. Choose **Form definition settings** (**Edit** menu).

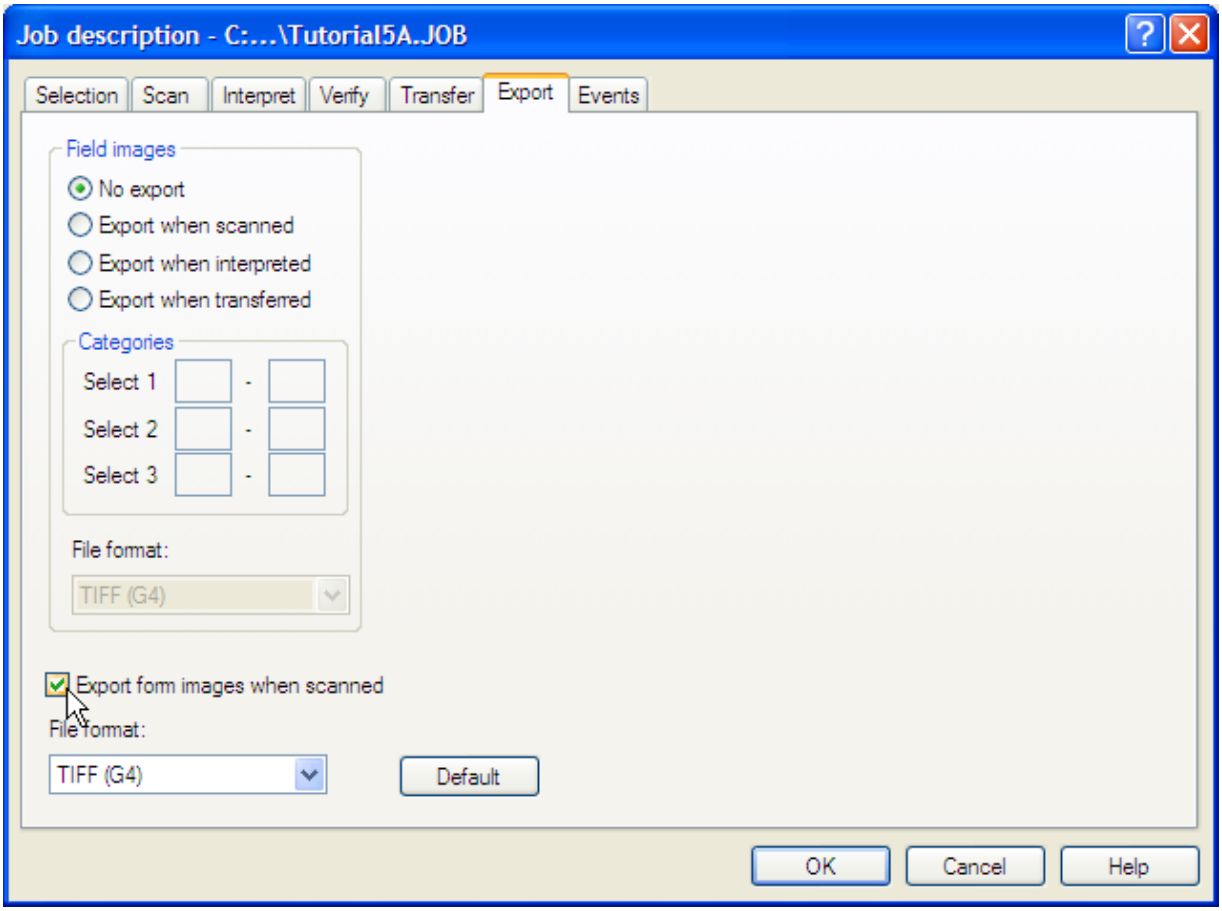
3. In the **Form definition settings** dialog, select **Export forms to image files** (you may have to click **Advanced** first if the dialog is not fully expanded) if it is not already selected. It is selected by default for new form definitions but may not be if the form definition was imported from a previous version of FORMS.



4. Click **OK** to close the **Form definition settings** dialog.

5. Resave the form definition by clicking the **Save** button on the toolbar.

6. Before you scan the forms, open the job description and click the **Export** tab.



7. Select **Export form images when scanned**.

8. Close the **Job description** dialog and save the job description when prompted.

9. Scan the forms as usual. In addition to scanning and identifying the forms, Scan saves a copy of each form as a TIFF image in FORMS' `Img` directory.

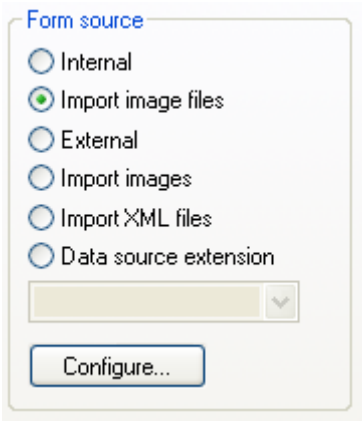
10. Process the scanned forms as usual with the other production modules.

11. You want to skip the Scan module the next time you process these forms. You have to therefore tell Interpret to interpret the image files that Scan just created instead of forms from FORMS' database, which is where the form images go the first time you scan the paper forms.

    Before you interpret the forms the second time, open the job description.

12. Click the **Interpret** tab.

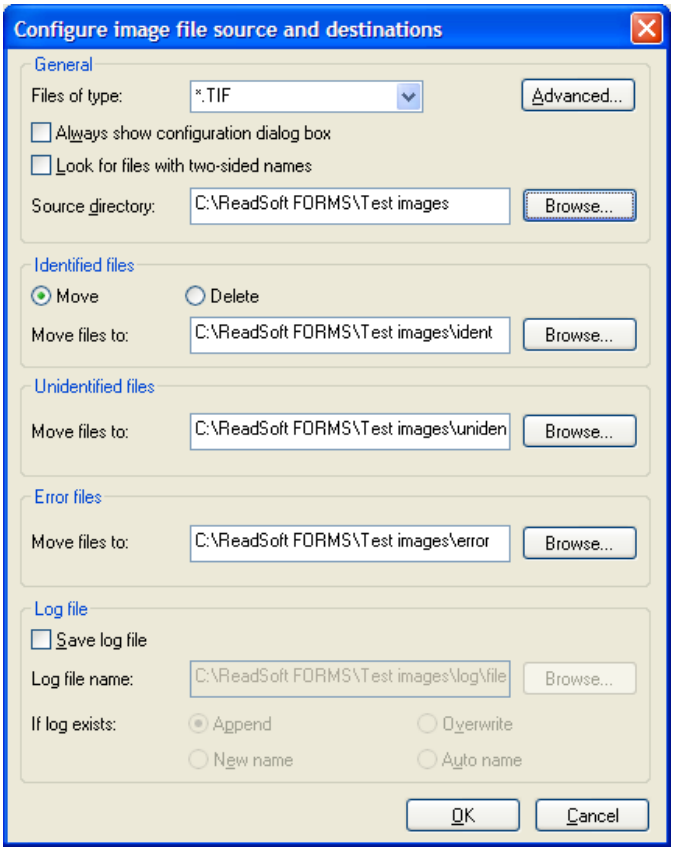
13. Choose **Import image files** as the form source.



14. Click **Configure**. The **Configure image file source and destinations** dialog is displayed.

15. In the **General** group box, click **Browse**, navigate to the folder you are using to test the images (such as `C:\ReadSoft FORMS\Test images\Img`, for example) and click **OK**. FORMS automatically enters the destination subdirectories for identified files, unidentified files, and error files processed in the Interpret module.

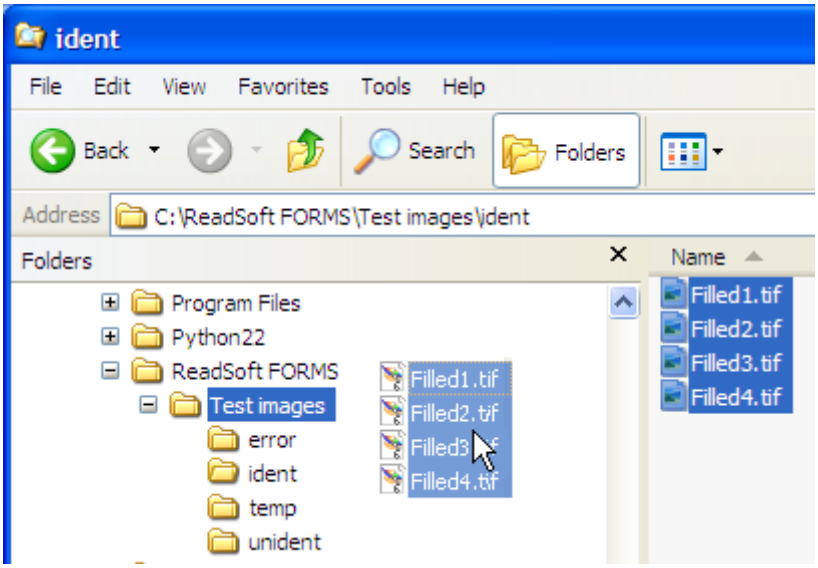
    Note: Remember, it does not matter where you create or specify this folder, but if **Export form images when scanning** is selected on the **Export** tab in the job, you cannot use the folder that is specified for images in `ehlocal.ini`.



16. Click **OK** to close the **Configure image file source and destinations** dialog. (If you came directly to steps 12-16 from the section called "Create a job description," then read steps 18 and 19 below before returning to step 7 on page 76.)

17. Click **OK** to close the **Job description** dialog and save the job description when prompted.

18. Run the edited Interpret job. FORMS moves the image files from the `Img` directory to various subdirectories, depending on whether or not Interpret successfully unpacked and identified them. Normally, they are moved to `C:\Program Files\ReadSoft\FORMS\Img\ident` in Windows XP and Windows Server 2003 R2, or to `C:\Program Files\ReadSoft\FORMS\Img\ident` in newer operating systems.



19. Before each additional time you process the image files in the Interpret module, use Windows Explorer to move them from the destination directory or directories (`C:\ReadSoft FORMS\Test images\ident` in this example) back to their source directory (`C:\ReadSoft FORMS\Test images`).

Test the transaction description

When you process forms using FORMS, the result is the data output from those forms. The information above on testing the form definition describes one way you can optimize the accuracy of that data.

The content and format of that data is also important. For example, does the data match the way the forms were filled in? Does the output file have the desired layout?

Testing the transaction description reveals adjustments you can make that will result in output that is best suited to the target system.

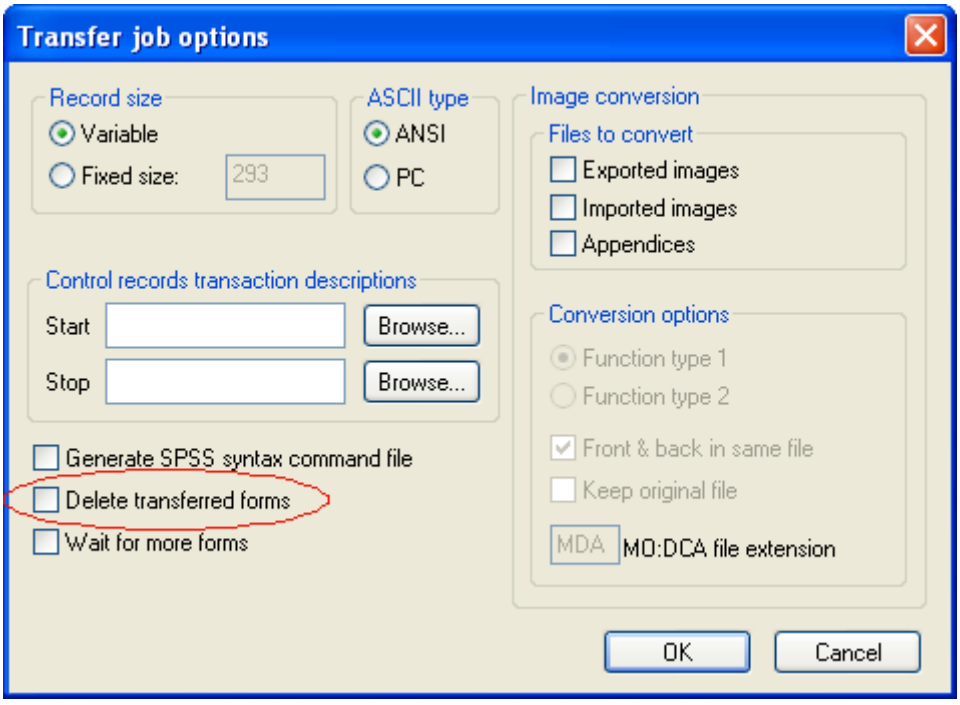
To test a transaction description, you must run forms in Transfer repeatedly, checking the transfer file after each Transfer job to evaluate the results.

This need not be as time-consuming as it might seem if you follow the instructions below, which allow you to transfer the same data repeatedly, without having to scan, interpret, and verify it.

1. Before you transfer the data, use Transfer or the Manager to open the job description you need to test.

2. Click the **Transfer** tab.

3. Click **Options**.



4. In the **Transfer job options** dialog, deselect **Delete transferred forms**. This prevents transferred data from being deleted from FORMS' database when you transfer it.

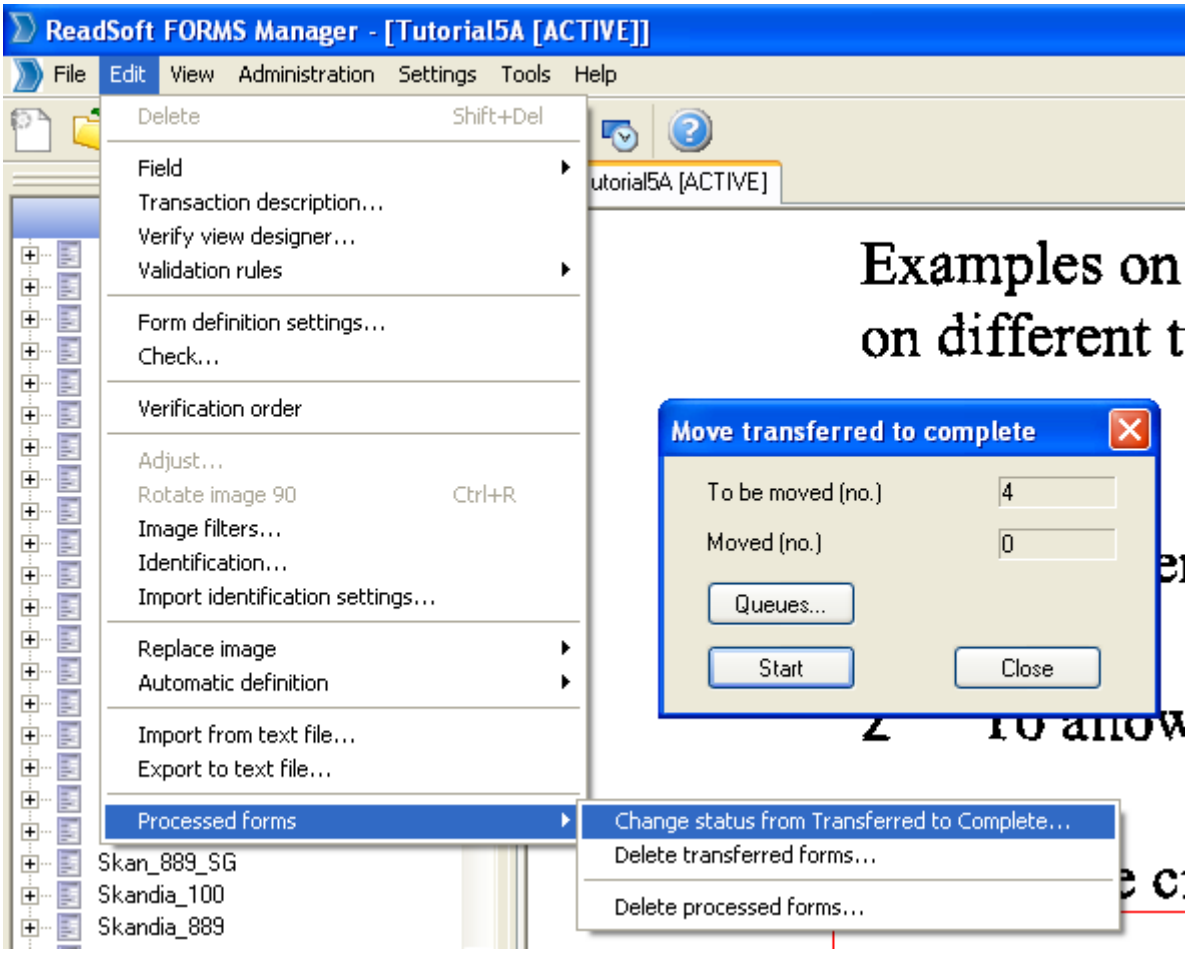
5. Click **OK** twice to close the **Transfer job options** dialog and the **Job description** dialog, and save the job description when prompted.

6. Run the Transfer job (that is, transfer the data from processed forms that have been verified at least once) as described on page 102 and check the result. See steps 4 and 5 on page 103 for details on how to locate and open the output file.

7. In the Manager, make any changes you wish to the transaction description and resave the form definition.

8. Still in the Manager, choose **Edit** ⇨ **Processed forms** ⇨ **Change status from Transferred to Complete**.

9. When the **Move transferred to complete** dialog is displayed, click **Start**.



This simply changes the status of processed forms in the forms database from *Transferred* to *Complete*. Forms with *Complete* status are ready to be transferred again.

10. Repeat steps 6 through 9 as necessary.

It is important that you reselect **Delete transferred forms** when you are finished testing, or use a different job description in production. Otherwise, data from transferred forms accumulates and takes up valuable space in the database. If you accidentally run many jobs without the **Delete transferred forms** option, you can delete them by choosing **Edit** ⇨ **Processed forms** ⇨ **Delete transferred forms** in the Manager module after the appropriate form definition has been opened.

All of the information above on testing is simply here to show you some of the ways that you can perform the final preparations before moving into production.

You have not yet learned the full range of possible settings that can be used. However, awareness of what to pay attention to allows you to begin understanding how settings can be used to make FORMS function as if it were designed specifically for your forms.

You have much to gain by testing. If it reveals anything that you want to change before using the form definition in production, then you have the chance to change it. At the same time, if the testing does not reveal anything that needs changing, you still go into production with the confidence that you have a valid form definition, transaction description, and job description.

5 Getting started: Production step by step

The previous chapter described the stages involved in preparing for production; that is, getting FORMS prepared to process your forms. This chapter describes the actual forms processing or *production.*

While you used the Manager for all of the tasks to prepare for production, the four steps in the production process are performed in the remaining four modules:

1. Scan and identify forms (Scan module) (page 87)
2. Interpret and validate forms (Interpret module) (page 92)
3. Verify forms (Verify module) page (94)
4. Transfer data from processed forms (Transfer module) (page 102)

Create some forms to use with the tutorial

If you based your form definition on an image file such as `Sample.tif`, you may skip this section, but make a note of the details regarding the use of filled-in image files in step number 7 on page 76.

If you based your form definition on a scanned form rather than an image file, you will need some actual filled-in forms as you proceed through this chapter.

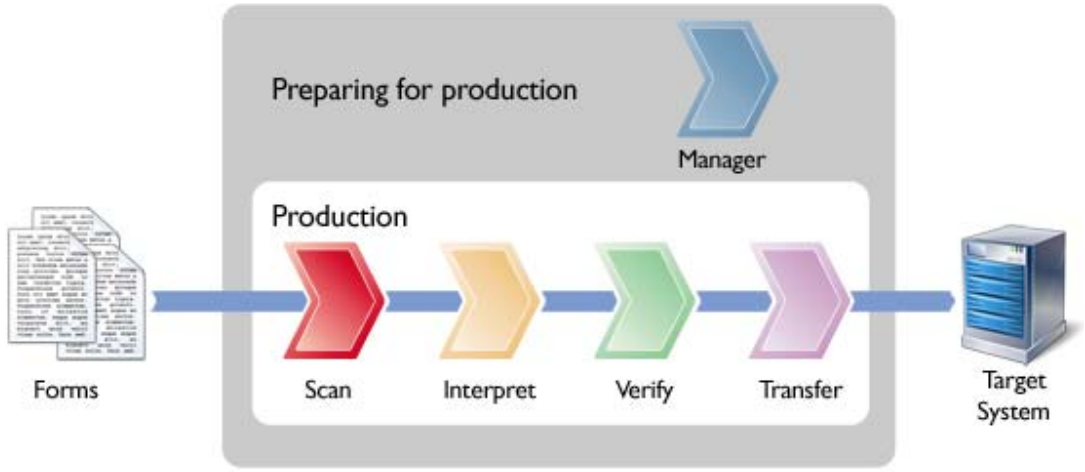
If you based your form definition on a scanned copy of `Sample.doc`, follow the instructions below:

1. Use Microsoft Word (version 6.0 or later) to open and print several copies of the file `Blank.doc`, if you have not already done so.
2. Fill in the copies of `Blank.doc` form as a customer might, keeping the format specifications and validation routines in mind (see below). Not all fields must be filled in. You may choose to make certain mistakes on purpose, to see how FORMS handles them.



No.	Field	Format/validation
1	CustNo	One uppercase letter followed by 5 digits.
2	Month Day Year	A number between 1 and 12. A number between 1 and 31. A 2-digit number.
3	Products	Circle one or more of the products.
4	Amount Total	A dollar amount between 4.99 and 49.99. A dollar amount that is not higher than 249.95.
5-7	Question5-7	Check one box in each row.
8	Suggestions	Write any text, or leave blank on selected forms.
9	Gift1 and Gift2	Any of the three-digit numbers 001, 005, 019, or 144, or their equivalents (01, 1, 05, 5, and 19).
10	NewAddr	Any address, or leave blank.

Scan and identify forms



You are now working in the Scan module.

In this step, paper forms are scanned and converted to electronic images using a scanner.

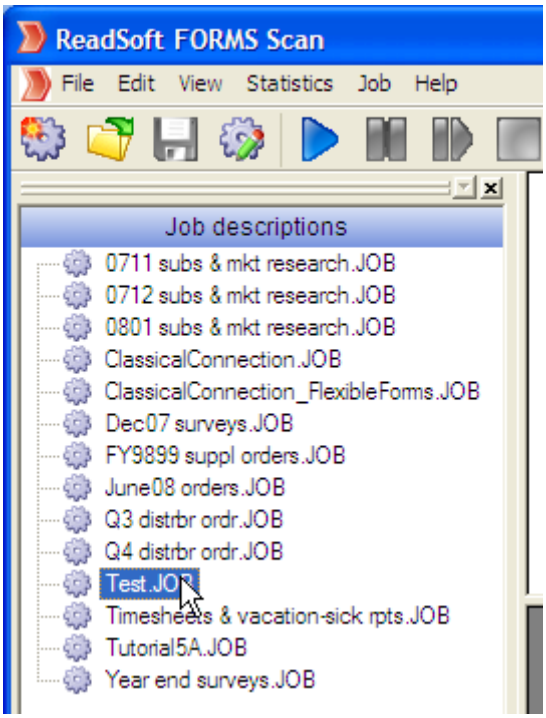
If you are using the TIFF files supplied with FORMS, skip this section and go to "Interpret and validate forms," on page 92.

1. Switch on the scanner before you turn your PC on. Otherwise, your PC's device driver may not find the scanner. (Windows 2000 users should be able to switch the scanner on without shutting down the PC.)

2. Start Scan by double-clicking the Scan icon or by double-clicking `Efscan.exe` in FORMS' `Bin` directory.

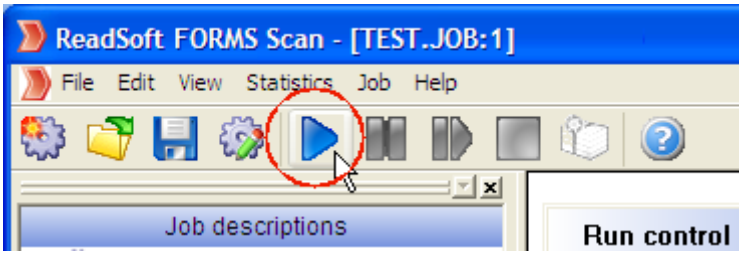
3. Open or create a *job description* to tell FORMS which forms to look for during scanning and how to handle the images. If you have followed the tutorial up to this point, open the job description you created according to the instructions on page 74.

   To open a job description, double-click the desired job under the **Job descriptions** object bar. If the **Job descriptions** object bar is not visible, then click the **Open** button on the toolbar, and double-click the desired job. (To make the object bar visible, choose **Object bars** on the **View** menu.)

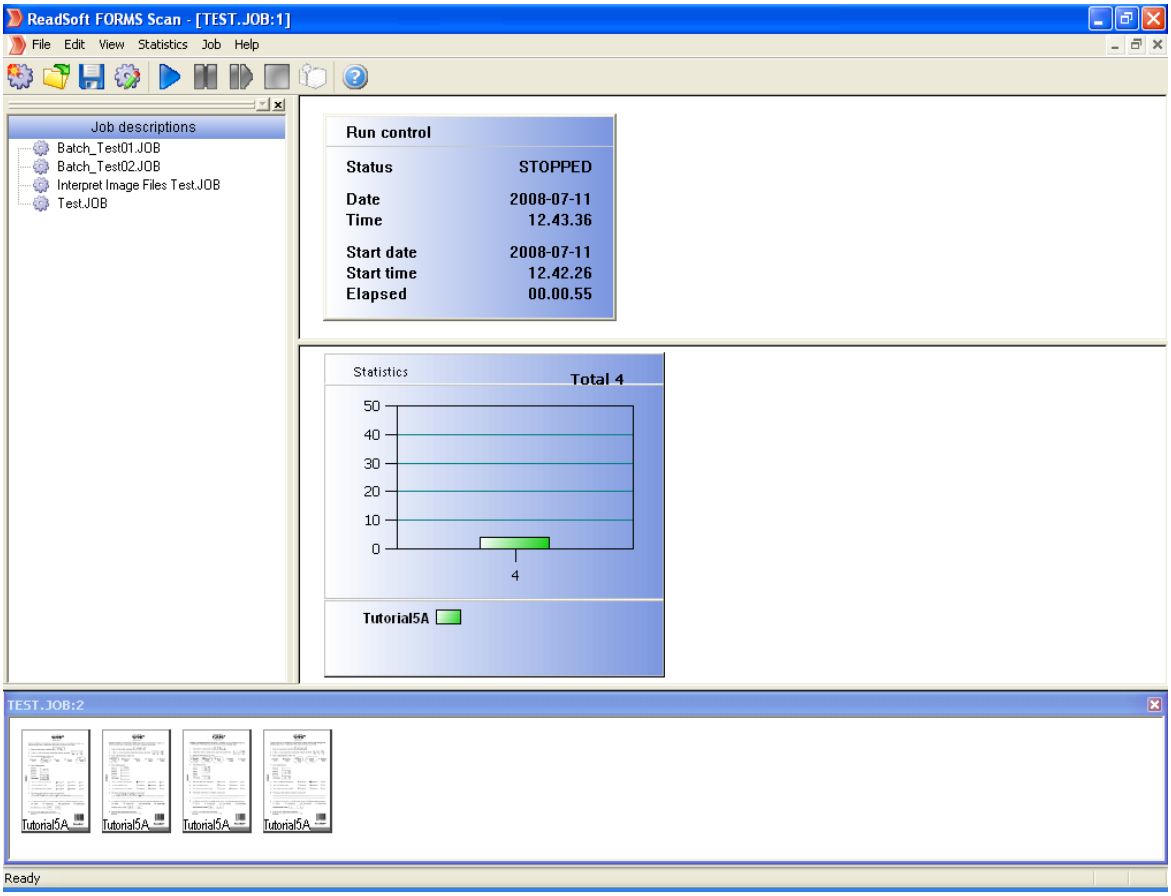


4. Place the forms as straight as possible in the document feeder.

5. When the scanner is ready, click the **Start** button on the toolbar.



   FORMS starts the scanner and begins receiving and identifying images of forms. The status, date, and time are displayed, as well as a bar chart indicating how many forms of each type have been scanned. Thumbnail images of each form appear at the bottom of the screen during scanning.

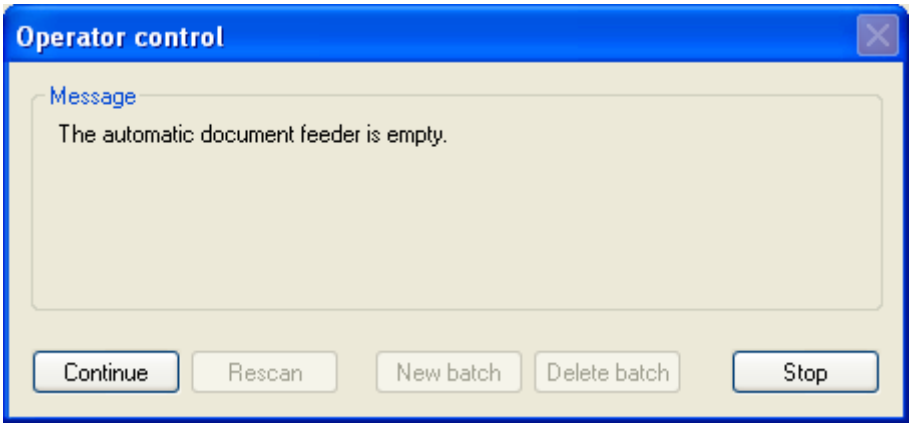


If you are running your forms in batches, FORMS runs until the specified number of forms has been scanned. If not, FORMS runs until the scanner's document feeder is empty.

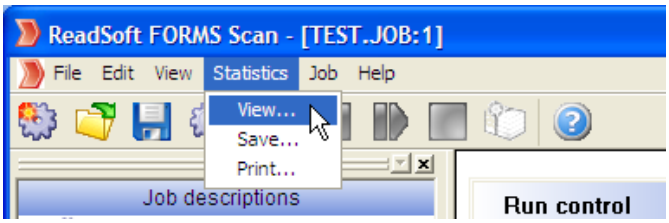
You can double-click any of the thumbnail images to enlarge an individual form and check its quality on-screen. See Help for details of operations you can perform from this view.

You can **Pause** or **Stop** the job at any time.

6. When the document feeder is empty or when the batch is finished, you can add more forms and restart the job by clicking **Continue** in the **Operator control** dialog, or you can click **Stop** to stop scanning.



7. When the job is finished, you can view statistics, which tell you how many form definitions were scanned, when they were scanned and how long it took.



   You can save statistics from the latest run to a file by choosing **Save** (**Statistics** menu). You can also print statistics that have been saved.

As forms are scanned, they are saved to FORMS' database unless you specifically discard them. This is why there is no **Save** or **Save as** command for scanned forms in Scan.

Certain circumstances may require your attention. A message displayed in an operator dialog describes the situation.

When the scanner needs attention

If the scanner is not connected or is switched off, connect or switch on the scanner, wait until it is ready, make sure there are forms in the document feeder, and start the job again.

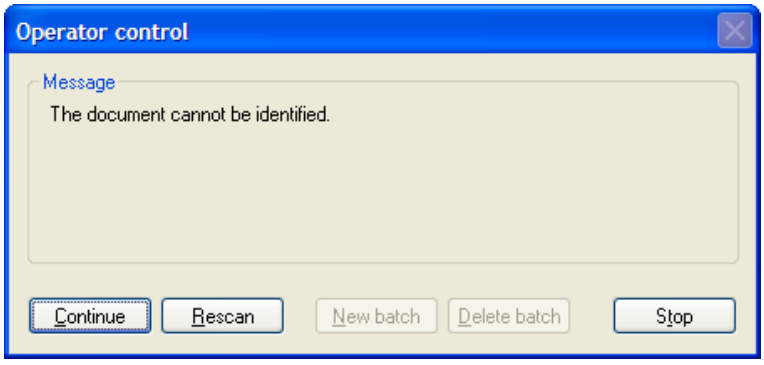
If a paper jam occurs, clear the paper path, clear the error status on the scanner (refer to the scanner's manual), and restart the process.

A form cannot be identified

A form cannot be identified, for example, if the form definition is not included in the job description, if there is too much skew (that is, if the form did not pass through the scanner straight enough), or if the form is dirty or torn.

What happens when a form cannot be identified? That depends on which options were selected in the Scan job description. Unless you selected **Accept unidentified forms** as a Scan job option, the information below applies. If you selected that option, see Help.

When a form cannot be identified, scanning stops, the unidentified form is shown on the screen, and the **Operator control** dialog is displayed with an error message.



The operator dialog showing the error message saying that a form cannot be identified.

This is to show you which form stopped the scanning process. Depending on the scanner card installed in your computer and the type of scanner you are using, the scanner *can* be several forms ahead of the Scan module. In other words, the scanner might have images stored in its internal buffer that FORMS has not yet received (see “Buffering” in Help). It is therefore important that you choose the correct course of action, particularly if you have a fast scanner.

Continue processes the images that entered the scanner’s buffer after the unidentified or rejected form, while **Rescan** causes FORMS to discard those images.

Again, if you choose to **Continue** scanning, only the unidentified or rejected form is ignored; forms that passed through the scanner after the unidentified form are saved and used. This is appropriate, for example, when an extra page has mistakenly been scanned. In this case, you would remove the actual unidentified form, since it is not supposed to be included in the job.

If you choose **Rescan**, all forms scanned after the unidentified form are ignored, as well. Choose **Rescan** if the sequence of the forms is of significance, such as if you are processing batches or sets (not covered in this book).

When you choose **Rescan**, follow these steps:

1. Use the image of the form to find the unidentified form. Remove this form, as well as all other forms lying on top of it (that is, scanned after it), from the scanner’s output tray.

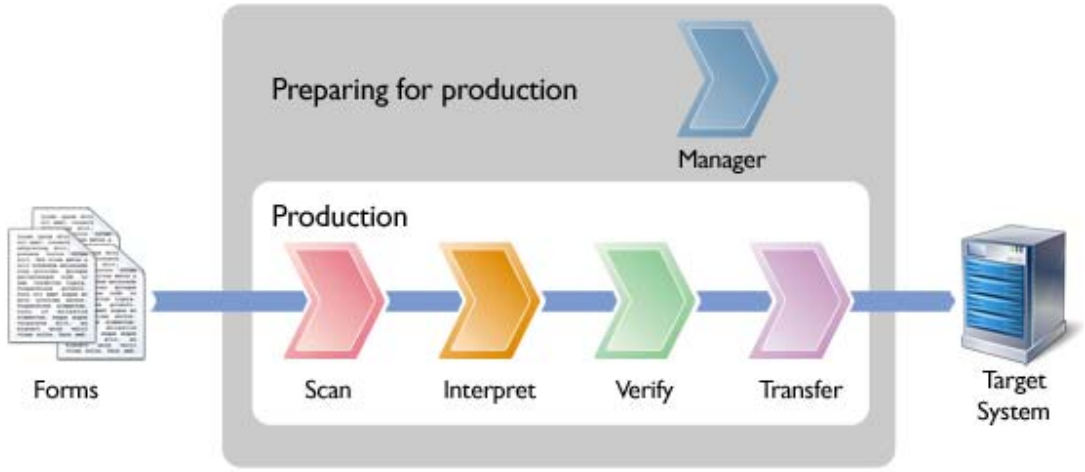
2. Fix the problem with the unidentified form (turning it right-side up, removing it, or whatever is necessary).

3. Place the stack in the document feeder in the correct order, with the unidentified form (if it is to be used) as the next form to be scanned.

4. Click **Rescan**.

If you choose **Stop**, both the unidentified form and any forms that passed through the scanner after the unidentified form are ignored.

Interpret and validate forms



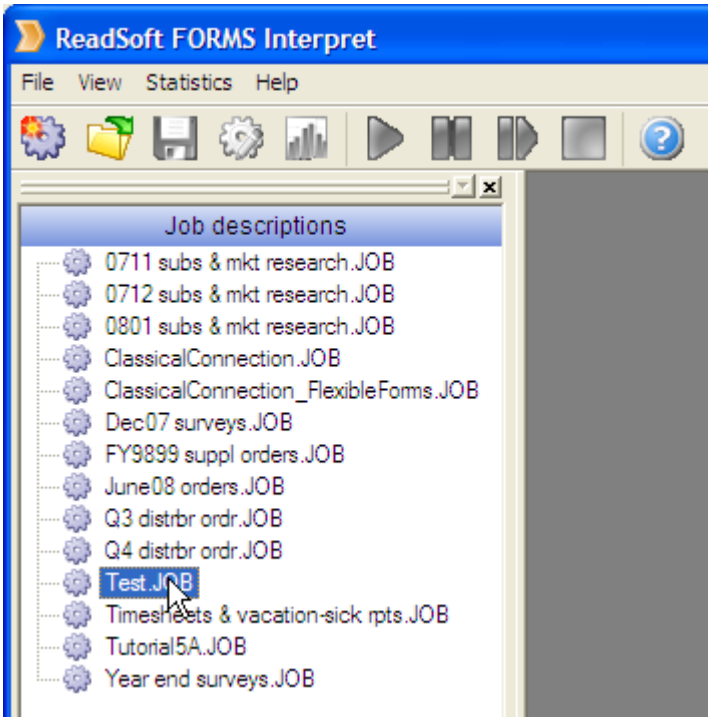
You are now working in the Interpret module.

The Interpret module both interprets the data from forms and validates it in accordance with the validation routines you specified in the form definition. Information is automatically stored in FORMS' database.

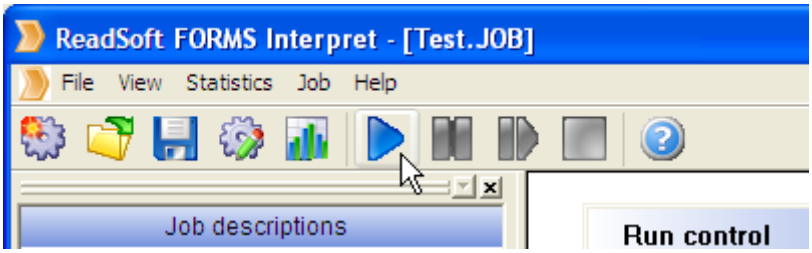
1. Start Interpret by double-clicking the Interpret icon or by double-clicking `Efitrp.exe` in FORMS' `Bin` directory.

2. Tell FORMS which forms to interpret and validate by opening a *job description*. If you followed the tutorial, open the job description you created according to the instructions on page 74. (If you are using the TIFF files supplied with FORMS, use the job description created according to the instructions on page 81, which imports the files you moved to the `Img` directory as described on page 76.)

   To open a job description, double-click the desired job under the **Job descriptions** object bar. If the **Job descriptions** object bar is not visible, then click the **Open** button on the toolbar, and double-click the desired job. (To make the object bar visible, choose **Object bars** on the **View** menu.)



3. Click the **Start** button on the toolbar.



Status, date, and time are displayed, as well as a bar chart indicating how many forms of each type are being processed. This is similar to the information displayed during scanning.

There are no operator dialogs that require your attention.

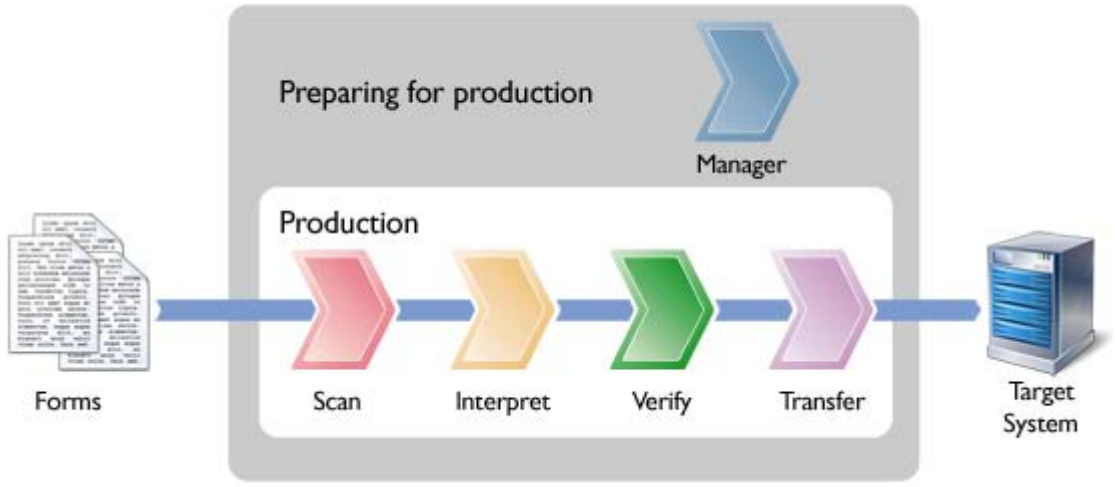
4. Unless you have selected **Wait** in the Interpret job description, Interpret runs until all forms of the specified types have been interpreted and validated or until you click the **Stop** button on the toolbar.

5. When the job is finished, you can view statistics, which tell you how many forms were interpreted, the current statuses of those forms, when they were interpreted, and how long it took. They also tell you how successful interpretation of the characters was.

   You can save statistics from the latest session to a file by choosing **Save** (**Statistics** menu). You can also print statistics that have been saved.

   As forms are interpreted, the results are automatically saved to FORMS' database.

Verify forms



You are now working in the Verify module.

Using Verify, you manually type data for forms with uninterpretable characters, missing data, or errors detected by validation routines. Fields containing information that must be typed manually (called a *field complement*), such as a field with a handwritten name and address, are completed in a special dialog.

1. Start Verify by double-clicking the Verify icon or by double-clicking `Efveri.exe` in FORMS' `Bin` directory.

2. Tell FORMS which forms you want to verify by opening or creating a *job description*. If you followed the tutorial, open the same job description you processed in the Interpret module.

   To open a job description, double-click the desired job under the **Job descriptions** object bar. If the **Job descriptions** object bar is not visible, then click the **Open** button on the toolbar, and double-click the desired job. (To make the object bar visible, choose **Object bars** on the **View** menu.)

   Though the standard job description settings are quite functional in most situations, you have many alternatives to choose from when you create a job description for Verify. You can limit your work to forms and fields with a certain status and choose from many options for how you want FORMS to interact with you during the process. You can work in *fast verify mode* or in *normal mode*.

   Some of these alternatives are described below. For details about others, see Help.

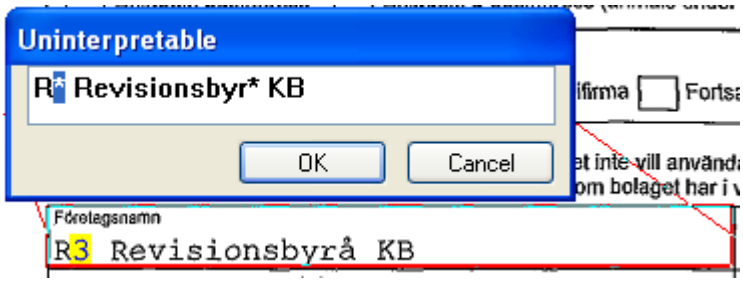
3. Click the **Start** button on the toolbar.

4. In Verify, images of the forms are shown with the data that FORMS interpreted for each field.

   Since there are many different kinds of errors and Verify settings, it is not practical to offer detailed instructions here; you must refer to Help for information about verifying various types of errors and statuses.

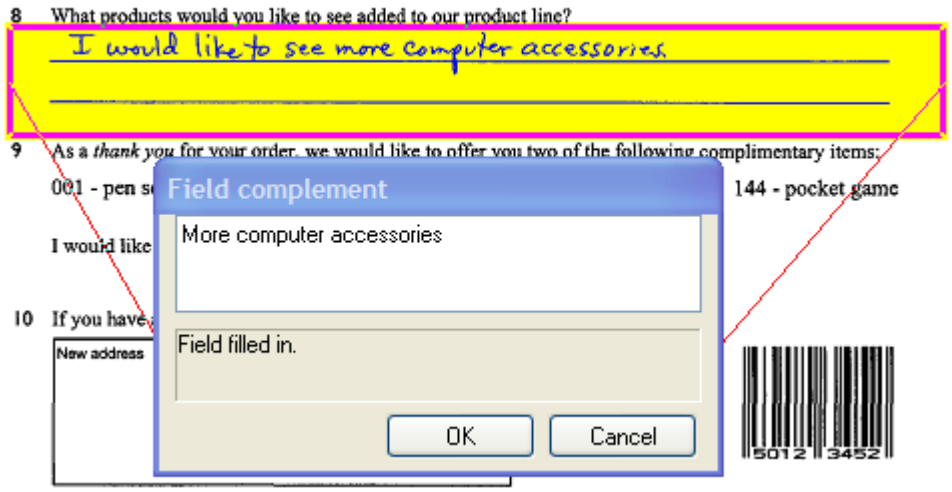
However, FORMS' user interface makes much of the work self-explanatory. Do not hesitate to start a job and get acquainted with Verify. The work usually involves the following activities:

- Replacing asterisks. Unknown characters are represented by asterisks (*) in a small dialog. You compare the contents of the dialog with the form displayed on the screen, type the missing character or characters, and press **RETURN** on your keyboard or click **OK** on the dialog.



Entering a character that FORMS could not interpret.

- Typing field complements. If you specified a field complement (see page 54), this is where you type the information from the image field.



Entering a field complement.

- Checking format and validation errors. If you specified any type of validation routine, and data that does not match the specifications was interpreted, FORMS shows you all the fields involved, along with a message describing the error and a dialog in which you can correct it. What you type in the dialog replaces data previously interpreted.

When you have removed all the errors from a form, FORMS gives it *Complete* status. This means that the data is ready to be transferred to the target system.

Each form that receives the status *Complete* exits the Verify flow. Verify continues to display forms until you have verified all the forms included in the job description.

You can **Pause** or **Stop** the job at any time. See "Getting around in Verify" below, which gives you some examples of how to navigate in Verify.

Tip: FORMS might not register mark fields for question 7 as filled in, particularly if they are filled in with thick check marks. This is because if you followed the instructions for creating the form definition, you set the option **Detect strikeouts** at 50%. If the check mark covers 50% or more of the field area, FORMS interprets it as not being selected (that is, as if the respondent crossed it out). You can see from this example that a strikeout level of 50% is too low.

To register a mark field as filled in, simply enter one of the values you specified when you defined the fields in question 7 (71, 72, or 73).

5. Verify runs until all forms specified in the job description have been verified or until you click the **Stop** button on the toolbar.

   When the job is finished or stopped, you can view statistics, which tell you the number of forms that were verified, as well as when they were verified and how long it took. They also tell you the current status of the forms you verified.

   You can save statistics from the latest session to a file by choosing **Save** (**Statistics** menu). You can also print statistics that have been saved.

Getting around in Verify

Scrolling and scaling an image

Use the **F3** and **F4** keys on your keyboard (or right-click anywhere on the form and choose **Enlarge** and **Reduce** on the shortcut menu), and the **CTRL**+arrow keys (or the scroll bars on the side and at the bottom of your screen), to adjust the image. Experiment to obtain the best view of the part of the form you are working with.

You can also press **F2** to get the **Normal** view without magnification or reduction.

Returning to the previous field

Click the **Previous field** button on the toolbar (or **CTRL+BACKSPACE**) to go back to the previous field. Changes made to the current field are discarded.



Skipping fields

Press **F10** to skip the field for the time being. The status of the field is not changed, so it reappears the next time you run the job.

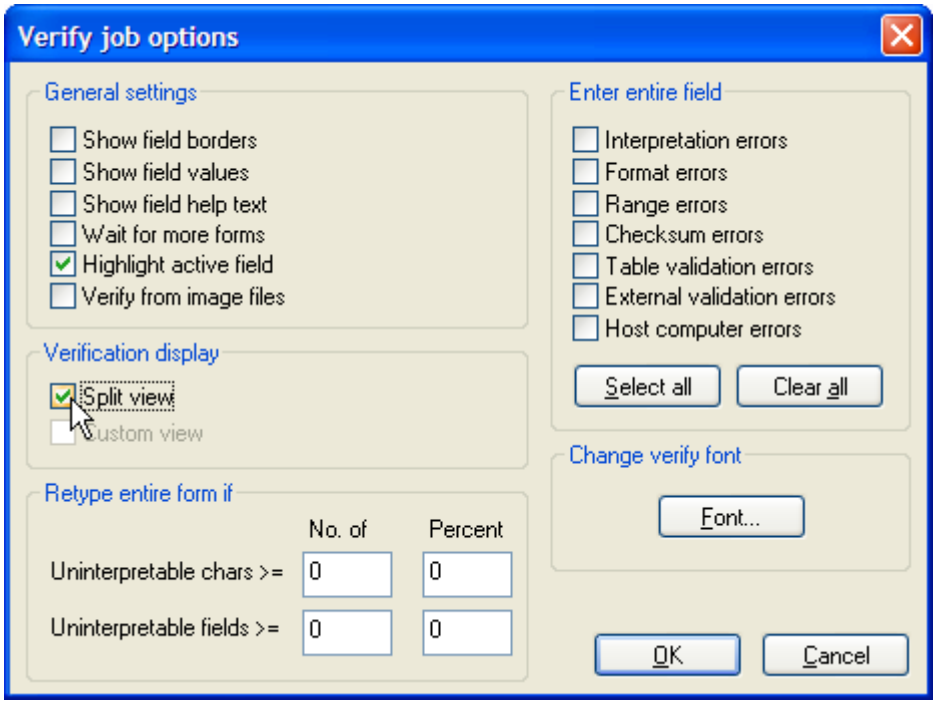
If you have gone back to the previous field, **F10** will also return you to the original field.

Alternative settings in Verify

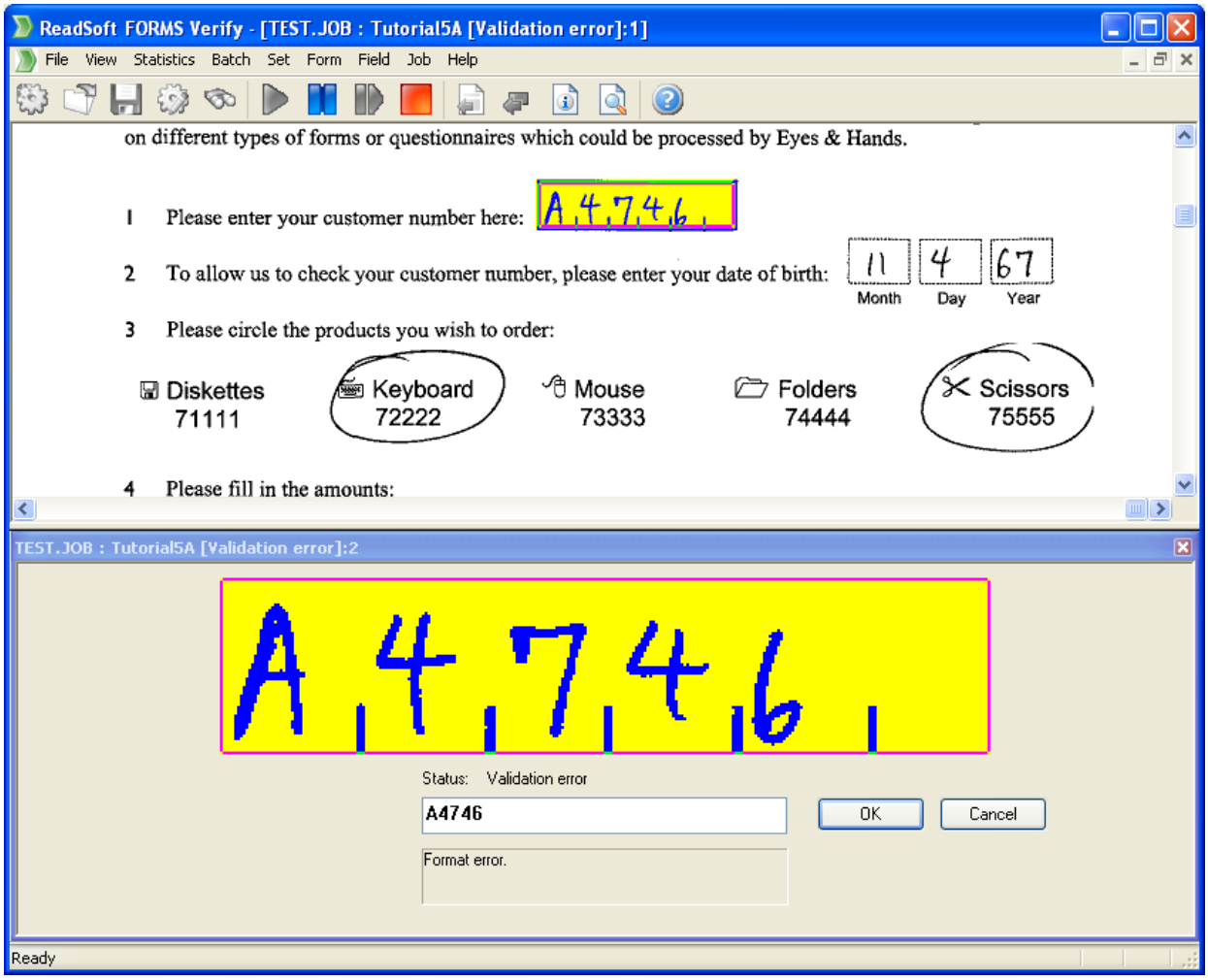
As mentioned earlier, there are many settings you can use to change the way data is presented to you on-screen in the Verify module. You can use the sample form to experiment with any of these. However, if you are doing this tutorial for the first time, we recommend that you run Verify and Transfer first, so that you get a feeling for how FORMS functions with default settings.

Afterwards, you can apply alternative settings in FORMS, and then reprocess your test forms (for more information on testing, see "Test the form definition and job description before production," which begins on page 78).

Using the split view

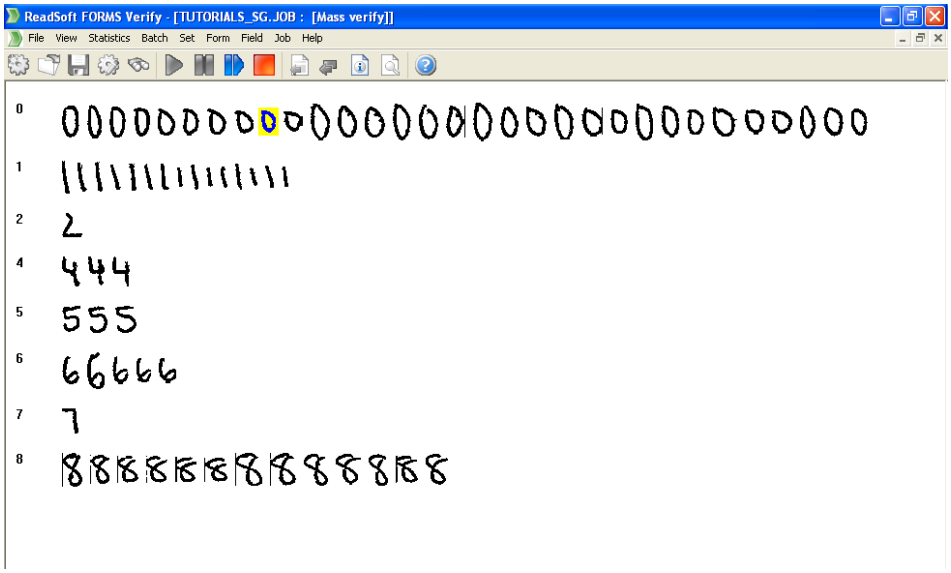


When you select **Split view** in the **Verify job options** dialog, FORMS creates a split screen in Verify, where the upper half of the screen displays the form and the lower half the fields.


ReadSoft FORMS Verify - [TEST.JOB : Tutorial5A [Validation error]:1]
File View Statistics Batch Set Form Field Job Help
on different types of forms or questionnaires which could be processed by Eyes & Hands.
1 Please enter your customer number here:
A,4,7,4,6,
2 To allow us to check your customer number, please enter your date of birth:
11 4 67
Month Day Year
3 Please circle the products you wish to order:
Diskettes 71111
Keyboard 72222
Mouse 73333
Folders 74444
Scissors 75555
4 Please fill in the amounts:
TEST.JOB : Tutorial5A [Validation error]:2
A,4,7,4,6,
Status: Validation error
A4746
OK
Cancel
Format error.
Ready

Mass verify

With mass verify, FORMS cuts out all interpreted characters from fields and shows them to you in rows in order from 0 to 9, A to Z, etc. You can select characters that appear to be misinterpreted by clicking them. If you change your mind, you can deselect any character by clicking it again.



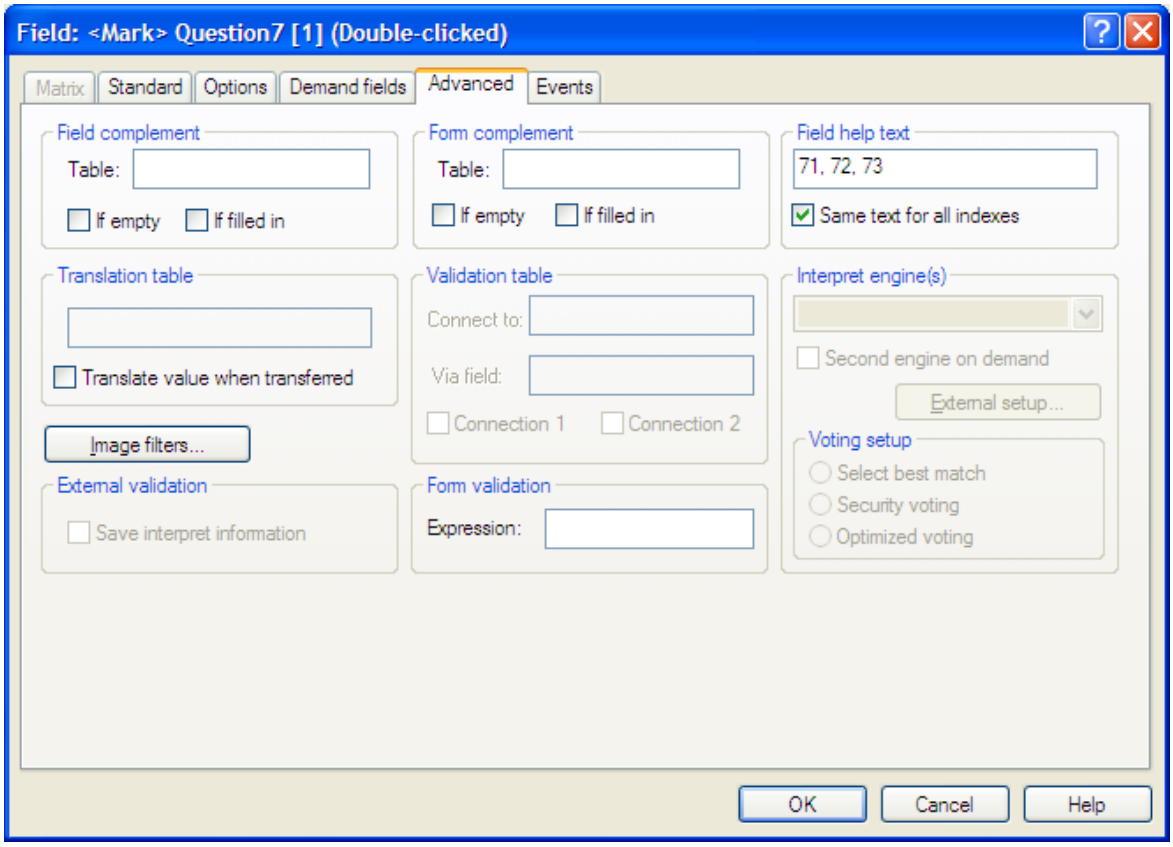
Character, mark, selection, and object fields can be mass verified. When you create a form definition, you select this setting on the **Options** tab of the **Field** dialog. **Mass verify** is selected by default in the Verify job (see Help for details).

Adding field help text

For character, mark, image, barcode, and object fields, you can type text in an edit box that you want to appear as a prompt or reminder when you verify a field. A typical field help text describes the field (for example, “This is a 7-digit customer number”) and helps you to verify the field correctly.

This is very useful when no information about the field content is available during verification.

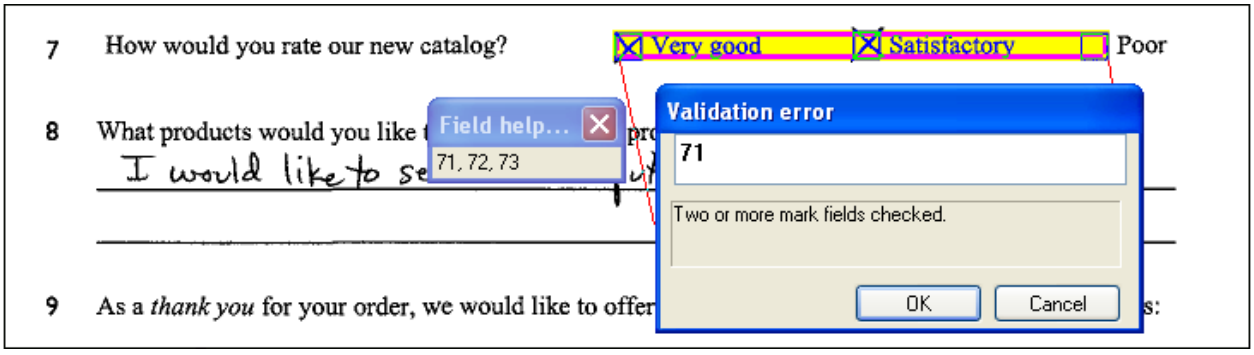
You type the field help text in the **Advanced** tab of the **Field** dialog when you create a form definition. Here is an example of field help text that could be entered for question 7 on the sample form:



In addition, select **Same text for all indexes**, so that the field help text is displayed for all of the mark fields in question 7.

You also have to select **Show field help text** in the **Verify job options** dialog to have the text displayed whenever you verify a field.

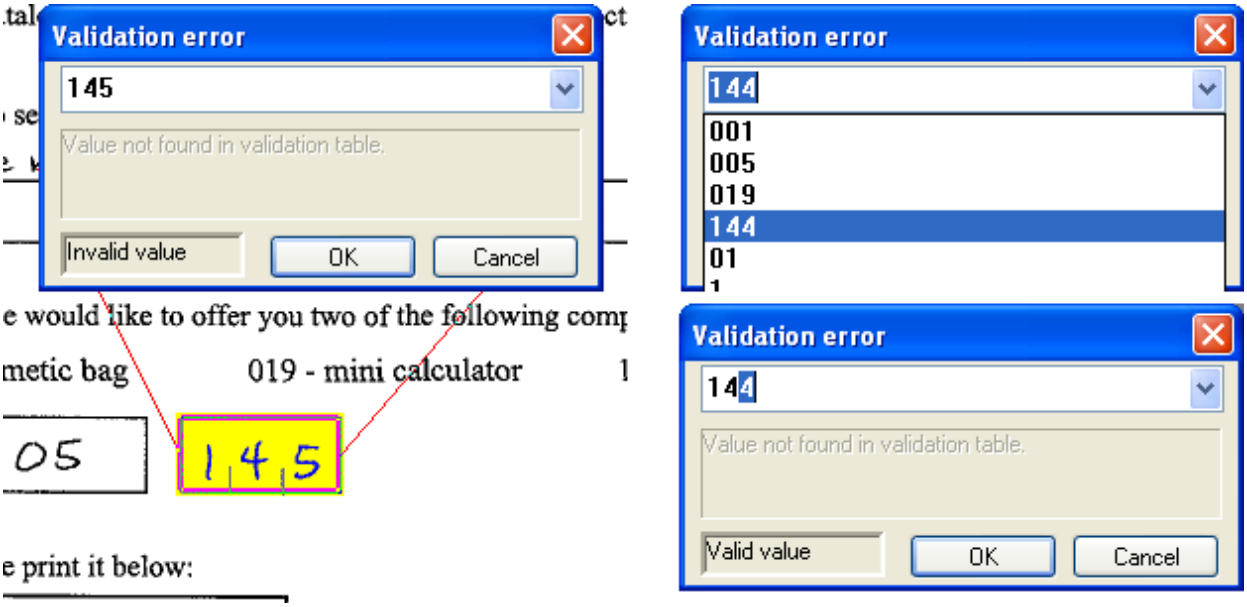
If there was a problem in the interpretation of the mark fields in question 7, for example, the **Field help text** box would remind the operator which value to enter for the correctly marked box on the form.



Field help text helps Verify operators to type in correct values.

Displaying table values

When you use Verify to validate field values that are supposed to be found in a table, the values in the table you specified are available in the **Validation error** dialog if **Show candidates in Verify** (**Field validation** group box) was selected as a setting for the field definition.



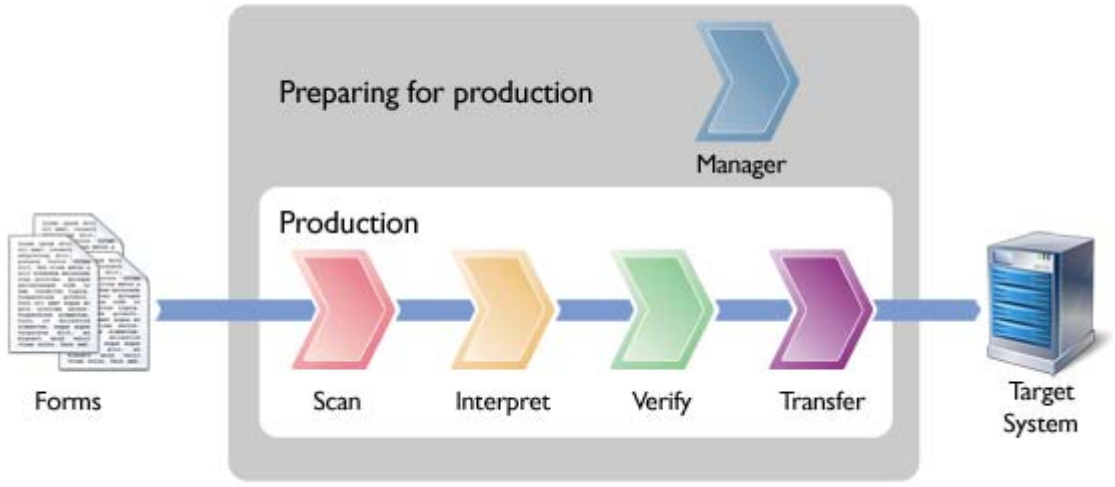
Show candidates in Verify *is a setting that helps Verify operators quickly find valid table values. The operator can either select the correct data from the list of valid values (top right) or simply begin typing the correct value while FORMS fills in the rest automatically (bottom right).*

When you begin typing a field value and the characters you type match a value contained in the corresponding table, the rest of the field value is filled in automatically by FORMS.

You can also click a list box arrow and choose a value from the table.

Again, refer to Help for more details about these and other alternative settings in the Verify module.

Transfer data from processed forms



You are now working in the Transfer module.

Finally, you use Transfer to transfer data from processed forms to another system. In many cases the data is simply written to a text file.

1. Start Transfer by double-clicking the Transfer icon or by double-clicking `Eftran.exe` in FORMS' `Bin` directory.

2. Open or create a *Transfer job description*, which tells FORMS which data from which forms to transfer, and to where. If you followed the tutorial, open the same job description you processed in the Verify module.

   To open a job description, double-click the desired job under the **Job descriptions** object bar. If the **Job descriptions** object bar is not visible, then click the **Open** button on the toolbar, and double-click the desired job. (To make the object bar visible, choose **Object bars** on the **View** menu.)

3. Click the **Start** button on the toolbar.

   Unless you have selected **Wait** in the Transfer job description, FORMS runs until all the data has been transferred from all the forms of the specified type with *Complete* status (which they always have after successful verification), or until you click the **Stop** button on the toolbar.

   The status, date, and time are displayed, as well as a bar chart indicating how many forms of each type have been transferred.

   When data is transferred to a file, there are no operator dialogs that require your attention. When data is transferred to the target system directly (via VBA, a DLL, or other interface), you can run Transfer and Verify at the same time, so that you can receive error messages from the target system and correct the problem in one process.

4. When the job is finished, you can view statistics, which tell you from how many forms data was transferred, when the transfer was performed, and how long it took. They also tell you the name of the file to which the data was saved. The default file name, `out.*`, is created in the `Transfer` subdirectory in FORMS' root directory.

   You can save statistics from the latest session to a file by choosing **Save** (**Statistics** menu). You can also print statistics that have been saved.

5. If data was transferred to a file (as it usually is), you can open and examine the text file using a text editor such as Notepad.

Note

After you finish running all of the production modules for the first time, you may want to adjust some of your settings. Then, you can repeat the production process (see "Test the form definition and job description before production" on page 78 for advice on how to run sample forms through the system repeatedly). This will give you a chance to experiment with options that you may find useful. For example, try any of the various settings available in the Verify module (see "Getting around in Verify" on page 96 and "Alternative settings in Verify" on page 97 for details).

Appendix

A VBA solution for unspecified text

An *event handler* executes a predefined VBA function, DLL function, or both.

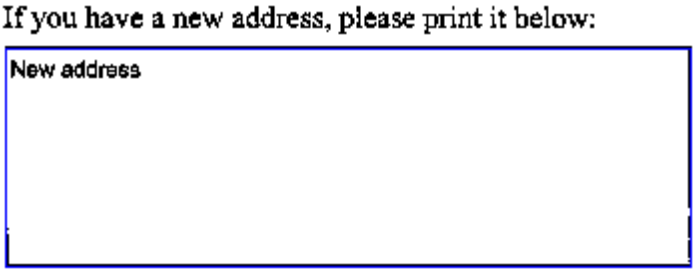
This appendix shows you how to use an event handler to define the new address field in question 10 on the sample form. You should have come to this page from the section, "A VBA solution for unspecified text," on page 60.

Pre-written VBA code for this solution is included in FORMS' `Tutorial` directory. If you are using the sample form, this code does not have to be modified. If not, a programmer can modify the example to suit your purposes.

The most noticeable difference between this alternative and the second one (defining an image field for unspecified text) is that instead of one field complement containing all the data, the end result is one or more fields containing the data.

Thus, rather than one field called NewAddr, you can follow the instructions below to create your own new address dialog containing four fields, one each for the street, city, state, and postal code. You will specify that whenever this field is filled in on the form, this custom dialog will be displayed in the Verify module.

1. Adjust the display using **F3** and **F4**, as well as **CTRL**+arrow keys, to find the **New address** box in question 10 on the sample form and double-click it.

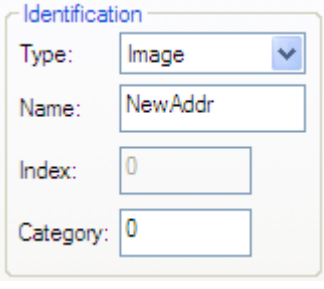


2. Double-click the field again (or press **F5**) to display the **Field** dialog.

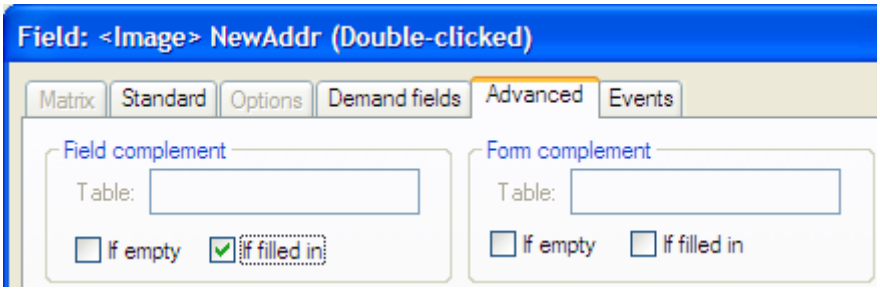
3. On the **Standard** tab, select **Image** as the field type. In this case, a filled-in image field will be used to trigger the VBA code.

4. Type a **Name** for the field (such as NewAddr).

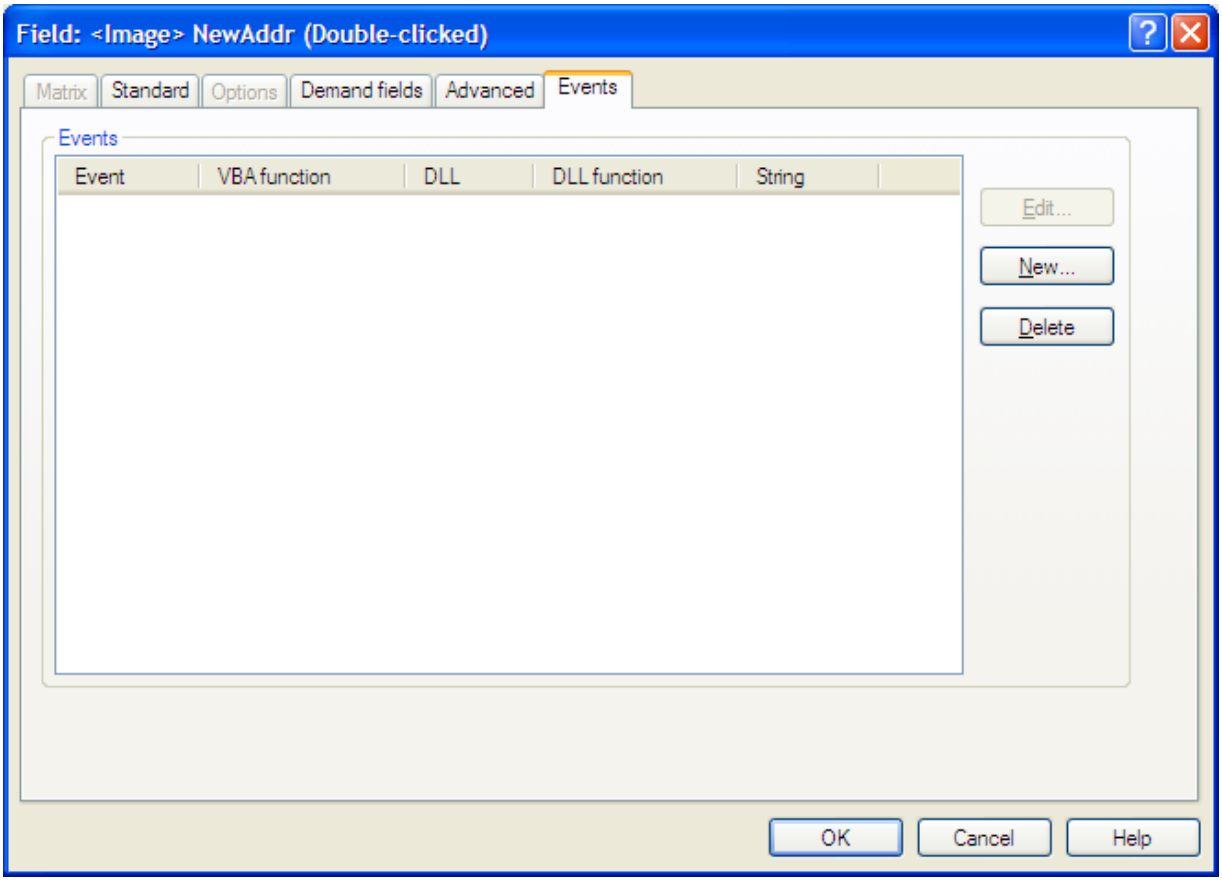
5. Click the **Advanced** tab at the top of the **Field** dialog.



6. In the **Field complement** group box, click **If filled in**.

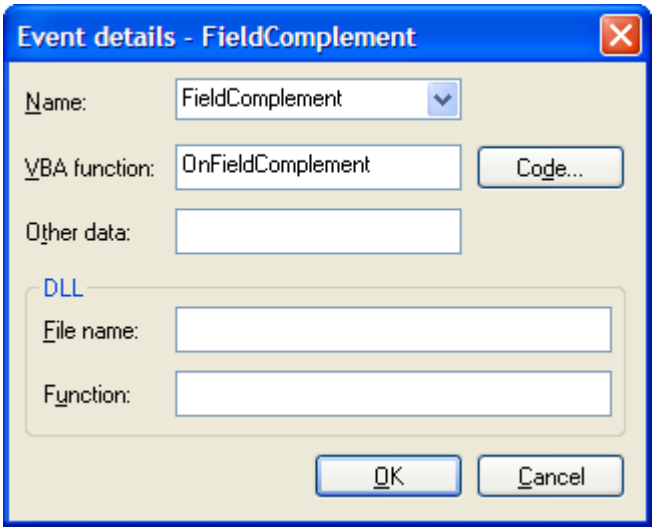


7. Click the **Events** tab at the top of the **Field** dialog.



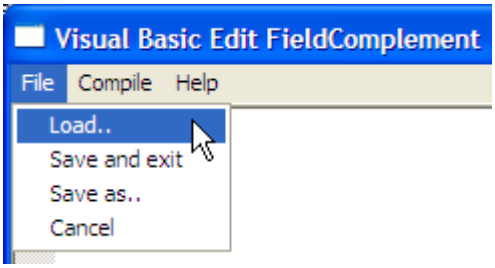
8. Click **New**.

9. Select FieldComplement from the **Name** box.

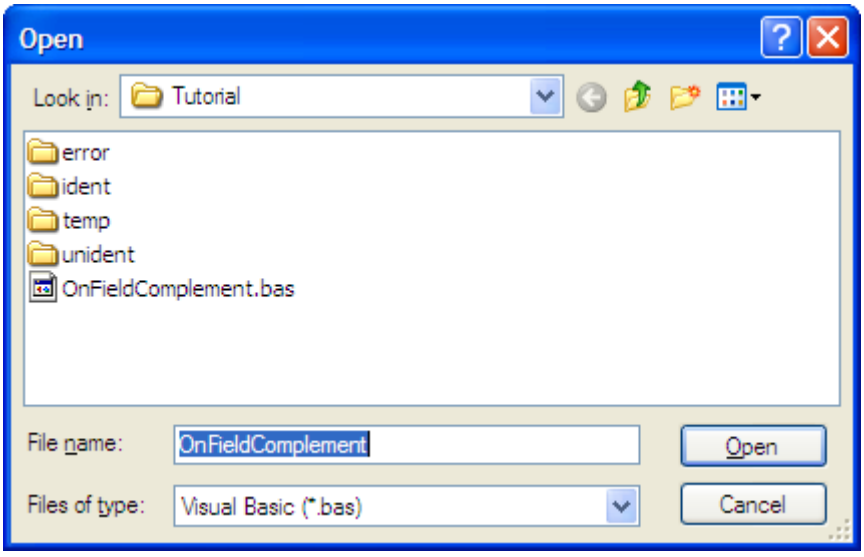


10. Click **Code**. The **Visual Basic Edit** window is displayed.

11. Choose **Load** (**File** menu) to import an event handler for the event.

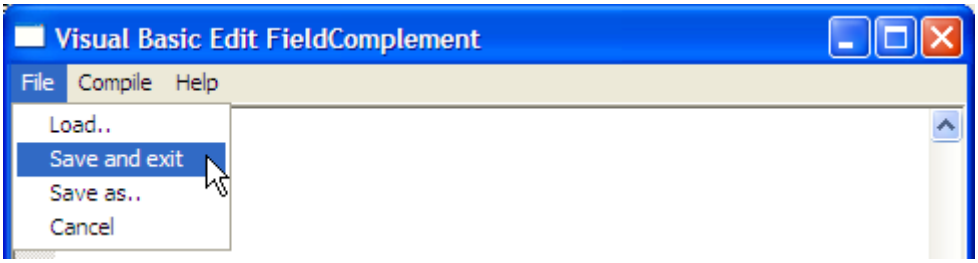


12. Navigate to FORMS' `Tutorial` directory and open `OnFieldComplement.bas`. The sample code is imported to the current event handler.



13. Choose **Compile** (**Compile** menu).

14. Choose **Save and exit** on the **File** menu.

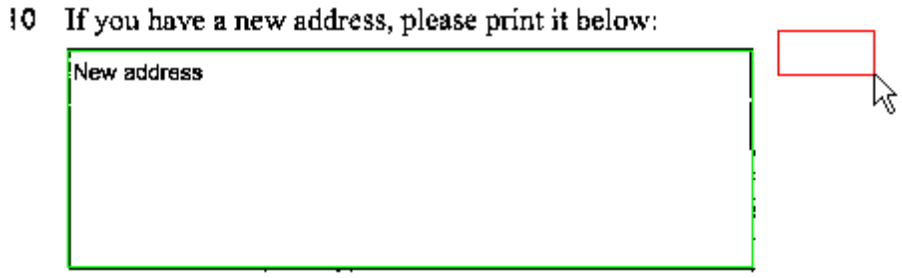


15. Click **OK** twice to close the **Event details** dialog and the **Field** dialog.

16. On a blank spot on the form (such as next to the NewAddr field), draw one small field; it will represent the street field in the custom dialog. The field you draw does not need to be large enough to contain any text; its only purpose is to create a place in FORMS to save data.

Note

Even though you are defining several of these fields, you must not define them together, as indexed fields. Define them one at a time.



17. Double-click the box you have just drawn (or press **F5**) to display the **Field** dialog. Select **Image** as the field type.

18. Type Street as the field's name (not STREET or any other variation), since this is exactly how the field is referred to in the sample VBA code.



19. Click **OK** to close the **Field** dialog.

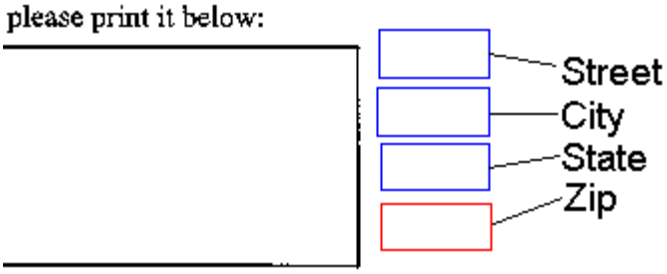
20. Draw another small field to represent the second line of text.

21. Press **F6** on your keyboard to **Define as previous**.

22. Name this second field City.

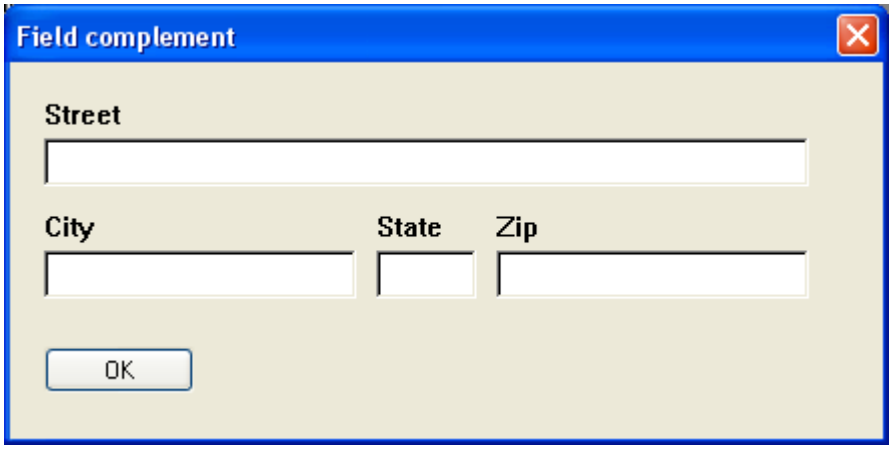
23. Click **OK** to close the **Field** dialog.

24. Define two additional image fields in the same way. Name them State and Zip.



25. Resave the form definition by clicking the **Save** button on the toolbar.

What happens when you use this solution is that when the NewAddr field in question 10 is filled in, the FieldComplement event is fired. Verify calls the code you imported, and the **Field Complement** dialog is displayed. (The **Field Complement** dialog, shown below, is just a sample, and a programmer can modify it to suit your purposes.) While looking at NewAddr, you type text or data for each field, and that information is displayed in the output file.



*The **Field complement** dialog is just a sample, and a programmer can modify it to suit your purposes.*

Note

Although defining event handlers can be part of the process of creating a form definition, we strongly recommend that you understand fully how this process works in FORMS before adding any external code. Refer to Help for more information about customization and integration.

If you have followed the tutorial up to this point, it is now time to return to “Step 7 – Specify options and validation routines,” on page 60.

Quick notes

Commonly used keyboard shortcuts

General shortcuts	
F1	Help
F2	Scale image (normal)
F3	Scale image (enlarge)
F4	Scale image (reduce)
CTRL + ↑	Scroll image up (Manager, Scan, and Verify)
CTRL + ↓	Scroll image down (Manager, Scan, and Verify)
CTRL + ←	Scroll image left (Manager, Scan, and Verify)
CTRL + →	Scroll image right (Manager, Scan, and Verify)
CTRL + N	Create new object (Manager) Create new job description (Scan, Interpret, Verify, Transfer)
CTRL + O	Open object (Manager) Open job description (Scan, Interpret, Verify, Transfer)
CTRL + P	Print
ALT + F4	Exit

Pre-production shortcuts (Manager)	
SHIFT + DEL	Delete object
F5	Define field
F6	Define as previous
F7	Copy field pattern
F9	Delete field
CTRL + S	Save form definition
CTRL + M	Read image file

Production shortcuts (Scan, Interpret, Verify, Transfer)	
CTRL + O	Open job description
CTRL + N	New job description
CTRL + E	Edit job description
CTRL + S	Save job description
CTRL + P	Print job description
ALT + S	Start job
ALT + P	Pause job
ALT + C	Continue (resume) job
ALT + T	Stop job
F12	Specify form definition (Scan only)
PGUP	Front page (Scan only)
PGDN	Back page (Scan only)

Production shortcuts (Verify only)	
F12	Specify form definition
ALT + F5	Inspect form
ALT + F7	Form complement
ALT + F9	Delete form
ALT + F10	Skip form
ALT + BACKSPACE	Previous form
CTRL + G	Go to form
CTRL + F5	Show image from file
F5	Edit field
F6	Field complement
F8	Clear field
F9	Delete field
F10	Skip field

Production shortcuts (Verify only)	
CTRL + F9	Delete index chain
CTRL + BACKSPACE	Previous field

Note

Each module has a toolbar with buttons for many of these functions, and more. To see what a button does, place the cursor on the button. A *tool tip* appears, giving an explanation of the function. At the same time, the status bar at the bottom of the screen displays a more detailed description of the button's function.

Five stages to prepare for production

1. Create a form definition (page 31)

   A. Adjust scanner settings (page 33)
      - Paper size
      - DPI (preferably 200)
      - Light and contrast

   B. Scan a blank form or read an image file (page 34)

   C. Save the form definition (page 35)

   D. Define 3 to 5 adjustment fields (characters, lines, etc., that are unique for this specific form) (page 35)

   E. Define a recognition field if necessary (use a preprinted row of characters on your form) (page 37)

   F. Define all other fields (page 38)
      - Field area (double-click or draw)
      - Type of field and field name
      - Format specification
      - Other settings, depending on the field (decimal, rotate, and so on)

   G. Specify options and validation routines (page 60)

   H. Check the validity of the form definition (page 66)

2. Create a standard transaction description (page 66)

   A. Open the transaction description dialog and view a standard transaction description (page 67)

   B. Remove unnecessary transaction fields (page 68)

   C. Edit transaction fields (page 69)

   D. Insert additional transaction fields (page 71)

   E. Change the order of transaction fields (page 72)

   F. Determine the physical layout of the output data (page 73)

   G. Save the standard transaction description and resave the form definition (page 73)

3. Activate and resave the form definition (page 74)

4. Create a job description (page 74)

5. Test the form definition and job description before production (page 78)



Four steps in the production process

1. Scan and identify forms (page 87)

    - Scan is used to scan filled-in paper forms that are based on one or more form definitions created earlier in the Manager. Data from the forms is captured and saved as images in a database.

    - You can check the quality of scanned forms by enlarging thumbnail images. The scanner may also require attention during scanning (for example, if a paper jam occurs or if a form is unidentified).

2. Interpret and validate forms (page 92)

    - The Interpret module both interprets the data from forms and validates that data in accordance with the validation routines you specified in the form definition. There is nothing for you to do once the job is started.

3. Verify forms (page 94)

    - Verify is used to check, supplement, and correct interpreted forms.

    - You can modify the order in which data is presented in Verify and also how data is viewed on-screen.

4. Transfer data from processed forms (page 102)

    - Transfer is used to transfer data from processed forms to a target system, which can be another application in the same computer as FORMS, or it can be in another computer.

Glossary

adjustment field. There are two kinds of adjustment fields: FORMS' traditional adjustment fields and flexible line maps. Traditional adjustment fields are recommended for most form definitions. They are fields containing a character, symbol, or pair of intersecting lines that is unique to a specific form (compared to other forms to be processed at the same time).

FORMS uses adjustment fields to identify and deskew forms and match them to the form definition. All the other fields on a form have a fixed position on the form relative to the adjustment fields.

appendix. A form or document that is attached to an identified form but is itself not identified. Its data is not processed but only saved as an image. Not the same as unidentified form.

archive transaction description. Specifies the format and layout of data you want FORMS to place in a separate transfer file. Archive transaction descriptions are saved to separate files and are optional in Transfer jobs.

background color. A color that you want FORMS to disregard when interpreting a field. The color is removed before interpretation.

barcode field. Contains a barcode to be interpreted.

batch. A stack of forms that you process together in order to (1) set up a scheme for locating forms after archiving; (2) divide forms into manageable quantities; and/or (3) summarize information from each batch in order to check the integrity of the captured data. You can either physically count the forms to be processed together and put them into the scanner's document feeder one stack at a time, or you can put as many forms as you like into the scanner's document feeder and let FORMS count them.

batch data. Values that describe a batch and each form in the batch, such as the batch number and the number of forms in the batch.

Batch data is contained in batch variables. There are standard batch data variables. You can edit these or specify your own in a batch report layout file.

batch definition. A batch definition is a template of sorts that allows you to easily group forms or sets that are to be processed together in batches. Once a batch has been specified in a batch definition, you can add it directly to a job description for processing. A batch definition contains information on the batch size, numbering, prefix (if any), and in which modules the processing of batches is locked.

batch report. A printout of *batch data* that is printed after a batch is scanned.

batch report layout file. A file that contains the batch variables. You use it to change the standard batch variables or specify your own batch variables.

black and white images. FORMS always interprets data from a black and white image, regardless of the number of colors on the original form or how it is displayed on your screen.



Black and white is not the same as grayscale, which typically includes black, white, and various shades of gray. When FORMS converts a color or grayscale image to black and white, dark colors are changed to black and light colors are changed to white.

If the displayed image is grayscale or color, you can adjust the display to see the black and white image that FORMS is interpreting.

buffering. While you are verifying one form, Verify is normally gathering data from the next and storing it in a buffer. If you want Verify to gather data from more than one form at a time, you can change the buffer size. Buffering affects how much data is displayed during mass verification and fast verification, and it reduces the number of calls to the database. If your Verify module is hesitating too much, verification speed might increase with buffering.

character field. An alphabetic or numeric field to be interpreted.

character set. A list of alphabetic, numeric and other characters that FORMS chooses from when interpreting data.

Character sets vary depending on the language to be interpreted. For example, the U.S. character set includes the letters A-Z and common special characters such as the dollar sign. European alphabets include additional letters and different special characters.

A default character set is chosen during installation, and the setting is saved in Ehglobal.ini. However, you can choose a different character set for a specific form definition in the Form definition settings dialog, and you can edit or even create your own character set.

Character sets are saved in Ehcharset.ini.

check digit, **checksum**. Some numbers, such as ID and product numbers, include a control digit that is based on the other digits in the number. A checking routine (usually Modulo 10) performed on the number practically eliminates the risk that digits in the number have been interpreted incorrectly.

comb marks, hash marks. Small vertical marks in a field that guide respondents' placement of handwritten characters, like this:



complement. Supplementary text (max. 512 characters) associated with a field or form that does not directly result from the field or form but is typed manually during verification.

- **field complement**. You can specify conditions under which you want to type text and associate it with a field. When the conditions are met, Verify prompts you to type a text string during the verification process. Then the field has both the value FORMS interpreted for it (unless you change or delete it) and the text associated with it.

- **form complement**. You can specify conditions under which you want to type text and associate it with the form as a whole (rather than a particular field). When the conditions are met, Verify prompts you to type a text string. Then the form has extra text assigned to it.

copy field. An area on the form that you want to save to an image file (in TIFF format) while scanning with Scan or during interpretation of imported image files. Copy fields must be defined last.

data field. Used to store additional information about the form.

database. A collection of saved data that can be read, compared, changed, and resaved. FORMS' databases include the following: form/set/batch definitions, processed forms/sets/batches, users, and tables.

description file. A text file with the extension .tab that FORMS uses to create form definitions automatically. It contains lines of text that describe one or more form definitions, one after the other. It is usually possible to generate a description file automatically using software designed for creating forms.

deskew. See **skew**.

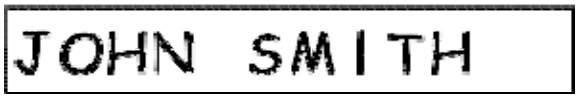
dropout color. A light, solid color (usually pastel green, red, or yellow) used on forms. It is dark enough to be visible to respondents but light enough to be treated as invisible (white) by scanners with this ability.

While filling in the field, respondents are constrained to the white areas of the box, which keeps the characters separate and clear. Characters that extend into the colored areas do not cause problems, because the scanner ignores the dropout color (that is, it sees the color as white), leaving FORMS with only the filled-in box.

What the filled-in field looks like:



What FORMS sees:



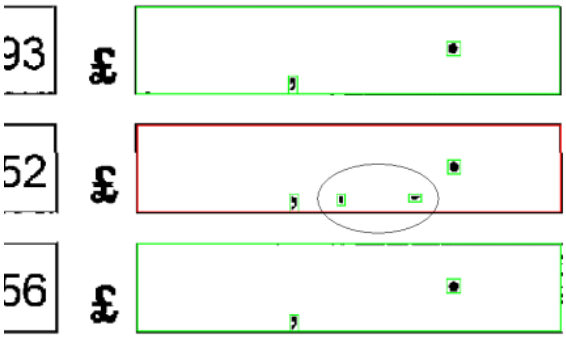
endorser. An endorser is a built-in printer in the scanner that prints a sequence number on each form as it is scanned. If your scanner has an endorser, you can enable it and print a number on each form as it is scanned. Scan can also save the number in FORMS' form database along with other data about the form. This creates a link between the physical form and the scanned form.

events and event handlers. As forms and sets move through the FORMS production modules (Scan, Interpret, Verify and Transfer), events are fired at predefined occasions. An event is a condition or the result of an activity at a specific point in a program.

When an event is fired, FORMS checks to see whether you have defined a handler for the event. An event handler causes execution of a predefined VBA function, DLL function, or both.

exclusion rectangles. If any marks remain within a field when you define it, they are surrounded by green rectangles (exclusion rectangles). Anything appearing within green rectangles is excluded from interpretation.

Sometimes stray marks or noise from the image are mistakenly excluded, like the two marks circled below. If you see anything like this in a field definition, go back to the form definition and delete the marks using FORMS' Erase command. If it is a color form, you can also use Edit color to define the noise as background color. (Right-click the field and select Edit color.)



extension. Extensions are COM (Component Object Model) objects that can be developed in any programming language the supports COM, such as Visual Basic, Java, and C++. With plug-ins, you have two sets of object hierarchies: One that is backward-compatible with earlier versions of FORMS, and a completely new one that offers state-of-the art capabilities, such as smart collections and type libraries.

fast verify. Fast verify and normal mode refer to the way fields are displayed on the screen while you verify forms. Both character fields and object fields can be fast-verified.

field. A box or other area on a form which contains information to be interpreted and processed by FORMS. Examples: A mark field which may contain a check mark to be registered, or a space for a product ID number to be read.

field complement. See **complement**.

field definition. Information for FORMS about where a field is located, what type of data it is expected to contain, how it should be validated, etc. Using the Manager, you create field definitions by marking the area containing the data, then using the **Field** dialog to provide FORMS with the rest of the information. Field definitions are part of form definitions.

field image. A representation of a field taken from a processed form, created during scanning and saved on a computer. FORMS only saves field images in a separate file when you choose to export fields from FORMS or when you define one or more copy fields on your form definition.

field name. You label each field definition with a name consisting of up to 11 characters. Every field name includes an index.

field number. Reflects the order in which you created field definitions on the form definition.

field validation. To validate a field means to check whether its interpreted value is acceptable based on criteria you indicated in the form definition. Field validations are also what we call the criteria themselves, in addition to the act of applying them.

foreground color. A color that you want FORMS to preserve in order to enhance the interpretation. This can include the color of the lines that surround the field (that is, those used by support lines and double-clicked fields) and tick marks, for example.

This does not include the color of the text to be interpreted. It just needs to be in contrast to the background color.

form. A structured document containing spaces for information. Examples: Questionnaires, surveys, mail order forms, tax return forms, checks, pay orders, consignment notes.

form complement. See **complement**.

form definition. A software-based description (like a template) of a form to be processed in FORMS. You create a form definition by using the Manager to scan a blank form, create field definitions and a standard transaction description, make other settings such as status, give the form definition a unique name, and save it to the form definition database.

form image. A representation of a form, created by scanning and saved on a computer. The Scan module saves complete form images in the following cases:

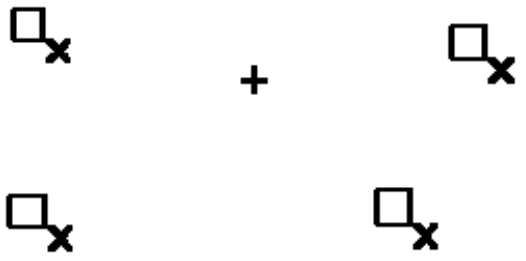
- When you choose to export forms from FORMS. Note that in order to do this, the export form options must be selected in both the Form definition settings and the Export page of the job description dialog.
- When you select one of the Store forms to files options in the Scan job description. Note that in this case the image files are deleted upon transfer.

The image of a form is stored separately from any data interpreted from the form, but the two are linked together by information in FORMS' form database.

Form images, both those exported by FORMS and those created by other means, can also be imported to FORMS.

format. See **specification**.

global field displacement. On a printed form it is a common shift in all field information that is caused by misalignment of the preprinted forms in the printer. In the example below, the field information has been displaced down and to the right.



grayscale image. FORMS always interprets data from a black and white image, regardless of the number of colors on the original form or how it is displayed on your screen.

Black and white is not the same as grayscale, which typically includes black, white, and various shades of gray. When FORMS converts a color or grayscale image to black and white, dark colors are changed to black and light colors are changed to white.

If the displayed image is grayscale or color, you can adjust the display to see the black and white image that FORMS is interpreting.

hash marks. See **comb marks**.

host system. See **target system**.

identification. Identification (recognition) of a form occurs when FORMS matches it to an existing form definition.

Interpretation is when FORMS converts the marks on a form into letters and numbers.

image database. Contains images of fields cut from processed forms. These images are always deleted as each form is transferred.

The image database does not contain images of entire forms, although such image files may exist elsewhere.

image field. Processed without any interpretation. Use them to provide an image of data you want to manually type as a field or form complement (for example, address changes or handwritten messages from customers) or as an image to support verification (for example, product names used to verify uninterpretable product codes).

image file. See **form image**.

imprinter. See **endorser**.

index. Every field name includes an index. FORMS gives solitary fields the index 0. Such fields are called non-indexed fields, though they actually have the index 0.

When several fields contain exactly the same type of information and are to be processed identically, you can define them in a series. Fields in a series receive the same name but different indexes, starting with 1 and increasing in the order the fields were marked or drawn. Such fields are referred to as indexed fields.

In certain cases, you must refer to a field by name. When you refer to an indexed field, the format is *Fieldname*[*index*]. Example: ProductID[3]

Interpret job description. Tells Interpret which forms to interpret and validate.

Interpetation. See **identification**.

job. A group of tasks performed by a specific module on a specific group of forms. A job is set up in a job description. The Scan, Interpret, Verify and Transfer modules perform jobs.

job description. A software-based collection of choices and settings for work to be done by Scan, Interpret, Verify, and Transfer, saved to a `*.job` file. It simplifies work in these production modules, since you set and save your choices, and then simply open the file and click the start button.



You can create all four types mentioned below using the Manager. You can also create them in the respective modules.

- **Scan job description**. Tells FORMS which forms to look for during scanning and how to handle the images.
- **Interpret job description**. Tells FORMS which forms to interpret and validate.
- **Verify job description**. Tells FORMS which forms with which status you want to verify. You can also choose from many options for how you want FORMS to interact with you during the verification process (such as **Use split view**).
- **Transfer job description**. Tells FORMS which data from which forms to transfer, and to where.

mark field. A box that is marked (ticked or checked) to select a choice on a form.

mass verify. A field attribute (a field option you can select when you create the field definition) which allows characters from the interpreted field to be strung together with characters from other fields and displayed for your confirmation in Verify. Allows you to very quickly check FORMS' interpretation of characters.

Mass verify is also a form status and a field status. There is a detailed explanation of form and field statuses in Help.

matrix. A collection of two or more character fields whose exact position cannot be defined.

noise. Speckles, or groups of black pixels surrounded by white pixels, on a scanned image.

object field. For a machine-printed string appearing in an area you cannot define precisely. A typical example would be an adhesive label, sticker or stamp affixed to a form.

output file. See transfer file.

plug-in. See **extensions**.

production modules. The four production modules are Scan, Interpret, Verify and Transfer.

queue. A group of forms, sets, or batches that is split off from the main production flow. You use queues to group items so that they can be processed by a production module according to queue number.

A queue is not a list of items in sequence, and queues are not a sequencing method.

recognition field. A field containing a preprinted string of characters which identify the form as compared to other forms. The preprinted string is erased from the field definition so that it can be found and interpreted when forms are processed. A recognition string is a form number.

record. Data resulting from one processed form or set.

Scan job desription. Tells Scan which forms to look for during scanning and how to handle the images.



SCSI. Small Computer Systems Interface, a standard interface used to connect peripheral equipment (hard disks, CD-ROM, scanners) to PCs. SCSI allows direct connection to the internal data bus, resulting in a very fast data transfer rate.

Selection field. A selection field is an area containing one or more words, characters, or symbols that respondents either circle or mark through (tick) to indicate a response.

A typical example is when respondents are asked how often they do something, and the answer is provided by circling one of the responses. The area containing the alternatives is the selection field. Each alternative within the selection field is an object within the selection field.

series of fields. See **index**.

set. A set is two or more form definitions that are scanned, interpreted, and verified in a group and where the sequence of the forms is important. The forms in a set are treated as one logical unit. Examples: Market research questionnaires consisting of several pages, and forms with enclosures.

set definition. A software-based description of how you want FORMS to process two or more forms as one logical unit. You use the Manager to choose which form definitions are to be included in the set and then indicate the set logic. Then you give the set a name and save it to FORMS' set database. A questionnaire-type set can also have its own transaction description.

set logic. Specifies which, how many, and in what sequence forms can appear in a set. It also determines when a new set begins (called a *set break*).

Skew. The slanting of a scanned image caused when an form is not aligned properly as it passes through the scanner. Using adjustment fields that are placed properly in the form definition, FORMS can normally deskew (that is, straighten) and identify images of forms that are skewed up to 10 degrees, which handles most situations.

snippet. The image of an individual field from a form.

Individual forms that FORMS displays are normally a composite consisting of a template plus snippets. The template, or background, is the one upon which the form definition is based. Snippets are cut from the actual scanned form (or interpreted image), based on the field definitions. They are saved in FORMS' image database and pasted onto the template for viewing. FORMS works this way to save disk space.

specification (of format). One element of most field definitions. Describes how many and what type of characters are expected to appear in a field.

start transaction description. Specifies the format and layout of data you want FORMS to place at the beginning of the transfer file (the same file that data from the standard transaction is being transferred to). Start transaction descriptions are optional in Transfer jobs.

status. A form definition can be *Active*, *Inactive*, or *Deactivated*. Basically, inactive form definitions are in the process of being defined; active form definitions are in production (that is, being processed); and deactivated form definitions are not currently in use but are protected from being altered.

stop transaction description. Specifies the format and layout of data you want FORMS to place at the end of the transfer file (the same file that data from the standard transaction is being transferred to). Stop transaction descriptions are optional in Transfer jobs.



Every form being processed by FORMS also has a status, as does each individual filled-in field on each form. There is a detailed explanation in Help.

substitution rate (error rate). During interpretation, the percentage of characters that are classified with the status Certain but are actually interpreted incorrectly.

support lines. Straight lines along one or more sides of a field that FORMS can use to delineate a field. Interpretation results are improved when you use the **Support lines** option instead of drawing the field area by hand.

system variable. A variable represents a value that is not currently known. Most system variables represent data that stems from an aspect of forms processing, rather than a value from a specific form.

A system variable is usually a hash mark plus a word describing what the variable represents. You use these in transaction descriptions.

Example: #Date represents the date that a form is scanned.

table. In FORMS, you can use tables for validation, translation, and detecting if supplementary manual data entry is required. You can use a single table for all of these purposes, and you can use a single table with different form definitions. You use the Manager to create each table, give it a unique name, and save it to FORMS' database. You can edit tables at any time, either manually or automatically with information from a file.

target system. Transfer transfers data from processed forms to your target system, which makes use of that data or gives you access to it. The target system can be another application in the same computer as FORMS, or it can be in another computer.

transaction. A package of information that FORMS transfers to another computer system, either directly (via DLL or other interface) or by saving it to a transfer file. A transaction usually contains many records whose format is specified by one or more transaction descriptions.

transaction description. A digital template containing transaction fields which specify the content, format and layout of transactions. Using the Manager, you create at least one transaction description for each form definition (and in some cases for set definitions). There are three types of transaction descriptions:

- **standard transaction description**. Specifies the format and layout of the data from processed forms or sets that FORMS will transfer to a target system, regardless of the transfer method. Every Transfer job requires a standard transaction description.

- **start or stop transaction description**. Specifies the format and layout of data you want FORMS to place at the beginning/end of the transfer file (the same file that data from the standard transaction is being transferred to). Start and stop transactions are optional for Transfer jobs.

- **archive transaction description**. Specifies the format and layout of data you want FORMS to place in a separate transfer file. Archive transaction descriptions are optional for Transfer jobs.

transaction field. One line in a transaction description that specifies one item of data that is included in the transaction. That item of data usually consists of a value interpreted from a processed form, a system variable, or headers or other text.



transfer file. After forms have been processed in Scan, Interpret, and Verify, you use Transfer to transfer data from the forms to the target system. This is normally how the forms database is emptied. If you choose to transfer data to an output file, then the transfer file is where Transfer puts the data. Transaction descriptions control the content and layout of that data.

Transfer job description. Tells Transfer which data from which forms to transfer, as well as where to transfer the data.

unidentified form. A form that FORMS failed to identify as belonging to one of the form definitions included in the Scan or Interpret job description.

user definition. The information included for each user defined in the system including their login and complete names, password, and access rights.

Verify job description. Tells Verify which forms you want to verify according to their statuses. You can also adjust many settings that determine how the program interacts with the user during the verification process. For example, you can choose to use the fast verify mode or the normal mode to correct various types of errors.

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